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Note: *This application note was derived from a manuscript originally prepared for publication in EDN Magazine.*

APPENDIX A

ABC's of Probes – Tektronix, Inc.

This appendix, guest written by the engineering staff of Tektronix, Inc., is a distillation of their booklet, "ABC's of Probes". The complete booklet is available, at no charge,

through any Tektronix sales office or call 800-835-9433 ext. 170. For excellent technical background on probe theory see Reference 42.

PART I: UNDERSTANDING PROBES

The vital link in your measurement system

Probes connect the measurement test points in a DUT (device under test) to the inputs of an oscilloscope. Achieving optimized system performance depends on selecting the proper probe for your measurement needs.

Though you could connect a scope and DUT with just a wire, this simplest of connections would not let you realize the full capabilities of your scope. By the same token, a probe that is not right for your application can mean a significant loss in measurement results, plus costly delays and errors.

Why not use a piece of wire?

Good question: There are legitimate reasons for using a piece of wire or, more correctly, two pieces of wire; some low bandwidth scopes and special purpose plug-in amplifiers only provide binding post input terminals, so they offer a convenient means of attaching wires of various lengths.

DC levels associated with battery operated equipment could be measured. Low frequency (audio) signals from the same equipment could also be examined. Some high output transducers could also be monitored. However, this type of connection should be kept away from line-operated equipment for two basic reasons, safety and risk of equipment damage.

Safety: Attachment of hookup wires to line-operated equipment could impose a health hazard, either because the "hot" side of the line itself could be accessed, or because internally generated high voltages could be contacted. In both cases, the hookup wire offers virtually no operator protection, either at the equipment source or at the scope's binding posts.

Risk of Equipment Damage:

Two unidentified hookup wires, one signal lead and one ground, could cause havoc in line-operated equipment. If the "ground" wire is attached to **any** elevated signal in line-operated equipment, various degrees of damage will result simply because both the scope and the equipment are (or should be) on the same three-wire outlet system, and short-circuit continuity is completed through one common ground.

Performance Considerations:

In addition to the hazards just mentioned, there are two major performance limitations associated with using hookup wires to transfer the signal to the scope: circuit loading and susceptibility to external pickup.

Circuit Loading: This subject will be discussed in detail later, but circuit loading by the test equipment (scope-probe) is a combination of resistance and capacitance. Without the benefit of using an attenuator (10X) probe, the loading on the device under test (DUT) will be 1M ohm (the scope input resistance) and more than 15 picofarad (15pF), which is the typical scope input capacitance plus the stray capacitance of the hookup wire.

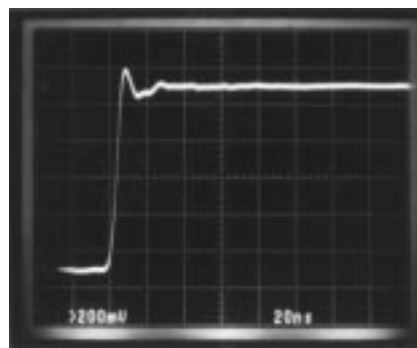


Figure 1-1

Figure 1-1 shows what a "real world" signal from a 500 ohm impedance source looks like when loaded by a 10M ohm, 10 pF probe:

the scope-probe system is 300MHz. Observed risetime is 6 nSec.

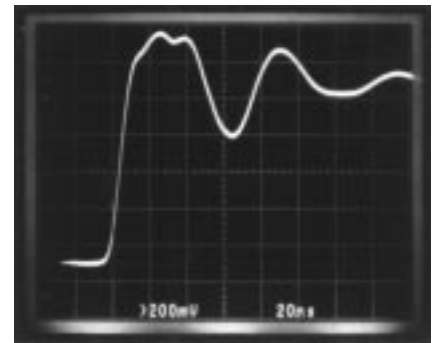


Figure 1-2

Figure 1-2 shows what happens to the same signal when it is accessed by two 2-meter lengths of hookup wire: loading is 1M ohm (the scope input resistance) and about 20 pF (the scope input capacitance, plus the stray capacitance of the wires). Observed risetime has slowed to 10 nSec and the transient response of the system has become unusable.

Susceptibility to External Pickup: An unshielded piece of wire acts as an antenna for the pickup of external fields, such as line frequency interference, electrical noise from fluorescent lamps, radio stations and signals from nearby equipment. These signals are not only injected into the scope along with the wanted signal, but can also be injected into the device under test (DUT) itself.

The source impedance of the DUT has a major effect on the level of interference signals developed in the wire. A very low source impedance would tend to shunt any induced voltages to ground, but high frequency signals could still appear at the scope input and mask the wanted signal. The answer, of course, is to use a probe which, in addition to its other features, provides coaxial shielding of the center conductor and virtual elimination of external field pickup.

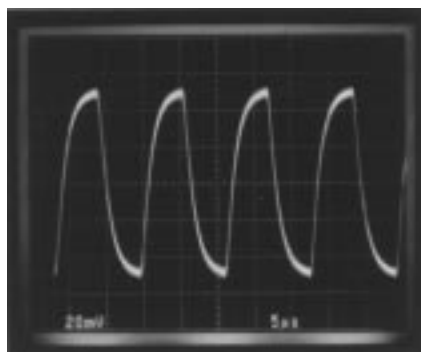


Figure 1-3

Figure 1-3 shows what a low level signal from a high impedance source (100mV from 100K ohm) looks like when accessed by a 300MHz scope-probe system. Loading is 10M ohm and 10 pF. This is a true representation of the signal, except that probe resistive loading has reduced the amplitude by about 1%; the observed high frequency noise is part of the signal at the high impedance test point and would normally be removed by using the BW (bandwidth) limit button on the scope. (See Figure 1-4.)

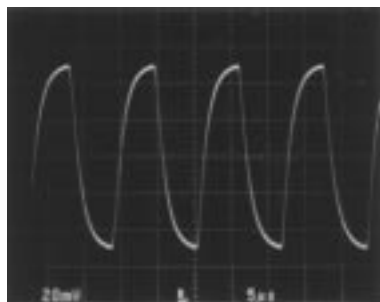


Figure 1-4

If we look at the same test point with our pieces of wire, two things happen. The amplitude drops due to the increased resistive and capacitive loading, and noise is added to the signal because the hookup wire is completely unshielded. (See Figure 1-5)

Most of the observed noise is line frequency interference from fluorescent lamps in the test area.

Probably the most annoying effect of using hookup wire to observe high frequency signals is its unpredictability. Any touching or rearrangement of the leads can produce different and nonrepeatable effects on the observed display.

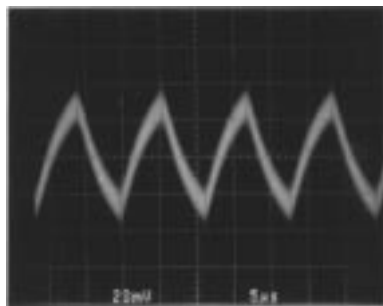


Figure 1-5

Benefits of using probes

Not all probes are alike and, for any specific application, there is no one ideal probe; but they share common features and functions that are often taken for granted.

Probes are convenient. They bring a scope's vertical amplifier to a circuit. Without a probe, you would either need to pick up a scope and attach it to a circuit, or pick up the circuit and attach it to the scope. Properly used, probes are convenient, flexible and safe extensions of a scope.

Probes provide a solid mechanical connection. A probe tip, whether it's a clip or a fine solid point, makes contact at just the place you want to examine.

Probes help minimize loading. To a certain extent, all probes load the DUT—the source of the signal you are measuring. Still, probes offer the best means of making the connections needed. A simple piece of wire, as we have just seen, would severely load the DUT; in fact, the DUT might stop functioning altogether.

Probes are designed to minimize loading. Passive, non-attenuating 1X probes offer the highest capacitive loading of any probe type—even these, however, are designed to keep loading as low as possible.

Probes protect a signal from external interference. A wire connection, as described earlier, in addition to loading the circuit, would act as an antenna and pick up stray signals such as 60Hz power, CBers, radio and TV stations. The scope would display these stray signals as well as the signal of interest from the DUT.

Probes extend a scope's signal amplitude-handling ability. Besides reducing capacitive and resistive loading, a standard passive 10X

(ten times attenuation) probe extends the on-screen viewability of signal amplitudes by a factor of ten.

A typical scope minimum sensitivity is 5V/division. Assuming an eight-division vertical graticule, a 1X probe (or a direct connection) would allow on-screen viewing of 40V p-p maximum. The standard 10X passive probe provides 400V p-p viewing. Following the same line, a 100X probe should allow 4kV on-screen viewing. However, most 100X probes are rated at 1.5kV to limit power dissipation in the probe itself.

Check the specs. Bandwidth is the probe specification most users look at first, but plenty of other features also help to determine which probe is right for your application. Circuit loading, signal aberrations, probe dynamic range, probe dimensions, environmental degradation and ground-path effects will all impact the probe selection process, as discussed in the pages that follow.

By giving due consideration to probe characteristics that your application requires, you will achieve successful measurements and derive full benefit from the instrument capabilities you have at hand.

How probes affect your measurements

Probes affect your measurements by loading the circuit you are examining. The loading effect is generally stated in terms of impedance at some specific frequency, and is made up of a combination of resistance and capacitance.

Source Impedance. Obviously, source impedance will have a large impact on the net effect of any specific probe loading. For example, a device under test with a near zero output impedance would not be affected in terms of amplitude or risetime to any significant degree by the use of a typical 10X passive probe. However, the same probe connected to a high impedance test point, such as the collector of a transistor, could affect the signal in terms of risetime and amplitude.

Capacitive Loading. To illustrate this effect, let's take a pulse generator with a very fast risetime. If the initial risetime was assumed to be zero ($t_r = 0$), the output t_r of the generator would be limited by the

associated resistance and capacitance of the generator. This integration network produces an output rise time equal to $2.2 RC$. This limitation is derived from the universal time-constant curve of a capacitor.

Figure 1-6 shows the effect of internal source resistance and capacitance on the equivalent circuit. At no time can the output risetime be faster than $2.2 RC$ or 2.2 nSec.

If a typical probe is used to measure this signal, the probe's specified input capacitance and resistance is added to the circuit as shown in Figure 1-7.

Because the probe's 10M ohm resistance is much greater than the generator's 50 ohm output resistance, it can be ignored.

Figure 1-8 shows the equivalent circuit of the generator and probe, applying the $2.2 RC$ formula again. The actual risetime has slowed from 2.2 nSec. to 3.4 nSec.

Percentage change in risetime due to the added probe tip capacitance:

$$\% \text{ change} = \frac{tr_2 - tr_1}{tr_1} \times 100 = \frac{3.4 - 2.2}{2.2} \times 100 = 55\%$$

Another way of estimating the affect of probe tip capacitance on a source is to take the ratio of probe tip capacitance (marked on the probe compensation box) to the known or estimated source capacitance.

Using the same values:

$$\frac{C_{\text{probe tip}}}{C_1} \times 100 = \frac{11 \text{ pF}}{20 \text{ pF}} \times 100 = 55\%$$

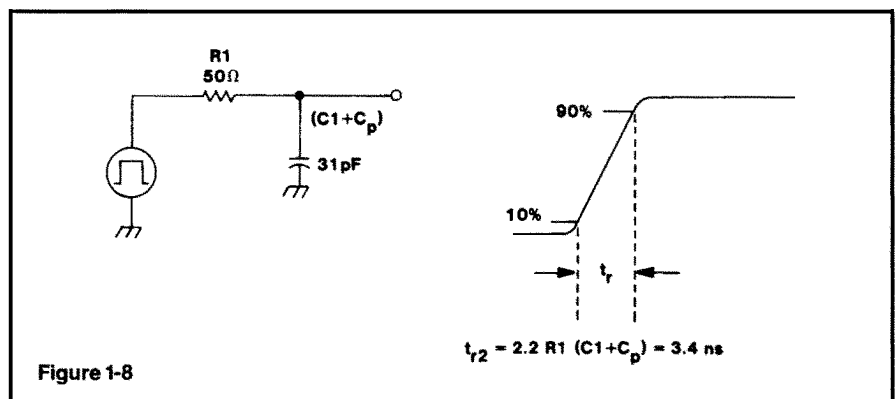
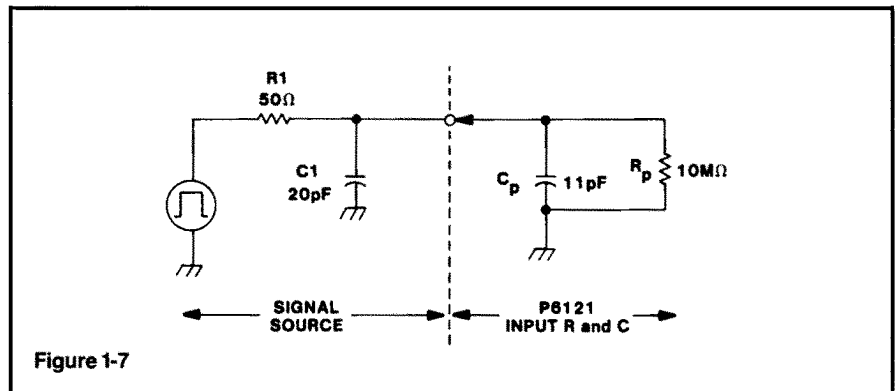
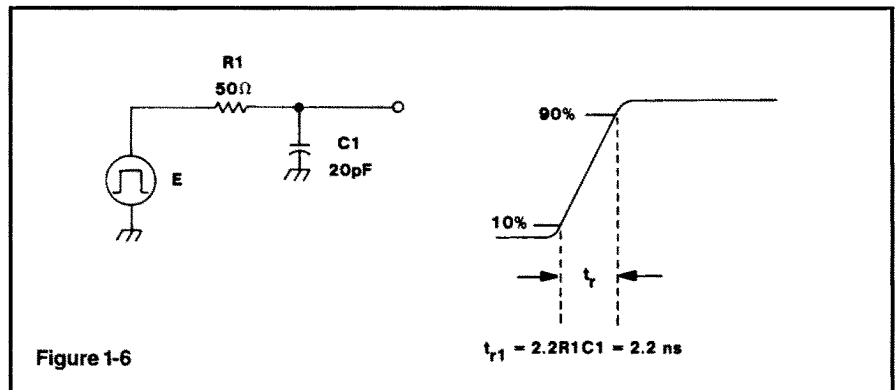
To summarize, any added capacitance slows the source risetime when using high impedance passive probes. In general, the greater the attenuation ratio, the lower the tip capacitance. Here are some examples:

Probe	Attenuation	Tip Capacitance
Tektronix P6101A	X1	54 pF
Tektronix P6105A	X10	11.2 pF
Tektronix P6007	X100	2 pF

Capacitive Loading: Sinewave.

When probing continuous wave (CW) signals, the probe's capacitive reactance at the operating frequency must be taken into account.

The total impedance, as seen at the probe tip, is designated R_p and is a function of frequency. In addition to the capacitive and resistive elements, designed-in inductive elements serve to offset the pure capacitive loading to some degree.



Curves showing typical input impedance vs frequency, or typical X_p and R_p vs frequency are included in most Tektronix probe instruction manuals. Figure 1-9A shows the typical input impedance and phase relationship vs frequency of the Tektronix P6203 Active Probe. Note that the 10 K Ω input impedance is maintained to almost 10 MHz by careful design of the associated resistive, capacitive and inductive elements.

Figure 1-9B shows a plot of X_p and R_p vs frequency for a typical 10 M Ω passive probe. The dotted line (X_p) shows capacitive reactance vs frequency. The total loading is again offset by careful design of the associated R, C and L elements.

If you do not have ready access to the information and need a worst-case guide to probe loading, use the following formula:

$$X_p = \frac{1}{2\pi FC}$$

X_p = Capacitive reactance (ohms)
 F = Operating frequency
 C = Probe tip capacitance (marked on the probe body or compensation box.)

For example, a standard passive 10M ohm probe with a tip capacitance of 11 pF will have a capacitive reactance (X_p) of about 290 ohm at 50MHz.

Depending, of course, on the source impedance, this loading could have a major effect on the

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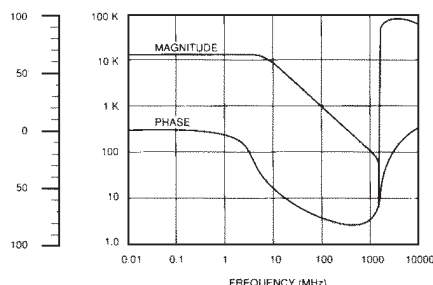


Figure 1-9A. Typical Input Impedance vs Frequency for the Tektronix P6203 Active Probe

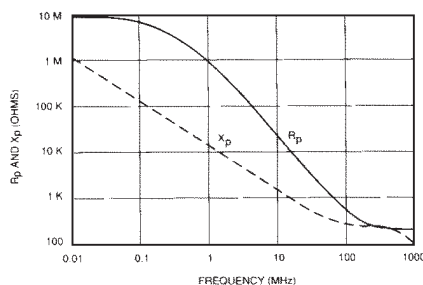


Figure 1-9B. X_p and R_p vs Frequency for a Typical 10 MΩ Passive Probe

signal amplitude (by simple divider action), and even on the operation of the circuit itself.

Resistive Loading. For all practical purposes, a 10X, 10M ohm passive probe has little effect on today's circuitry in terms of resistive loading, however, they do carry a trade-off in terms of relatively high capacitive loading as we have previously discussed.

Low Z Passive Probes. A "Low Z" passive probe offers very low tip capacitance at the expense of relatively high resistive loading. A typical 10X "50 ohm" probe has an input C of about 1 pF and a resistive loading of 500 ohm: Figure 1-10 shows the circuit and equivalent model of this type of probe.

This configuration forms a high frequency 10X voltage divider because, from transmission line theory, all that the 450 ohm tip resistor "sees" looking into the cable is a pure 50 ohm resistance, no C or L component. No low frequency compensation is necessary because it is not a capacitive divider. Low Z probes are typically high bandwidth (up to 3.5GHz and risetimes to 100 pS) and are best suited for making rise-time and transit-time measurements. They can, however, affect the pulse amplitude by simple resistive divider action between the source and the load (probe). Because of its resistive loading effects, this type of probe

performs best on 50 ohm or lower impedance circuits under test.

Note also that these probes operate into 50 ohm scope inputs only. They are typically teamed up with fast (500MHz to 1GHz) real time scopes or with scopes employing the sampling principle.

Bias-Offset Probes. A Bias/Offset probe is a special kind of Low Z design with the capability of providing a variable bias or offset voltage at the probe tip.

Bias/Offset probes like the Tektronix P6230 or P6231 are useful for probing high speed ECL circuitry, where resistive loading could upset the operating point. These special probes are fully described in Part 3; under Advanced Probing Techniques.

The Best of Both Worlds. From the foregoing, it can be seen that the totally "non-invasive" probe does not exist. However, one type of probe comes close—the active probe.

Active probes are discussed in the Tutorial section, but in general, they provide low resistance loading (10M ohm) with very low capacitive loading (1 to 2 pF). They do have trade-offs in terms of limited dynamic range, but under the right conditions, do indeed offer the best of both worlds.

Bandwidth. Bandwidth is the point on an amplitude versus frequency curve where the measurement system is down 3dB from a starting (reference) level. Figure 1-11 shows a typical response curve of an oscilloscope system.

Scope vertical amplifiers are designed for a Gaussian roll-off at the high end (a discussion of Gaussian response is beyond the scope of this primer). With this type of response, risetime is approximately related to bandwidth by the following equation:

$$Tr = \frac{.35}{BW} \quad \text{or, for convenience:} \quad \text{Risetime (nanoseconds)} = \frac{350}{\text{Bandwidth (MHz)}}$$

It is important to note that the measurement system is -3dB (30%) down in amplitude at the specified bandwidth limit.

Figure 1-12 shows an expanded portion of the -3dB area. The horizontal scale shows the input frequency derating factor necessary to obtain accuracies better than 30% for a specific bandwidth scope. For example, with no derating, a "100MHz" scope will have up to a 30% amplitude error at 100MHz (1.0 on the graph). If this scope is to have an amplitude accuracy better than 3%, the input frequency must be limited to about 30MHz (100MHz X .3).

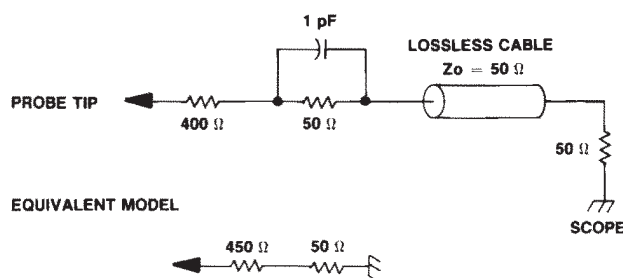


Figure 1-10

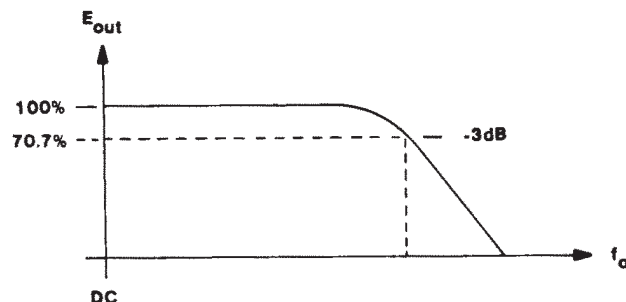


Figure 1-11

For making amplitude measurements within 3% at a specific frequency, choose a scope with at least four times the specified bandwidth as a general rule of thumb.

Probe Bandwidth. All probes are ranked by bandwidth. In this respect, they are like scopes or other amplifiers that are ranked by bandwidth. In these cases we apply the square root of the sum of the squares formula to obtain the "system risetime." This formula states that:

$$\text{Risetime system} = \sqrt{\text{Tr}_2^2 \text{ displayed} + \text{Tr}_2^2 \text{ source}}$$

Passive probes do not follow this rule and should not be included in the square root of the sum of the squares formula.

Tektronix provides a probe bandwidth ranking system that specifies "the bandwidth (frequency range) in which the probe performs within its specified limits. These limits include: total aberrations, risetime and swept bandwidth."

Both the source and the measurement system shall be specified when checking probe specifications (see Test Methods, this page).

In general, a Tektronix "100MHz" probe provides 100MHz performance (–3dB) when used on a compatible 100MHz scope. In other words, it provides full scope bandwidth **at the probe tip**.

However, not all probe/scope systems can follow this general rule. Refer to the sidebar, "Scope Bandwidth at the Probe Tip?"

Figure 1-13 shows examples of Tektronix scopes and their recommended passive probes.

Test Methods: As with all specifications, matching test methods must be employed to obtain specified performance. In the case of bandwidth and risetime measurements, it is essential to connect the probe to a properly terminated source. Tektronix specifies a 50 ohm source terminated in 50 ohm, making this a 25 ohm source impedance. Furthermore, the probe must be connected to the source via a proper probe tip to BNC adaptor. (Figure 1-14).

Figure 1-14 shows an equivalent circuit of a typical setup. The displayed risetime should be a 3.5 nSec or faster.

Figure 1-15 shows an equivalent circuit of a typical passive probe connected to a source.

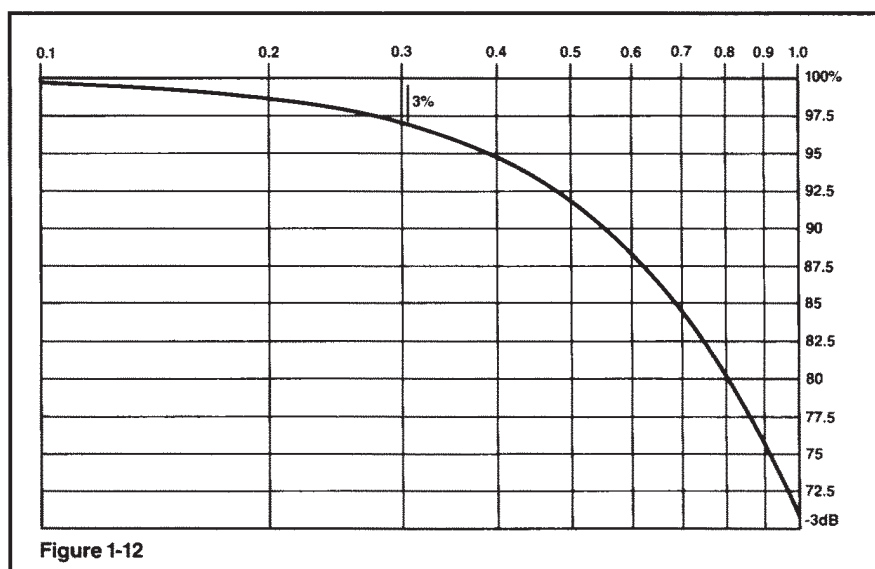


Figure 1-12

SCOPE	BW (1 M Ω input)	PROBE	BW	SYSTEM
2235	100	P6109	150	100
2245A	100	P6109	150	100
2246A	100	P6109	150	100
2445B	150	P6133	150	150
		Opt 25		
485	350	P6106A	250	250
2465B	400	P6137	400	400
2467B	400	P6137	400	400

Figure 1-13

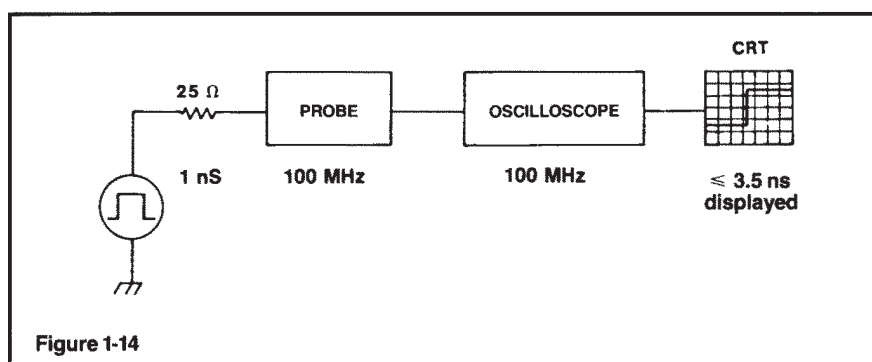


Figure 1-14

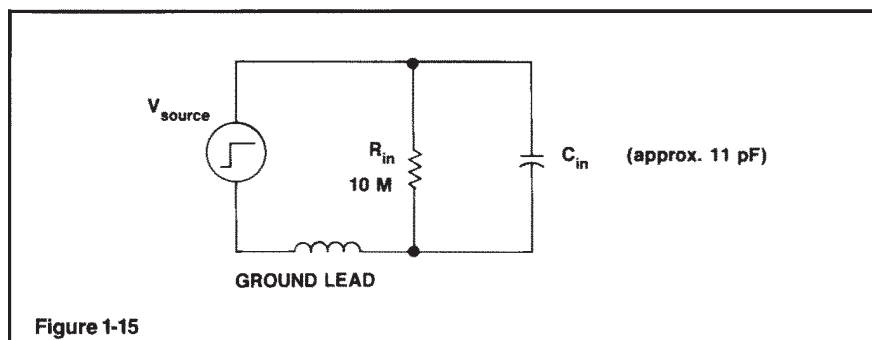


Figure 1-15

Scope Bandwidth at the Probe Tip?

Most manufacturers of general-purpose oscilloscopes that include standard accessory probes in the package, promise and deliver the advertised scope bandwidth **at the probe tip**.

For example, the Tektronix 2465B 400 MHz Portable Oscilloscope and its standard accessory P6137 Passive Probes deliver 400 MHz (-3db) at the probe tip.

However, not all high performance scopes can offer this feature, even when used with their recommended passive probes. For example, the Tektronix 11A32 400 MHz plug-in has a system bandwidth of 300 MHz when used with its recommended P6134 passive probe. This is simply because even the highest impedance passive probes are limited to about 300 to 350 MHz, while still meeting their other specifications.

It is important to note that the above performance is only obtainable under strictly controlled, and industry recognized conditions; which states that the signal must originate from a $50\ \Omega$ back-terminated source ($25\ \Omega$), and that the probe must be connected to the source by means of a probe tip to BNC (for other) adaptor.

This method ensures the shortest ground path and necessary low impedance to drive the probe's input capacitance, and to provide the specified bandwidth at the signal acquisition point, the probe tip.

Real-world signals rarely originate from $25\ \Omega$ sources, so less than optimum transient response and bandwidth should be expected when measuring higher impedance circuits.

How ground leads affect measurements

A ground lead is a wire that provides a local ground-return path when you are measuring any signal. An inadequate ground lead (one that is too long or too high in inductance) can reduce the fidelity of the high frequency portion of the displayed signal.

What grounding system to use.

When making **any** measurement, some form of ground path is required to make a basic two-terminal connection to the DUT. If you want to check the presence or absence of signals from low-frequency equipment, **and** if the equipment is line-powered and plugged into the same outlet system as the scope, then the common 3-wire ground system provides the signal ground return. However, this indirect route adds inductance in the signal path—it can also produce ringing and noise on the displayed signal and is not recommended.

When making any kind of absolute measurement, such as amplitude, risetime or time delay measurements, you should use the shortest grounding path possible, consistent with the need to move the probe among adjacent test points. The ultimate grounding system is an in-circuit ECB (etched circuit board) to probe tip adaptor. Tektronix can supply these for either miniature, compact or subminiature probe configurations.

Figure 1-15 shows an equivalent circuit of a typical passive probe connected to a source. The ground lead L and C_{in} form a series resonant circuit with only $10\text{M}\ \Omega$ for damping. When hit with a pulse, it will ring. Also, excessive L in the ground lead will limit the changing current to C_{in} , limiting the risetime.

Without going into the mathematics, an 11pF passive probe with a 6-inch ground lead will ring at about 140MHz when excited by a fast pulse. As the ring frequency increases, it tends to get outside the passband of the scope and is greatly attenuated. So to increase the ring frequency, use the shortest ground lead possible and use a probe with the lowest input C .

Probe Ground Lead Effects. The effect of inappropriate grounding methods can be demonstrated several ways. Figs. 1-16A, B and C show the effect of a 12-inch ground lead when used on various bandwidth scopes.

In Figure 1-16A, the display on the 15MHz scope looks OK because the ringing aberrations are beyond the passband of the instrument and are greatly attenuated. Figs. 1-16B and C show what the same signal looks like on 50MHz and 100MHz scopes.

Even with the shortest ground lead, the probe-DUT interface has the **potential** to ring. The potential to ring depends on the **speed** of the step function. The ability to **see** the resultant ringing oscillation depends on the scope system bandwidth.

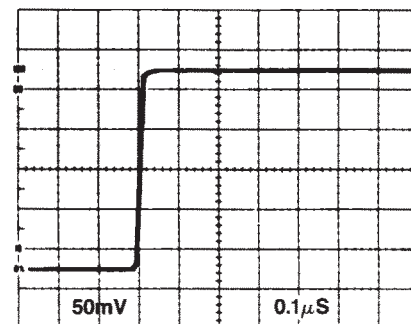


Figure 1-16A

Scope BW = 15MHz
Ground lead 12 inches

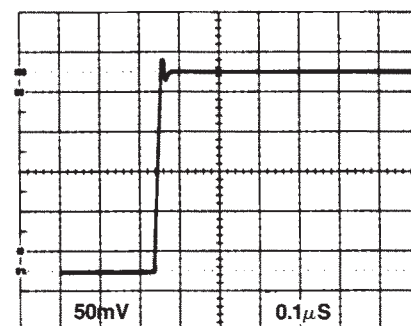


Figure 1-16B

Scope BW = 50MHz
Ground lead 12 inches

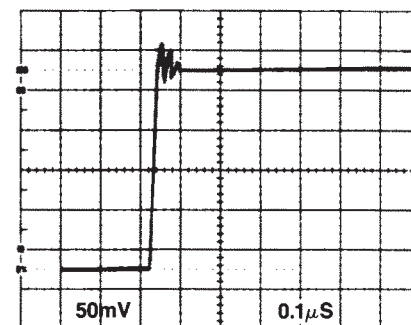


Figure 1-16C

Scope BW = 100MHz
Ground lead 12 inches

Figs. 1-17A through F show the effects of various grounding methods and ground lead lengths on the display of a very fast pulse. This is the most critical way of looking at ground lead effects: we used a fast pulse, with a risetime of about 70 pico seconds and a fast (400 MHz) scope with a matching P6137 probe.

Fig. 1-17A shows the input pulse under the most optimum conditions when using 50 ohm coax cable. Scope: the Tektronix 2465B with 50 ohm input and 50 ohm cable from a 50 ohm source. Displayed risetime is < 1 nSec.

Fig. 1-17B shows the same signal when using the scope-probe combination under the most optimum conditions. A BNC to probe adaptor or an in-circuit test jack provides a coaxial ground that surrounds the probe ground ring. This system provides the shortest probe ground connection available. Displayed risetime is < 1 nSec.

Figures 1-17C through E show the effects of longer ground leads on the displayed signal. Fig. 1-17C shows the effect of a short semi-flexible

ground connection, called a "Z" lead. Finally, Fig. 1-17F shows what happens when no probe ground lead is used.

How probe design affects your measurements

Probes are available in a variety of sizes, shapes and functions, but they do share several main features: a probe head, coaxial cable and either a compensation box or a termination.

The probe head contains the signal-sensing circuitry. This circuitry may be passive (such as a 9-M ohm resistor shunted by an 11 pF capacitor in a passive voltage probe or a 125-turn transformer secondary in a current probe); or active (such as a source follower or Hall generator) in a current probe or active voltage probe.

The coaxial cable couples the probe head output to the termination. Cable types vary with probe types.

The termination has two functions:

- to terminate the cable in its characteristic impedance.
- to match the input impedance of the scope.

The termination may be passive or active circuitry. For easy connection to various test points, many probes feature interchangeable tips and ground leads.

A unique feature of most Tektronix probes is the Tektronix-patented coaxial cable that has a resistance-wire center conductor. This distributed resistance suppresses ringing caused by impedance mismatches between the cable and its terminations when you're viewing fast pulses on wideband scopes.

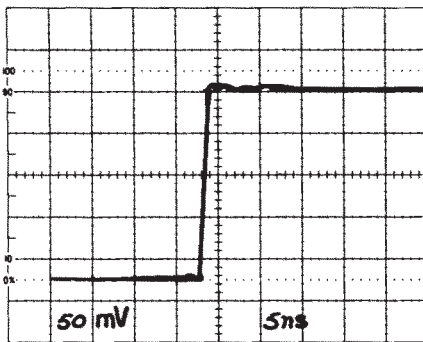


Figure 1-17A
50 ohm Source/Cable/2465B/50 ohm input

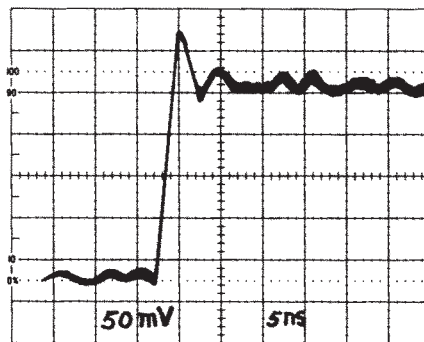


Figure 1-17C
P6137 - Probe/Z Ground Tr = 1.5 nS

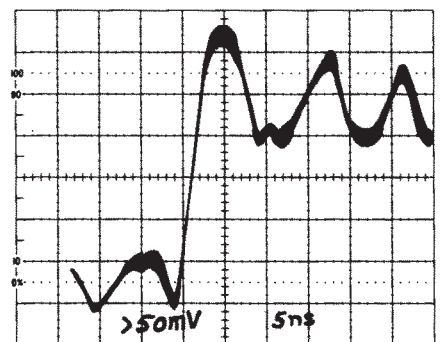


Figure 1-17E
P6137 - Probe/6" Gnd Lead Tr = 4 nS

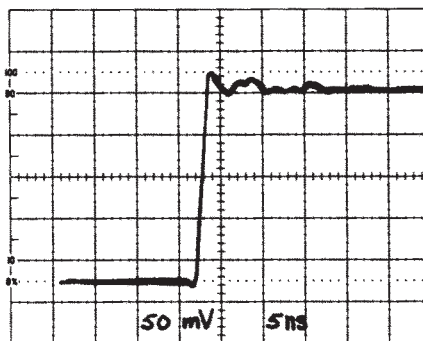


Figure 1-17B
P6137-BNC/Probe Adaptor Tr = < 1 nS

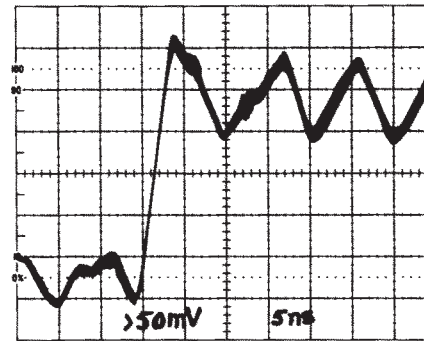


Figure 1-17D
P6137 - Probe/3" Gnd Lead Tr = 4 nS

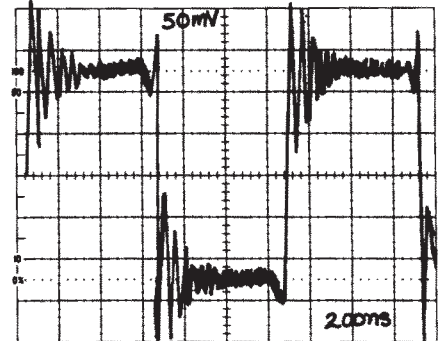


Figure 1-17F
No Ground Lead

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PART II: EFFECTS OF PROBE COMPENSATION — UNDERSTANDING PROBES

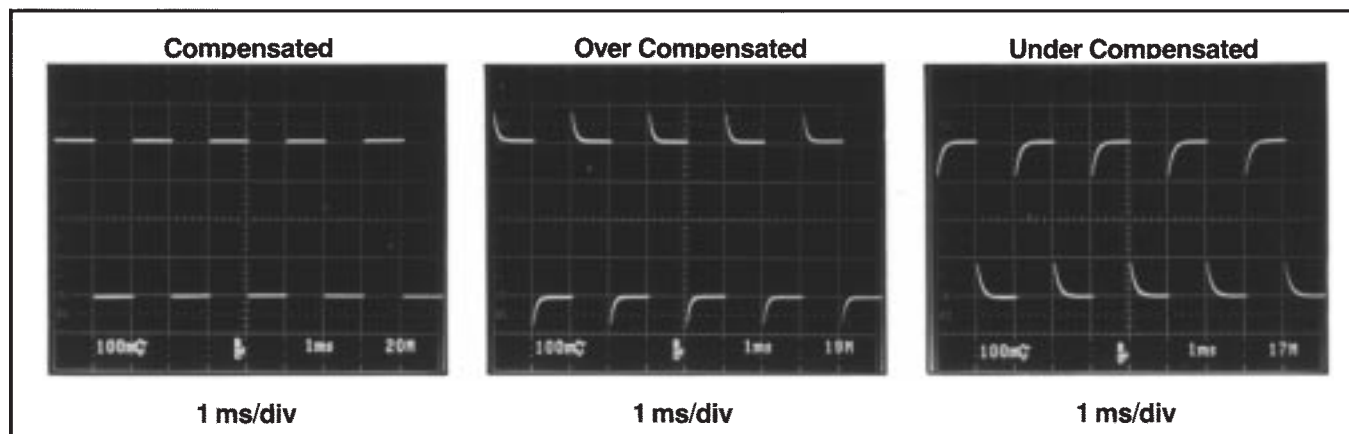


Figure 2-1. Shows the display associated with correctly and incorrectly compensated probes.

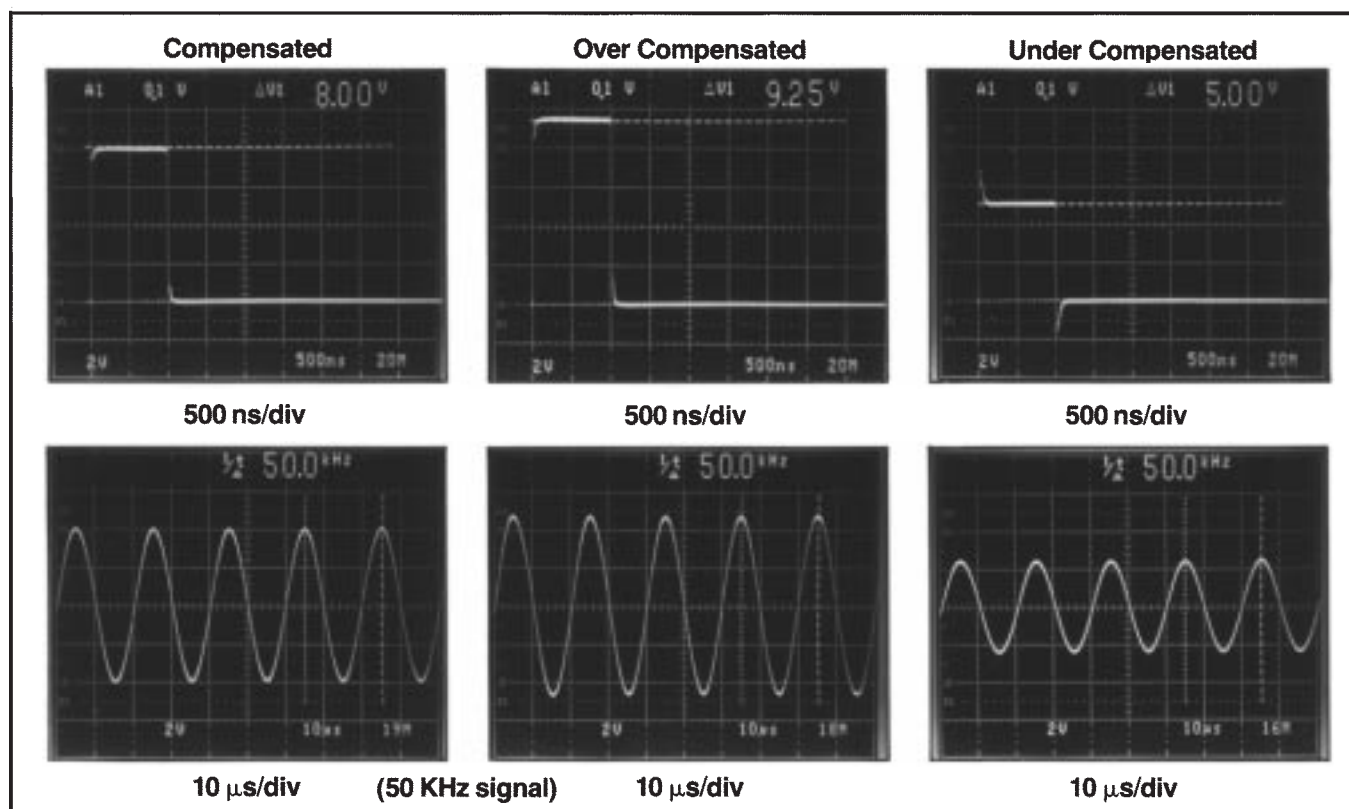


Figure 2-2. Shows the effects on faster pulses and sinewaves when an incorrectly compensated probe is used. Note that the much faster sweep rates used to correctly view these waveforms does not warn the user of an adjustment problem.

Tips on using probes

Compensating the probe. The most common mistake in making scope measurements is forgetting to compensate the probe. Improperly compensated probes can distort the waveforms displayed on the scope. The probe should be compensated as it will be used when you make the measurement.

The basic low frequency compensation (L.F. comp.) procedure is simple:

- Connect the probe tip to the scope CALIBRATOR (refer to Scope Calibrator Outputs.)
- Switch the channel 1 input coupling to dc.
- Turn on the scope and move the CH1 VOLTS/DIV switch to pro-

duce about four divisions of vertical display.

- Set the sweep rate to 1mSec/div. (for line-driven calibrators see Scope Calibrator Outputs.)
- Use a non-metallic alignment tool to turn the compensation adjust until the tops and bottoms of the square-wave are flat.

PART III: ADVANCED PROBING TECHNIQUES

Introduction:

In Part III we will examine some of the more advanced probing techniques associated with accessing high frequency and complex signals, such as fast ECL, waveforms offset from ground, and true differential signals.

Most of the techniques to be described follow recommended practices outlined throughout this Booklet, and to a large extent involve proper grounding techniques.

Workers in the audio and relatively low frequency fields may wonder what all the fuss is about, and may comment "I don't have any of these problems," or "I can't see any difference when I use different ground lead lengths, or even when I leave the ground lead completely off?"

In order to see aberrations caused by poor grounding techniques, two conditions must exist:

1. The scope system bandwidth must be great enough to handle the high frequency content existing at the probe tip.
2. The input signal must contain enough high frequency information (fast risetime) in order to cause ringing and aberrations due to poor grounding techniques.

To illustrate these points, a 20 MHz scope was used to access a 1.7 nS pulse by using a standard passive probe with a 6" ground lead.

NOTE: A fast scope can be made into a slow scope simply by pushing the Bandwidth Limit (B/W Limit) button?

We used a 350 MHz scope with a 20 MHz B/W Limit function.

Figure 3-1 shows the resultant clean displayed pulse with a risetime of about 20 nS (17.5 MHz).

This display does not represent conditions actually existing at the probe tip, because the 20 MHz measurement system cannot "see" what's really happening.

Figure 3-2 shows what the probe tip signal really looks like when a 350 MHz scope is used under the same conditions (B/W Limit off).

The observed risetime has improved to about 2 nS, but we have serious problems with ringing and aberrations, caused by incorrect grounding techniques.

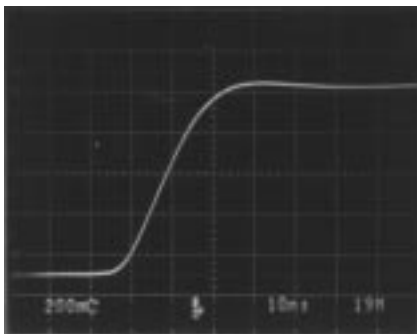


Figure 3-1. Resultant clean, but incorrect display caused by inadequate scope system bandwidth.

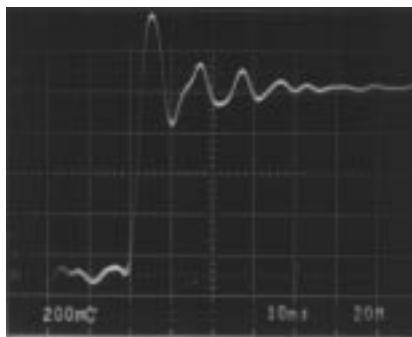


Figure 3-2. The same input signal as shown in figure 3-1, but accessed by a 350 MHz system bandwidth scope (same 6" ground lead).

The problem can now be seen because the scope system bandwidth is great enough to pass and display all the frequency content existing at the probe tip.

To further stress the points about high frequency content and scope system bandwidth, let's assume an input pulse with a risetime of about 20 nS. If the signal is accessed by the same probe /6" ground lead /350 MHz system, it would look very much like the display in figure 3-1.

There would be no frequency content higher than 17.5 MHz (20 nS Tr). The 6" ground lead would not ring, and would therefore be the correct choice for accessing this relatively slow signal.

In the following sections we discuss how to recognize signal acquisition problems, and how to avoid them.

Techniques for probing ECL, high speed 50 Ω environments, and accessing true differential signals are also discussed.

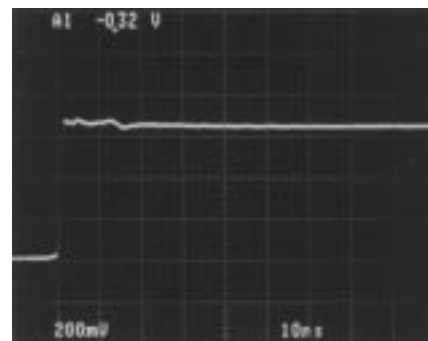


Figure 3-3. 1 nS Tr pulse accessed via an ECB to Probe Tip Adaptor (test point)

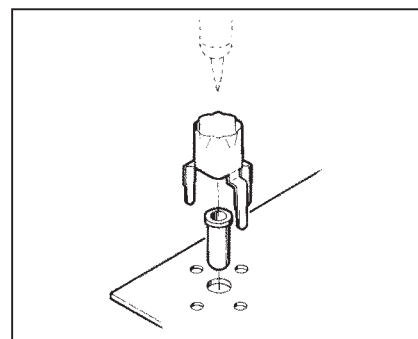


Figure 3-4. Typical ECB to Probe Tip Adaptor installation

Probe Ground Lead Effects. In Part I we discussed the basic need for probe grounding, and showed several different ways of looking at the effects of correct, and incorrect probe grounding.

In this section, we will expand upon these techniques and show how to identify problem areas.

When a probe (high Z, low Z, passive or active) is connected to the circuit under test via an ECB to Probe Tip Adaptor (test point), the coaxial environment existing at the probe tip is extended through the adaptor to the signal pick-off point, and to the ECB ground plane (or device ground).

Figure 3-3 shows what a typical 1 nS Tr pulse looks like when a suitable probe is connected to the circuit via an ECB to Probe Tip Adaptor.

Figure 3-4 shows a typical ECB to Probe Tip Adaptor (test point) installation.

These test points are available in three sizes to accept miniature, compact or sub-miniature series probes.

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If a flexible ground lead is used in place of the ECB to Probe Tip Adaptor, the 1 nS Tr input step (with high frequency content up to 350 MHz) will cause the ground lead to ring at a frequency determined by the ground lead inductance and the probe tip and source capacitance.

Figure 3-5 shows the effect of using a 6" ground lead to make the ground connection.

The ring frequency for the 6" ground lead/probe tip C combination is 87.5 MHz. This signal is injected in series with the wanted signal and appears at the probe tip, as shown in figure 3-6.

Unfortunately, the problem is not this simple.

The probe's coaxial environment has been disrupted at the signal acquisition point by ground lead inductance, and is no longer correctly terminated (for high speed signal acquisition).

This abrupt transition leaves the probe's outer shield susceptible to ring frequency injection (the ground lead inductance is in series with the outer braid)

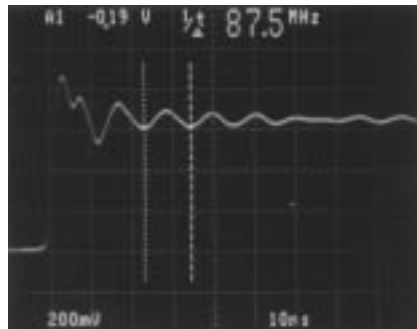


Figure 3-5. Effect of a 6" ground lead on a 1 nS Tr input step.

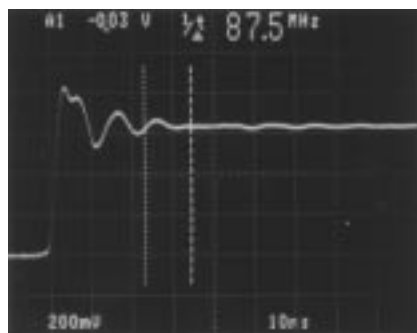


Figure 3-7. The same setup as in figure 3-5, except that the probe cable has been repositioned, and a hand has been placed over part of the probe cable.

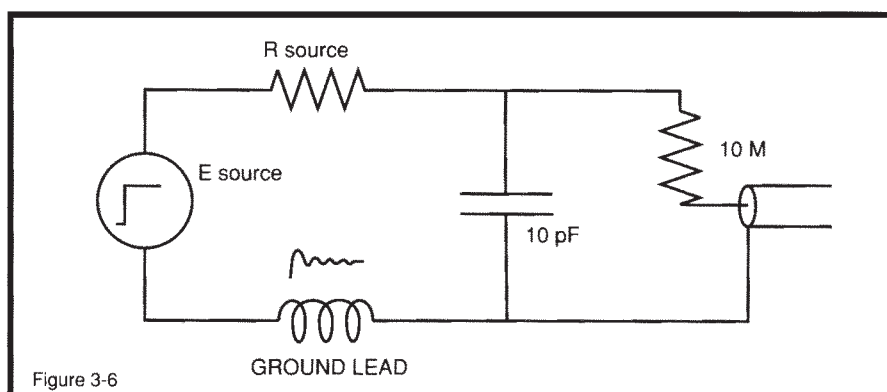


Figure 3-6. Equivalent circuit, ground lead inductance (excess inductance).

The now unterminated probe cable system develops reflections, which intermix with the ring frequency and the signal to produce a multitude of problems and unpredictable results.

Herein lies the key to the identification of ground lead problems.

Figure 3-7 shows exactly the same setup as in figure 3-5, except that the probe cable has been moved, and a hand has been placed over part of the probe cable.

KEY: If touching or moving the probe cable produces changes in the display, you have a probe grounding problem.

A correctly grounded (terminated) probe should be completely insensitive to cable positioning or touch.

Ground Lead Length. All things being equal, the shortest ground lead produces the highest ring frequency.

If the lead is very short, the ring frequency might be high enough to be outside the passband of the scope, and/or the input frequency content may not be high enough to stimulate the ground lead's resonant circuit.

In all cases, the shortest ground lead should be used, consistent with the need for probe mobility.

If possible, use 3" or shorter ground leads, such as the Low Impedance Contact (Z Lead). These are supplied with the Tektronix P613X and P623X family of probes.

One final note. The correct probe grounding method depends on the signal's high frequency content, the scope system bandwidth, and the need for mobility between test points.

A 12" ground lead may be perfect for many lower frequency applica-

tions. It will provide you with extra mobility, and nothing will be gained by using shorter leads.

If in doubt, apply the cable touch test outlined previously.

Ground Loop Noise Injection.

Another form of signal distortion can be caused by signal injection into the grounding system.

This can be caused by unwanted current flow in the ground loop existing between the common scope and test circuit power line grounds, and the probe ground lead and cable.

Normally, all these points are, or should be at zero volts, and no ground current will flow.

However, if the scope and test circuit are on different building system grounds, there could be small voltage differences, or noise on one of the building ground systems.

The resulting current flow (at line frequency or noise frequency) will develop a voltage drop across the probe cable's outer shield, and be injected into the scope in series with the desired signal.

Inductive Pickup in Ground Loops. Noise can enter a common ground system by induction into long 50 Ω signal acquisition cables, or into standard probe cables.

Proximity to power lines or other current-carrying conductors can induce current flow in the probe's outer cable, or in standard 50 Ω coax. The circuit is completed through the building system common ground.

Prevention of Ground Loop Noise Problems. Keep all signal acquisition probes and/or cables away from sources of potential interference.

Verify the integrity of the building system ground.

If the problem persists, open the ground loop:

1. By using a Ground Isolation Monitor like the Tektronix A6901.
2. By using a power line isolation transformer on either the test circuit or on the scope.
3. By using an Isolation Amplifier like the Tektronix A6902B.
4. By using differential probes (see Differential Measurements).

NOTE: Never defeat the safety 3-wire ground system on either the scope or on the test circuit.

Do not "float" the scope, except by using an approved isolation transformer, or preferably, by using the Tektronix A6901 Ground Isolation Monitor.

The A6901 automatically reconnects the ground if scope ground voltages exceed ± 40 V.

Induced Noise in Probe Ground Leads. The typical probe ground lead resembles a single-turn loop antenna when it is connected to the test circuit.

The relatively low impedance of the test circuit can couple any induced voltages into the probe, as shown in figure 3-8.

High speed logic circuits can produce significant electro-magnetic (radiated) noise at close quarters.

If the probe ground lead is positioned too close to certain areas on the board, interference signals could be picked up by the loop antenna formed by the probe ground lead, and mix with the probe tip signal.

Question: Is this what my signal really looks like?

Moving the probe ground lead around will help identify the problem.

If the **noise level** changes, you have a ground lead induced noise problem.

A more positive way of identification is to disconnect the probe from the signal source and clip the ground lead to the probe tip.

Now use the probe/ground lead as a loop antenna and search the board for radiated noise.

Figure 3-9 shows what can be found on a logic board, **with the probe tip shorted to the ground lead.**

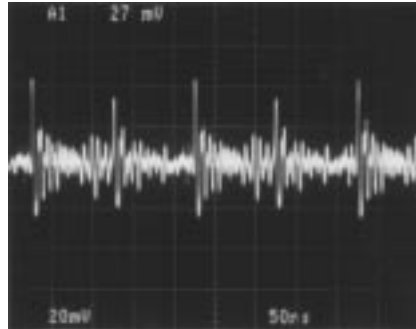


Figure 3-9. Induced noise in the probe ground loop (tip shorted to the ground clip).

This is radiated noise, induced in the single-turn loop and fed to the probe tip.

The significance of any induced or injected noise increases with reduced working signal levels, because the signal to noise ratio will be degraded. This is especially true with ECL, where signal levels are 1 V or less.

Prevention: If possible, use an ECB to Probe Tip Adaptor (test point). If not, use a Z Lead or short flexible ground lead.

Also, bunch the ground lead together to make the loop area as small as possible.

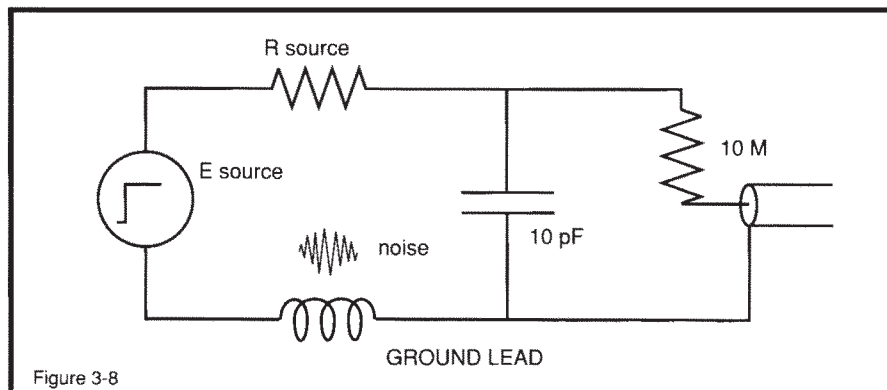


Figure 3-8.

Figure 3-8. Equivalent Circuit. Ground Lead Induced Noise.

Bias Offset Probes. A Bias/Offset probe is a special kind of Low Z design with the capability of providing a variable bias or offset voltage at the probe tip.

Bias/Offset probes like the Tektronix P6230 and P6231 are useful for probing high speed ECL circuitry, where resistive loading could upset the operating point.

They are also useful for probing higher amplitude signals (up to ± 5 V), where resistive loading could affect the DC level at some point on the waveform.

Bias/Offset probes are designed with a tip resistance of 450Ω (10X). When these probes are connected into a 50Ω environment, this loading results in a 10% reduction in peak to peak source amplitude. This round-figure loading is more convenient to handle than that produced by a standard 500Ω (10 X) Low Z probe, which would work out at 9.09% under the same conditions.

It is important to note that bias/offset probes always present a 450Ω resistive load to the source, regardless of the bias/offset voltage selected.

The difference between bias/offset and standard Low Z probes lies in their ability to null current flow **at some specific and selectable point** on the input waveform (within ± 5 V).

To see how bias/offset probes work, let's take a typical $10 \times 500 \Omega$ Low Z probe and connect it in the circuit shown in figure 3-10.

By taking a current flow approach we find that at one point on the waveform the source voltage is -4 V, therefore the load current will be;

$$I = ER = 4/R_s + R_p + R_{\text{scope}} = 4/550 = 7.27 \text{ mA.}$$

Therefore the voltage drop across the 50Ω source resistance (R_s) will be;

$$E = IR = 7.27 \times 10^{-3} \times 50 = 0.363 \text{ V}$$

And the measured pulse amplitude will be $-4 - 0.363 = 3.637$ V (E dut), or about 9% down from its unloaded state.

If we substitute the 500Ω Low Z probe with a 450Ω bias/offset probe, the circuit will look like figure 3-11.

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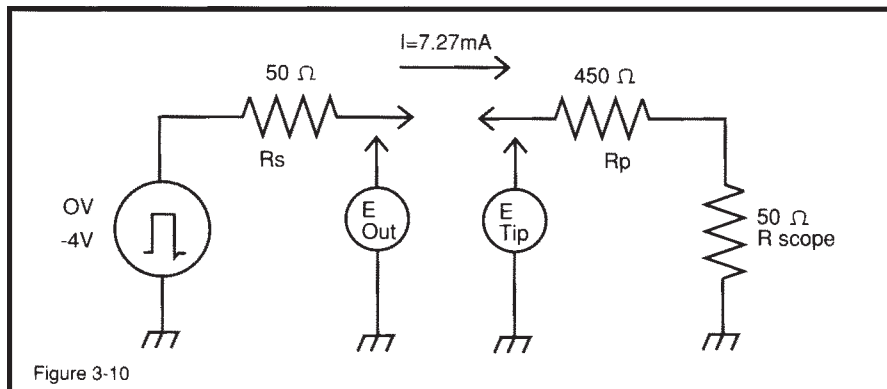


Figure 3-10. Low Z 10X 500 Ω probe connected to a 50 Ω source.

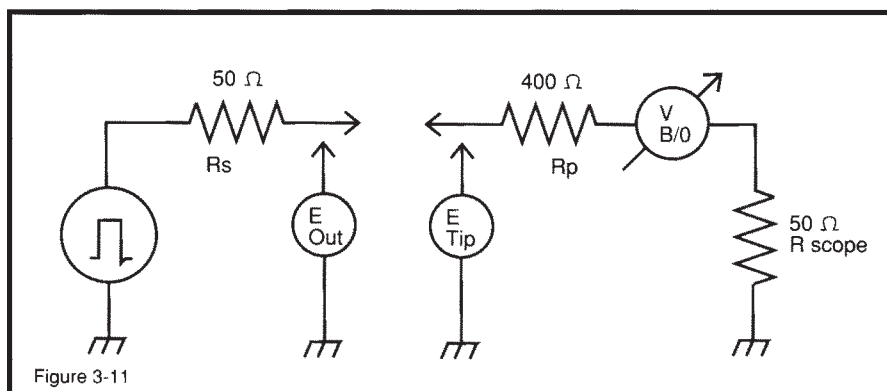


Figure 3-11. A 450 Ω Bias/Offset probe connected to a 50 Ω source.

With the bias/offset adjusted for 0 V, the effect on the circuit will be similar to a 500 Ω Low Z probe, except for the small resistive change.

Figure 3-12 shows the source waveform acquired by a 10 M Ω probe.

Figure 3-13 shows the effect on the waveform when the 450 Ω probe is added.

As expected, the pulse amplitude has reduced from -4 V to 3.60 V , or exactly 10% down.

Figure 3-14 shows the effect of adjusting the offset to -4 V . The -4 V bias opposes the signal at the -4 V level and results in zero current flow, and the source is effectively unloaded **at this point**.

However, when the signal returns to ground level, there is a 4 V differential between the top of the pulse and the bias/offset source. Current will flow, and Ohms Law will dictate that the top of the pulse will go negative by -40 mV (10%).

Sometimes it is desirable to adjust the offset mid-way between the peak to peak excursions. This distributes the effect of resistive loading between the two voltage swings.

Figure 3-15 shows the effect of adjusting the bias/offset to -2 V . Current flow will be the same for both signal swings, and they will be equally down by 5%, for a total of 10%.

Summary:

1. Bias/Offset probes can be adjusted (within $\pm 5\text{ V}$) to provide zero resistive (effective) loading at one selected point on the input waveform.
2. Bias/Offset probes can be used to simulate the effect of pull-up or pull-down voltages (within $\pm 5\text{ V}$) on the circuit under test (voltage source impedance is $450\ \Omega$).
3. Bias/Offset probes always present a total resistive load of $450\ \Omega$, and reduce the peak to peak amplitude of $50\ \Omega$ sources by 10%.
4. For simplicity, we have ignored the effects of capacitive loading. Typically, Bias/Offset probes have less than 2 pF tip C.

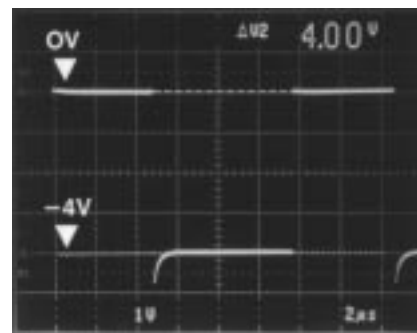


Figure 3-12. Unloaded negative-going 4 V pulse acquired by a 10 M Ω probe.

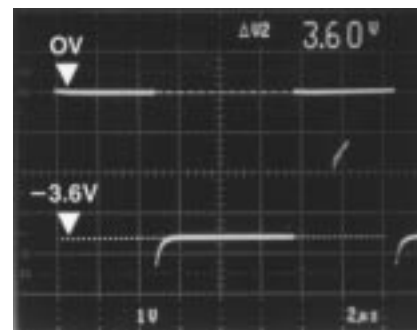


Figure 3-13. Effect of connecting a 450 Ω Bias/Offset probe (offset = 0 V). Minus level has been reduced by 10%.

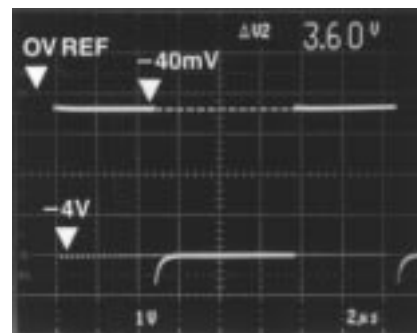


Figure 3-14. Bias/Offset adjusted for -4 V . Signal current at the -4 V level is zero. Current flow at ground level is maximum. Peak to peak amplitude remains the same (10% down).

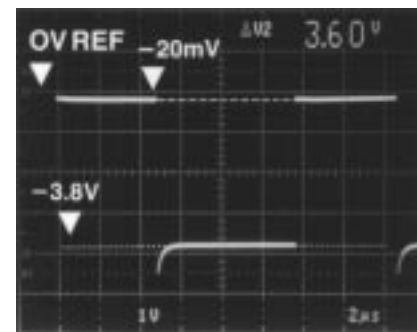


Figure 3-15. Bias/Offset adjusted for -2 V . Load current distributed between the negative and positive-going swings. Peak to peak amplitude remains the same (10% down).

Bias/Offset probes like the Tektronix P6230 or P6231 have bandwidths to 1.5 GHz, 450 Ω input R, and 1.3 pF (P6230), or 1.6 pF (P6231) input C.

They provide offset voltages of ± 5 V DC, and function with 1 M Ω or 50 Ω input systems (P6231, 50 Ω only).

The P6230 obtains operating power, either from the scope itself, or from the Tektronix 1101A or 1102 Power Supply.

The P6231 is designed to operate with the Tektronix 11000 Series scopes, and obtains operating power and bias/offset from the scope. Offset is selectable from the mainframe touch screen.

Differential Probing Techniques.

Accessing small signals elevated from ground, either at an AC level or a combination of AC and DC, requires the use of differential probes and a differential amplifier system.

One of the problems associated with differential measurements is the maintenance of high common mode rejection ratio (CMRR) at high common mode frequencies.

Poor common mode performance allows a significant portion of the common (elevated) voltage to appear across the differential probe's inputs. If the common mode voltage is pure DC, the result may only be a displayed baseline shift. However, if the common mode voltage is AC, or a combination of AC and DC, a significant portion may appear across the differential input and will mix with the desired signal.

Figure 3-16 shows the basic items necessary to make a differential measurement.

In this example two similar but un-matched passive probes are used. The probe ground leads are usually either removed or clipped together. They are **never** connected to the elevated DUT (device under test).

CMRR depends upon accurate matching of the probe-pair's electrical characteristics, including cable length. System CMRR can be no better than the differential amplifier's specifications, and in all cases, CMRR degrades as a function of frequency.

Figure 3-17 shows a simplified diagram of a DUT with a pulsed output of 1 V p-p floating on a 5 V p-p 20 MHz sinewave.

CMRR at 20 MHz is a poor 10:1 because of the un-matched probes.

Observed signal. (referred to probe input) = 1 V p-p pulse + (5 V p-p sine/10) = 1 V p-p pulse + 0.5 V p-p sine.

Figure 3-18a shows what the displayed waveform might look like under the conditions shown in figure 3-17.

In comparison, figures 3-18b and 3-18c show what the displayed signal might look like at CMRR's of 100:1 and 1000:1.

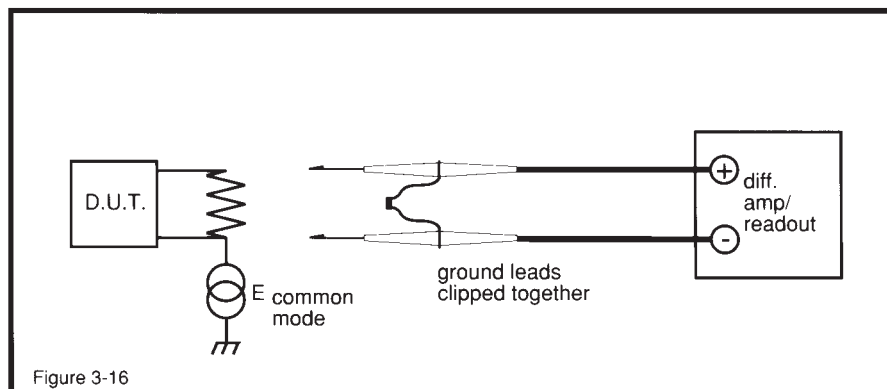


Figure 3-16. Basic connections to a device under test to make a differential measurement.

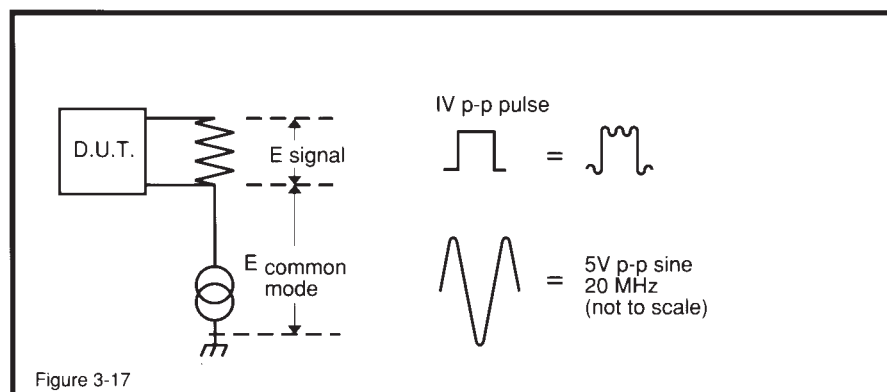
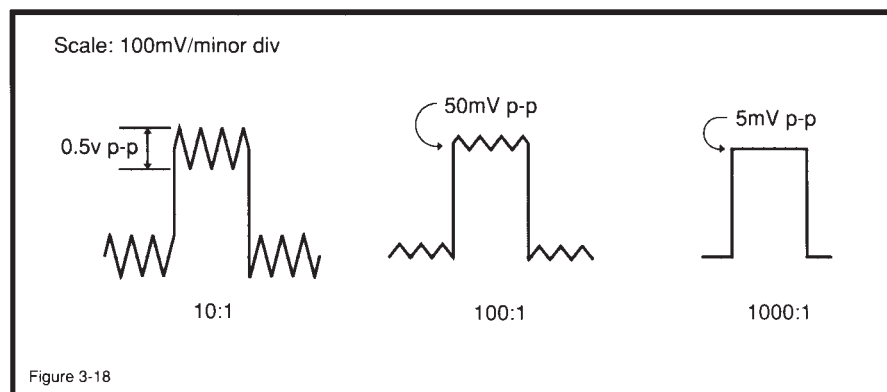


Figure 3-17. Simplified diagram. Elevated DUT. Common mode rejection is 10:1 at 20 MHz.



Figures 3-18, a, b and c. Displayed waveforms from the circuit shown in figure 3-17 at CMRR's of 10:1, 100:1 and 1000:1.

APPENDIX B

Measuring Amplifier Settling Time

High resolution measurement of amplifier settling at high speed is often necessary. Frequently, the amplifier is driven from a digital-to-analog converter (DAC). In particular, the time required for the DAC-amplifier combination to settle to final value after an input step is especially important. This specification allows setting a circuit's timing margins with confidence that the data produced is accurate. The settling time is the total length of time from input step application until the amplifier output remains within a specified error band around the final value.

Figure B1 shows one way to measure DAC amplifier settling time. The circuit uses the false sum node technique. The resistors and amplifier form a summing network. The amplifier output will step positive when the DAC moves. During slew, the oscilloscope probe is bounded by the diodes, limiting voltage excursion. When settling occurs, the summing node is arranged so the oscilloscope probe voltage should be zero. Note that the resistor divider's attenuation means the probe's output will be one-half the actual settled voltage.

In theory, this circuit allows settling to be observed to small amplitudes. In practice, it cannot be relied upon to produce useful measurements. Several flaws exist. The oscilloscope connection presents problems. As probe capacitance rises, AC loading of the resistor junction will influence observed settling waveforms. The 20pF probe shown alleviates this problem but its 10X attenuation

sacrifices oscilloscope gain. 1X probes are not suitable because of their excessive input capacitance. An active 1X FET probe might work, but another issue remains.

The clamp diodes at the probe point are intended to reduce swing during amplifier slewing, preventing excessive oscilloscope overdrive. Unfortunately, oscilloscope overdrive recovery characteristics vary widely among different types and are not usually specified. The diodes' 600mV drop means the oscilloscope may see an unacceptable overload, bringing displayed results into question (for a discussion of oscilloscope overdrive considerations, see the Tutorial Section on Oscilloscopes). With the oscilloscope set at 1mV per division, the diode bound allows a 600:1 overdrive. Schottky diodes can cut this in half, but this is still much more than any real-time vertical amplifier is designed to accommodate.¹ The oscilloscope's overload recovery will completely dominate the observed waveform and all measurements will be meaningless.

One way to achieve reliable settling time measurements is to clip the incoming waveform in *time*, as well as amplitude. If the oscilloscope is prevented from seeing the waveform until settling is nearly complete, overload is avoided. Doing this requires placing a switch at the settle circuit's output and controlling it with an input-triggered, variable delay. FET switches are not suitable because of their gate-source capacitance. This capacitance will allow gate drive artifacts to corrupt the oscilloscope display,

Note 1: See Reference 3 for history and wisdom about vertical amplifiers.

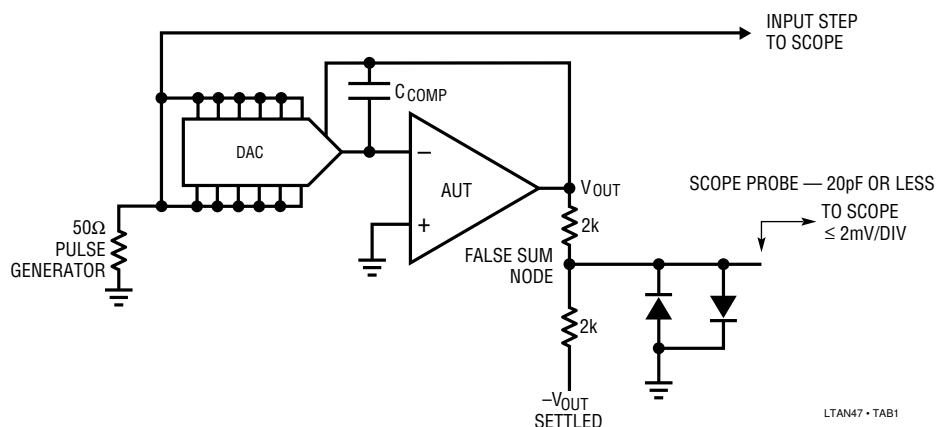


Figure B1. One (Not Very Good) Way to Measure DAC-Op Amp Settling Time

producing confusing readings. In the worst case, gate drive transients will be large enough to induce overload, defeating the switch's purpose.

Figure B2 shows a way to implement the switch which largely eliminates these problems. This circuit allows settling within 1mV to be observed. The Schottky sampling bridge is the actual switch. The bridge's inherent balance, combined with matched diodes and very high speed complementary bridge switching, yields a clean switched output. An output buffer stage unloads the settle node and drives the diode bridge.

The operation of the DAC-amplifier is as before. The additional circuitry provides the delayed switching function, eliminating oscilloscope overdrive. The settle node is buffered by A1, a unity gain broadband FET input buffer with 3pF input capacitance and 350MHz bandwidth. A1 drives the Schottky bridge. The pulse generator's output fires the 74123 one shot. The one shot is arranged to produce a delayed (controllable by the 20k potentiometer) pulse whose width (controllable by the 5k potentiometer) sets diode bridge on-time. If the delay is set appropriately, the oscilloscope will not see any input until settling is nearly complete, eliminating overdrive. The sample window width is adjusted so that all remaining settling activity is observable. In this way the oscilloscope's output is reliable and meaningful data may be taken. The one shot's output is level shifted by the Q1-Q4 transistors, providing complementary switching drive to the bridge. The actual switching transistors, Q1-Q2, are UHF types, permitting true differential bridge switching with less than 1ns of time skew.² The bridge's output may be observed directly (by oscilloscopes with adequate sensitivity) or A2 provides a times 10 amplified version. A2's gain of 20 (and the direct output's $\div 2$ scaling) derives from the 2k-2k settle node dividers attenuation. A third output, taken directly from A1, is also available. This output, which bypasses the entire switching circuitry, is designed to be monitored by a sampling oscilloscope. Sampling oscilloscopes are inherently immune to overload.³ As such, a good test of this settling time test fixture (and the above statement) is to compare the signals displayed by the sampling 'scope and the Schottky-bridge-aided real time 'scope. As an *additional* test, a completely different method of measuring settling time (albeit considerably more complex) described by Harvey⁴ was also employed. If all three approaches

represent good measurement technique and are constructed properly, results should be identical.⁵ If this is the case, the identical data produced by the three methods has a high probability of being valid. Figures B3, B4 and B5 show settling time details of an AD565A DAC and an LT1220 op amp. The photos represent the sampling bridge, sampling 'scope and Harvey methods, respectively. Photos B3 and B5 display the input step for convenience in ascertaining elapsed time. Photo B4, taken with a single trace sampling oscilloscope (Tektronix 1S1 with P6032 cathode follower probe in a 556 mainframe) uses the leftmost vertical graticule line as its zero time reference. All methods agree on 280ns to 0.01% settling (1mV on a 10V step). Note that Harvey's method inherently adds 30ns, which must be subtracted from the displayed 310ns to get the real number.⁶ Additionally, the shape of the settling waveform, in every detail, is identical in all three photographs. This kind of agreement provides a high degree of credibility to the measured results.

Some poorly designed amplifiers exhibit a substantial thermal tail after responding to an input step. This phenomenon, due to die heating, can cause the output to wander outside desired limits long after settling has apparently occurred. After checking settling at high speed, it is always a good idea to slow the oscilloscope sweep down and look for thermal tails. Often the thermal tail's effect can be accentuated by loading the amplifier's output. Such a tail can make an amplifier appear to have settled in a much shorter time than it actually has.

To get the best possible settling time from any amplifier, the feedback capacitor, C_F , should be carefully chosen. C_F 's purpose is to roll-off amplifier gain at the frequency which permits best dynamic response. The optimum value for C_F will depend on the feedback resistor's value and the characteristics of the source. DAC's are one of the most common sources and also one of the most difficult. DAC's current outputs must often be converted to a

Note 2: The Q1-Q4 bridge switching scheme, a variant of one described in Reference 14, was developed at LTC by George Feliz.

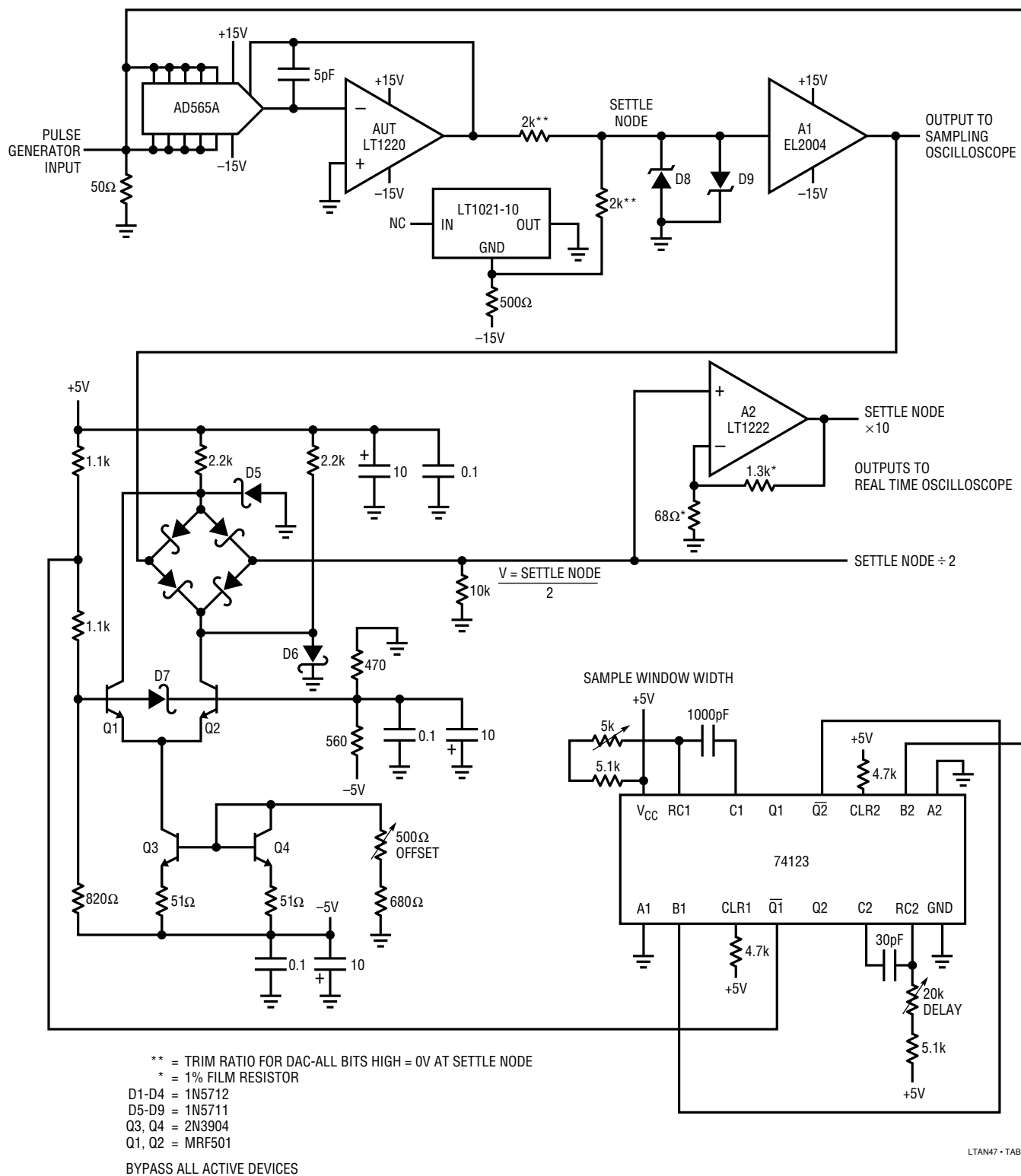
Note 3: See References 7, 8 and 18.

Note 4: See Reference 17.

Note 5: Construction details of the settling time fixture discussed here appear (literally) in Appendix F, "Additional Comments on Breadboarding". Also see the Tutorial Section on Breadboarding Techniques.

Note 6: See Reference 17.

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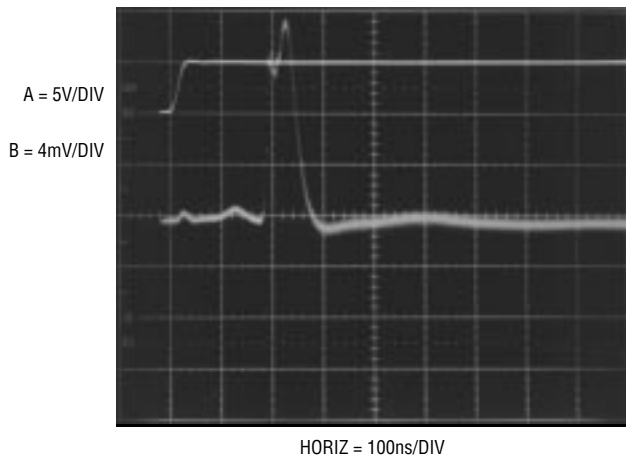


Figure B3. 280ns Settling Time as Measured by Figure B2's Circuit. Sampling Switch Closes Just Before Third Vertical Division, Allowing Settling Detail to be Observed Without Overdriving the Oscilloscope

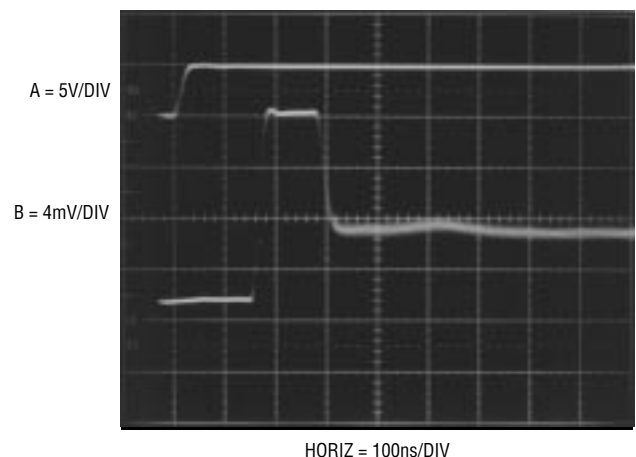


Figure B5. 280ns Settling Time as Measured by Harvey's Method. After Subtraction of this Method's Inherent 30ns Delay, Settling Time and Waveform Shape are Identical to Figures B3 and B4

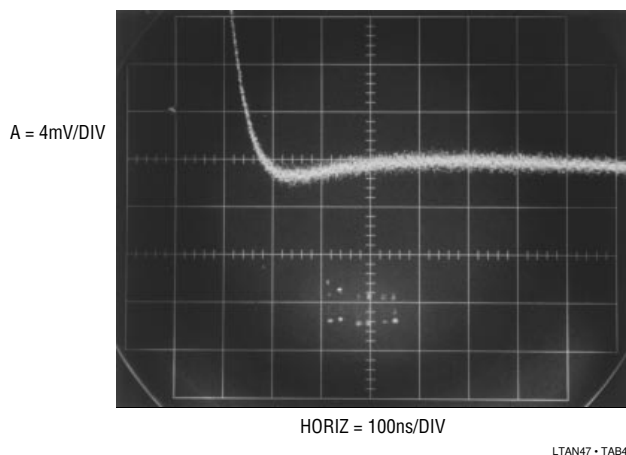


Figure B4. 280ns Settling Time as Measured at Figure B2's Sampling Oscilloscope Output by a Sampling 'Scope. Settling Time and Waveform Shape is Identical to Figure B3

voltage. Although an op amp can easily do this, care is required to obtain good dynamic performance. A fast DAC can settle to 0.01% in 200ns or less but its output also includes a parasitic capacitance term, making the amplifier's job more difficult. Normally, the DAC's current output is unloaded directly into the amplifier's summing junction, placing the parasitic capacitance from ground to the

amplifier's input. The capacitance introduces feedback phase shift at high frequencies, forcing the amplifier to hunt and ring about the final value before settling. Different DACs have different values of output capacitance. CMOS DACs have the highest output capacitance and it varies with code. Bipolar DACs typically have 20pF-30pF of capacitance, stable over all codes. Because of their output capacitance, DAC's furnish an instructive example in amplifier compensation. Figure B6 shows the response of an industry standard DAC-80 type and a relatively slow (for this publication) op amp. Trace A is the input, while Traces B and C are the amplifier and settle outputs, respectively. In this example no compensation capacitor is used and the amplifier rings badly before settling. In Figure B7, an 82pF unit stops the ringing and settling time goes down to 4μs. The overdamped response means that C_F dominates the capacitance at the AUT's input and stability is assured. If fastest response is desired, C_F must be reduced. Figure B8 shows critically damped behavior obtained with a 22pF unit. The settling time of 2μs is the best obtainable for this DAC-amplifier combination. Higher speed is possible with faster amplifiers and DACs but the compensation issues remain the same.

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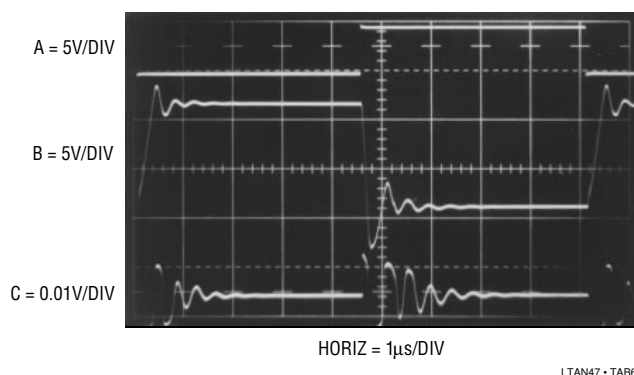


Figure B6. No Feedback Capacitor

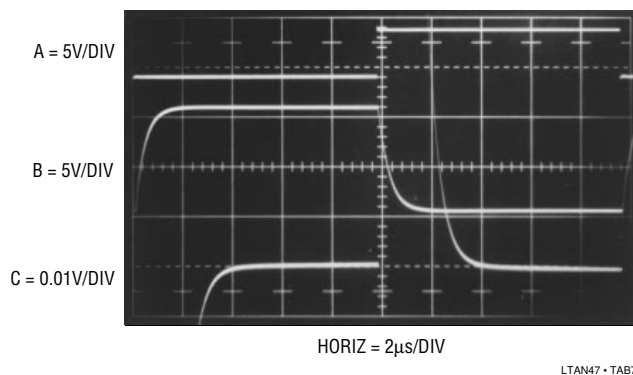


Figure B7. Relatively Large Feedback Capacitor

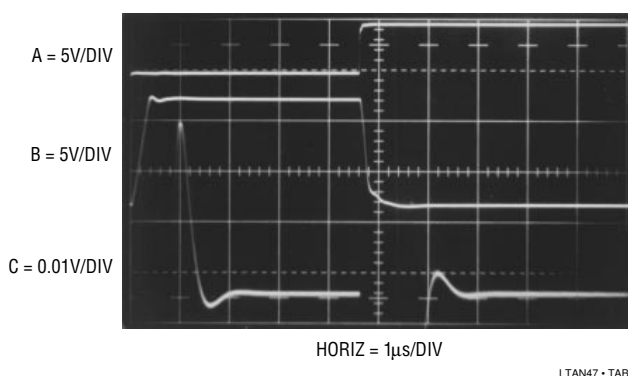


Figure B8. Reduced Feedback Capacitor Gives Fastest Settling

Figures B6-B8. Effects of Different Feedback Capacitors on a DAC-Op Amp Combination

APPENDIX C

The Oscillation Problem – Frequency Compensation Without Tears

All feedback systems have the propensity to oscillate. Basic theory tells us that gain and phase shift are required to build an oscillator. Unfortunately, feedback systems, such as operational amplifiers, have gain and phase shift. The close relationship between oscillators and feedback amplifiers requires careful attention when an op amp is designed. In particular, excessive input-to-output phase shift can cause the amplifier to oscillate when feedback is applied. Further, any time delay placed in the amplifier's feedback path introduces additional phase shift, increasing the likelihood of oscillation. This is why feedback loop enclosed stages can cause oscillation.

A large body of complex mathematics is available which describes stability criteria and can be used to predict stability characteristics of feedback amplifiers. For the most sophisticated applications, this approach is required to achieve optimum performance.

However, little has appeared which discusses, in practical terms, how to understand and address the issues of compensating feedback amplifiers. Specifically, a practical approach to stabilizing amplifier-power gain stage combinations is discussed here, although the considerations can be generalized to other feedback systems.

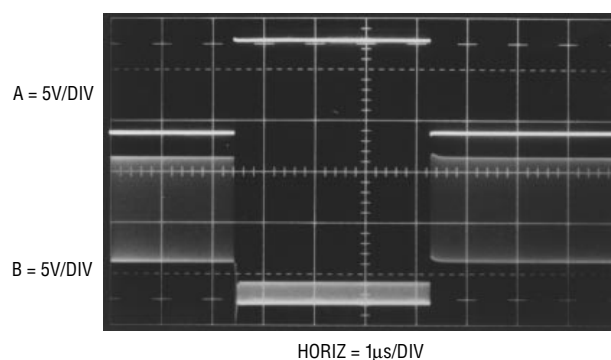
Oscillation problems in amplifier-power booster stage combinations fall into two broad categories; local and loop

oscillations. *Local* oscillations can occur in the boost stage, but should not appear in the IC op amp, which presumably was debugged prior to sale. These oscillations are due to transistor parasitics, layout and circuit configuration-caused instabilities. They are usually relatively high in frequency, typically in the 0.5MHz to 100MHz range. Usually, local booster stage oscillations do not cause loop disruption. The major loop continues to function, but contains artifacts of the local oscillation. Text Figure 101, repeated here as Figure C1 for reader convenience, furnishes an instructive example. The Q1, Q2 emitter follower pair has reasonably high f_t . These devices will oscillate if driven from a low impedance source (see insert, Figure C1 and References 43 and 44). The 100 Ω resistor and the ferrite beads are included to make the op amp's output look like a higher impedance to prevent problems. Q5 and Q6, also followers, have even higher f_t , but are driven from 330 Ω sources, eliminating the problem. The photo in Figure C2 shows Figure C1 following an input with the 100 Ω resistor and ferrite beads removed. Trace A is the input, while Trace B is the output. The resultant high frequency oscillation is typical of locally caused disturbances. Note that the major loop is functional, but the local oscillation corrupts the waveform.

Eliminating such local oscillations starts with device selection. Avoid high f_t transistors unless they are needed. When high frequency devices are in use, plan layout carefully. In very stubborn cases, it may be necessary to

lightly bypass transistor junctions with small capacitors or RC networks. Circuits which use local feedback can sometimes require careful transistor selection and use. For example, transistors operating in a local loop may require different f_t s to achieve stability. Emitter followers are notorious sources of oscillation and should never be directly driven from low impedance sources (again, see References 43 and 44).

Loop oscillations are caused when the added gain stage supplies enough delay to force substantial phase shift. This causes the control amplifier to run too far out of phase with the gain stage. The control amplifier's gain combined with the added delay causes oscillation. Loop oscillations are usually relatively low in frequency, typically 10Hz-1MHz.



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Figure C2. Local Oscillations Due to Booster Stage Instabilities

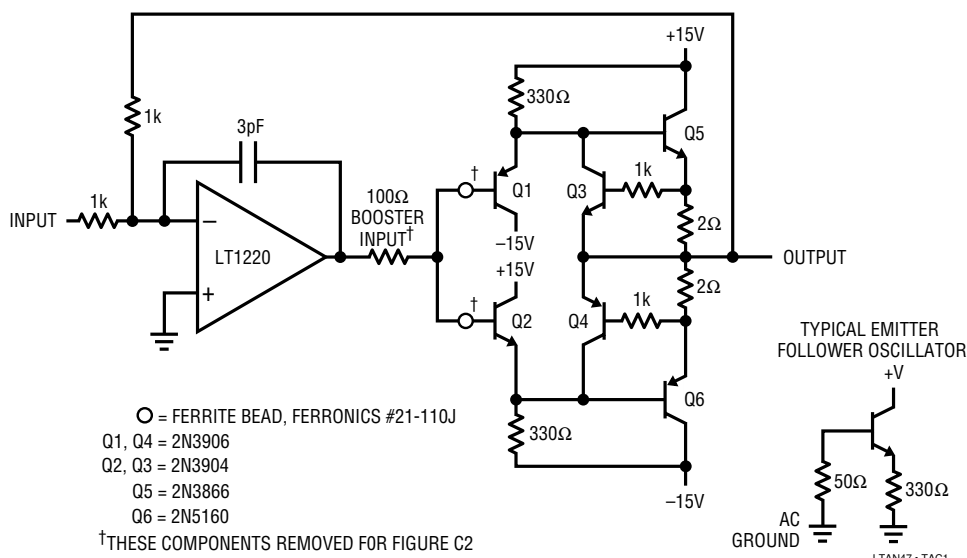


Figure C1. Figure 101's Booster Circuit with a Few Components Removed Begins Our Study of Loop Stability

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A good way to eliminate loop-caused oscillations is to limit the gain-bandwidth of the control amplifier. If the booster stage has higher gain-bandwidth than the control amplifier, its phase delay is easily accommodated in the loop. When control amplifier gain-bandwidth dominates, oscillation is assured. Under these conditions, the control amplifier hopelessly tries to servo a feedback signal which consistently arrives too late. The servo action takes the form of an electronic tail chase with oscillation centered around the ideal servo point.

Frequency response roll-off of the control amplifier will almost always cure loop oscillations. In many situations it is preferable to brute force compensation using large capacitors in the major feedback loop. As a general rule, it is wise to stabilize the loop by rolling off control amplifier gain-bandwidth. The feedback capacitor serves only to trim step response and should not be relied on to stop outright oscillation.

Figures C3 and C4 illustrate these issues. The LT1006 amplifier used with the LT1010 current buffer produces the output shown in Figure C4. As before, Trace A is the input and Trace B the output. The LT1006 has less than

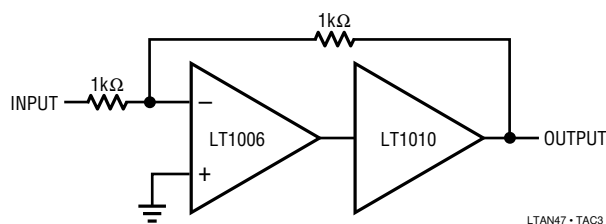


Figure C3. A Slow Op Amp and a Medium Speed Booster

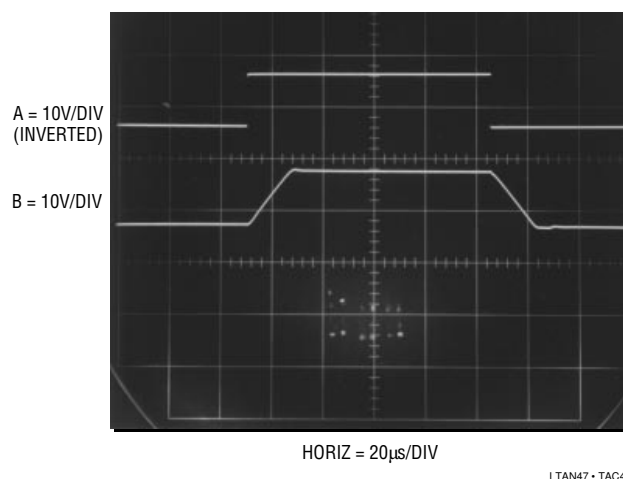


Figure C4. Loop Stability is “Free” When the Op Amp is Much Slower than the Booster

1MHz gain-bandwidth. The LT1010's 20MHz gain-bandwidth introduces negligible loop delay, and dynamics are clean. In this case, the LT1006's internal roll-off is well below that of the output stage and stability is achieved with no external compensation components. Figure C5 uses a 100MHz bandwidth LT1223 as the control amplifier. The associated photo (Figure C6) shows the results. Here, the control amplifier's roll-off is well beyond the output stage's, causing problems. The phase shift through the LT1010 is now appreciable and oscillations occur. Stabilizing this circuit requires degenerating the control amplifier's gain-bandwidth.

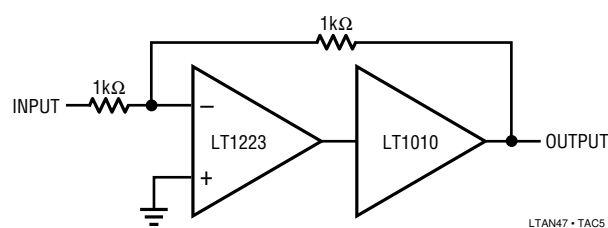


Figure C5. A Fast Op Amp and a Medium Speed Booster

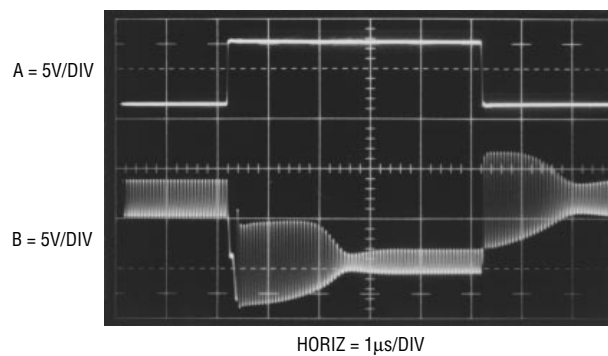


Figure C6. Loop Oscillation is “Free” When the Op Amp is Much Faster than the Booster

The fact that the slower op amp circuit doesn't oscillate is a key to understanding how to compensate booster loops. With the slow device, compensation is free. The faster amplifier makes the AC characteristics of the output stage become significant and requires roll-off components for stability. Practically, the LT1223's speed is simply too much for the LT1010. A somewhat slower amplifier is the way to go. Alternately, a faster booster may be employed. Figure C7 attempts this, but doesn't quite make it. Photo C8 is less corrupted, but 100MHz oscillation indicates the booster stage (borrowed from text Figure 101) is still too slow for the LT1223. Attempts to use another booster design in Figure C9 (similarly purloined from text Figure

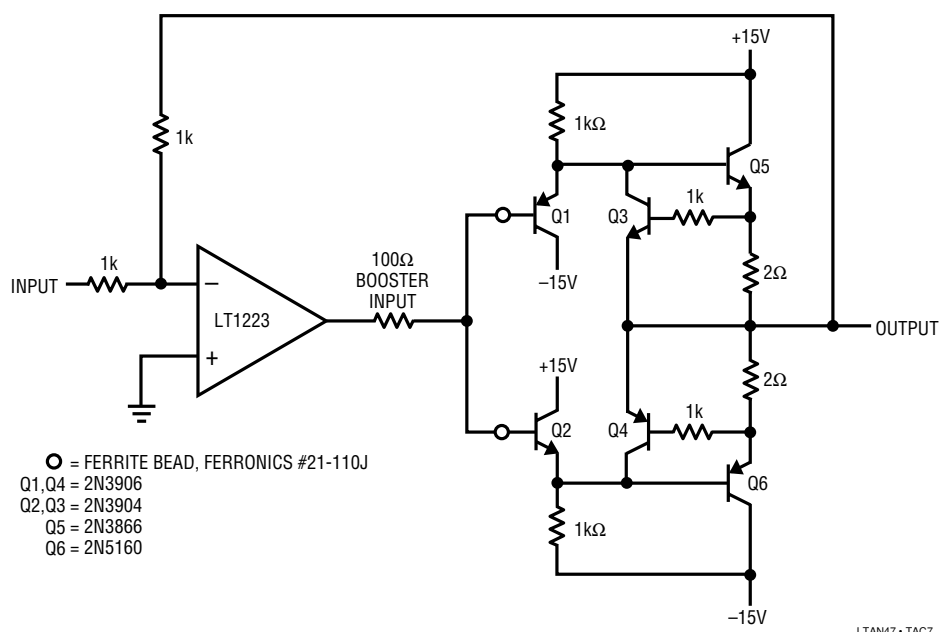


Figure C7. A Very Fast Amplifier with a Fast Booster

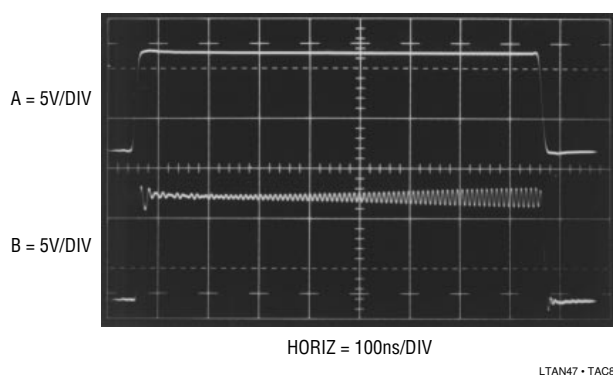


Figure C8. Figure C7's Booster is Not Quite Quick Enough to Prevent Loop Oscillation

104) fail for the same reason. Figure C10 shows 40MHz oscillation, indicative of this high power booster's slower speed.

Figure C11 has much more pleasant results. Here a 45MHz gain-bandwidth LT1220 has been substituted for the 100MHz LT1223 in Figure C7's circuit. The slower amplifier, combined with light local compensation, works well with the booster stage in its loop. Figure C12 shows a well controlled high speed output, nicely damped, with no sign of oscillations.

Power boosters are not the only things that can be placed within an amplifier's feedback loop. Text Figure 140's

current source, reproduced here as Figure C13, is an interesting variation. There is no power booster in the loop, but rather a 40MHz differential amplifier with a gain of 10. To stabilize the circuit the slowest amplifier in the 1190 family, the 50MHz LT1190, is chosen. The local 100pF feedback slows it down a bit more and the loop is fast and stable (Figure C14). What happens if we remove the 100pF feedback path? Figure C15 shows that the loop is no longer stable under this condition because the LT1190 control amplifier cannot servo the phase shifted feedback at higher frequency. Put that 100pF capacitor back in!

It's worth mentioning that similar results to those obtained back in Figure C3 are obtainable by substituting a very slow control amplifier (e.g., an LT1006 which has less than 1MHz gain-bandwidth). The slower amplifier would give "free" compensation, eliminating the necessity for the 100pF unit. However, the circuit's frequency response would be severely degraded.

Text Figure 142's high power current source furnishes further instruction. This loop contains the differential amplifier *and* a booster, seemingly making things even more difficult. Figure C16, recognizable as text Figure 142's high power current source with the 100pF local compensation removed, oscillates above 10MHz. Replacing the compensation restores proper response. Figure

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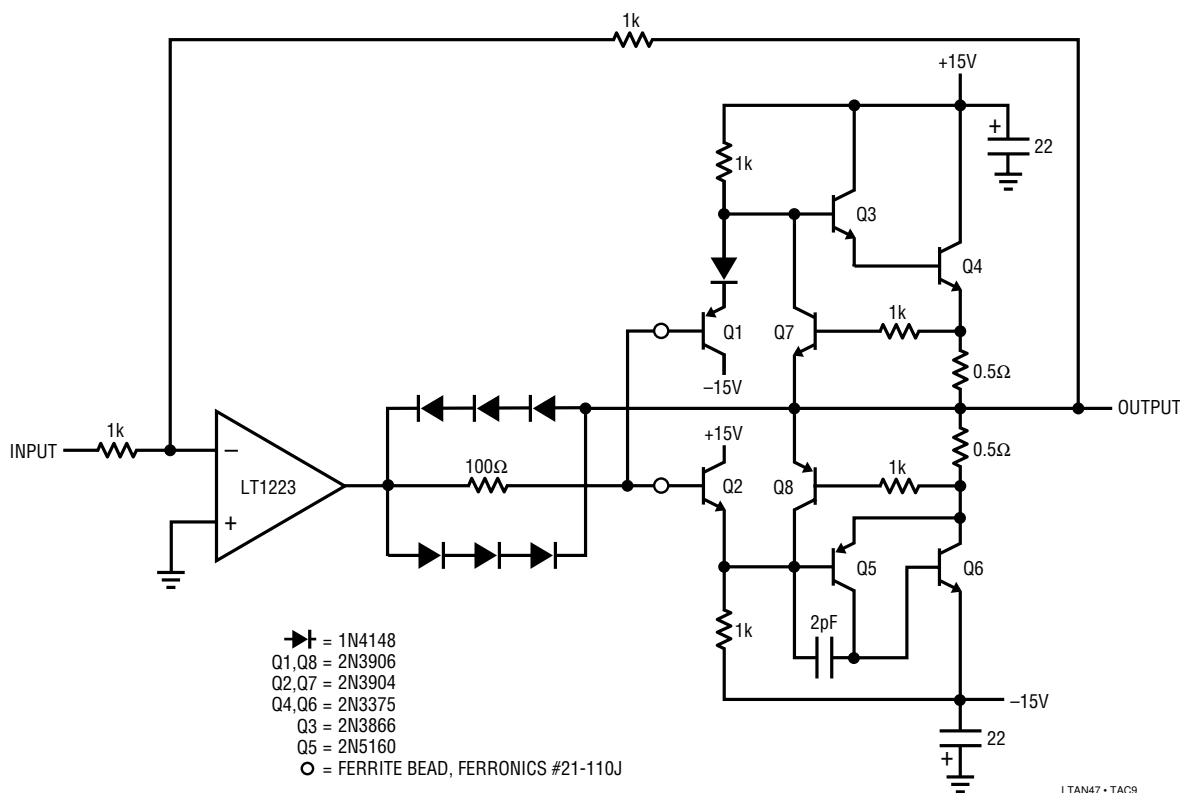


Figure C9. A Very Fast Amplifier with a Fast High Power Booster

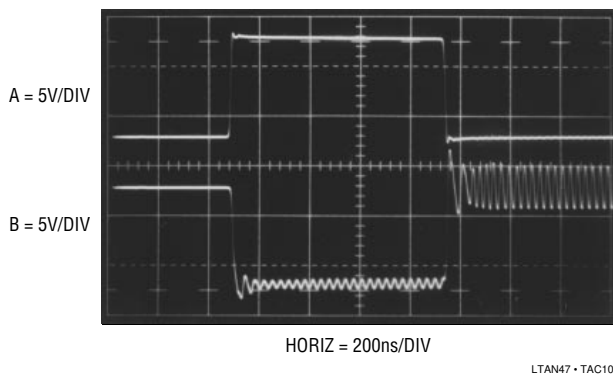


Figure C10. C9's High Power Booster is Fast But Causes Loop Oscillations

C17 shows the loop has no oscillations. What this tells us is that the control amplifier doesn't care just what generates the causal feedback between its input and output, so long as there isn't excessive delay. This circuit has a fairly busy feedback loop, but the control amplifier is oblivious to its bustling nature....unless you leave that 100pF feedback capacitor out!

When compensating loops like these, remember to investigate the effects of various loads and operating conditions. Sometimes a compensation scheme which appears fine gives bad results for some conditions. For this reason, check the completed circuit over as wide a variety of operating conditions as possible.

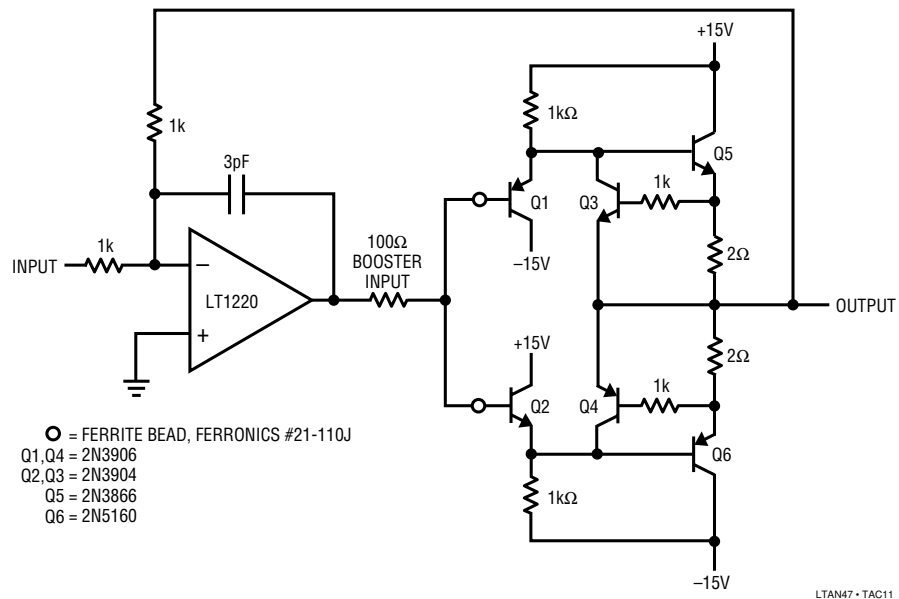


Figure C11. Figure 101 (Again) with 100Ω Resistor and Beads Reinstalled

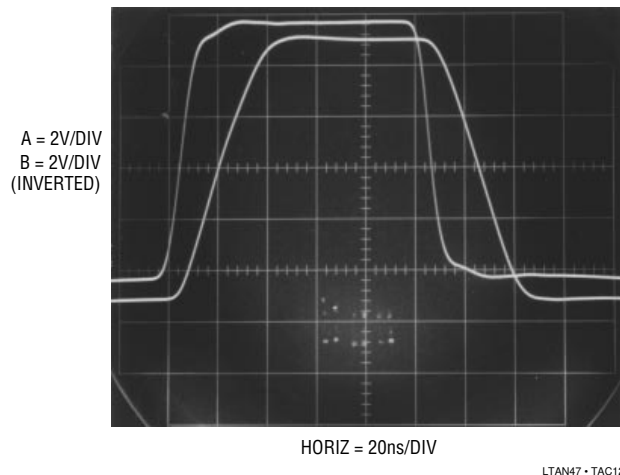


Figure C12. Lovely!

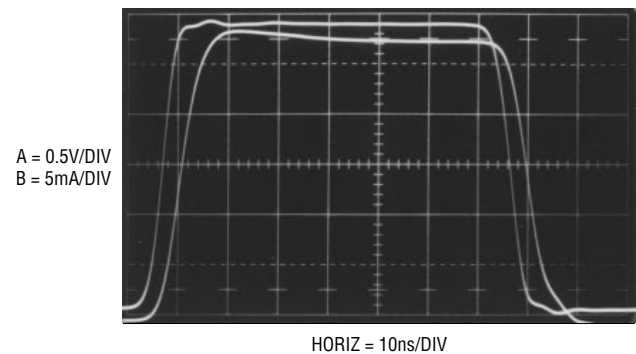


Figure C14. Response of the Current Source with the RC Components in Place

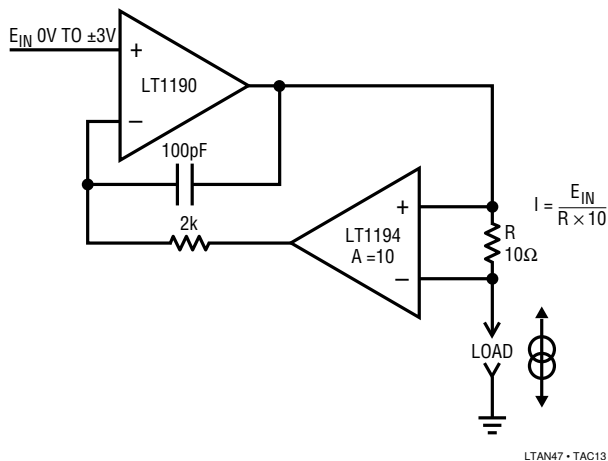


Figure C13. Figure 140's Current Source. What Do the RC Components Do?

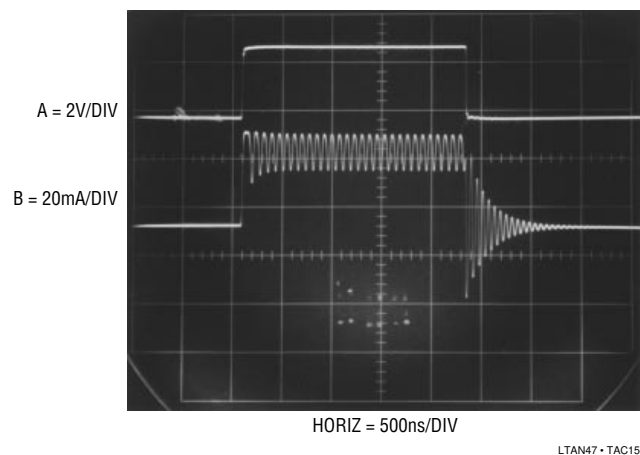


Figure C15. Removing the 100pF Capacitor Allows the Op Amp to See Phase Shifted Feedback, Causing Oscillation. Put that 100pF Back In!

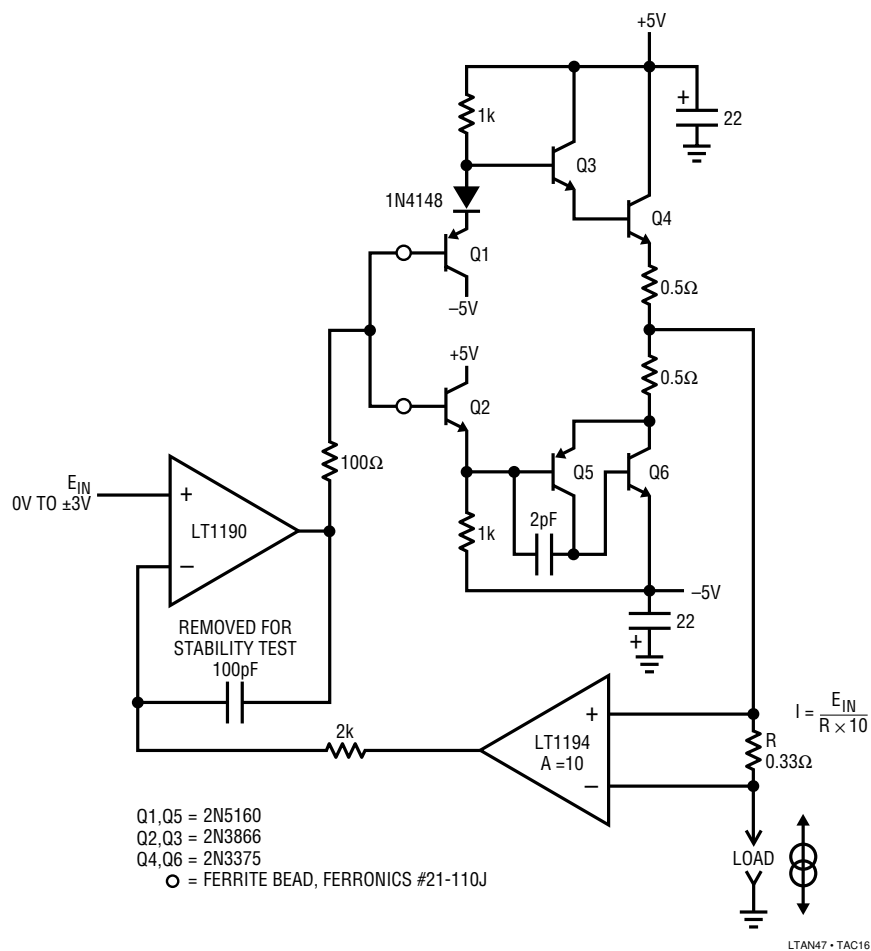


Figure C16. Text Figure 142's High Power Current Source. When the 100pF Capacitor is Removed, 10MHz Loop Oscillations Result

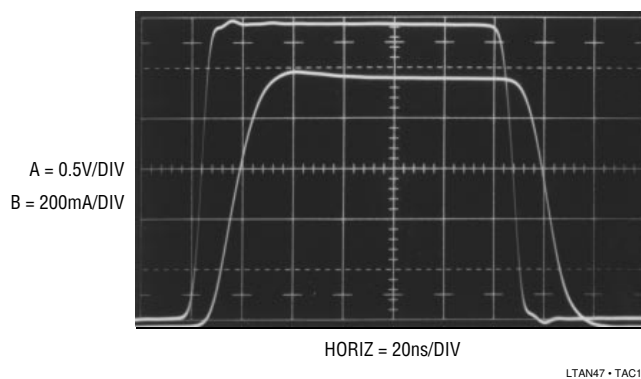


Figure C17. Much Better. Leave that 100pF Capacitor in There!

APPENDIX D

Measuring Probe-Oscilloscope Response

Verifying the rise time limit of wideband test equipment set-ups is a difficult task. In particular, the end-to-end rise time of oscilloscope-probe combinations is often required to assure measurement integrity. Conceptually, a pulse generator with rise times substantially faster than the oscilloscope-probe combination can provide this information. Figure D1's circuit does this, providing a 1ns pulse with rise and fall times inside 350ps. Pulse amplitude is 10V with a 50Ω source impedance. This circuit, built into a small box and powered by a 1.5V battery, provides a simple, convenient way to verify the rise time capability of almost any oscilloscope-probe combination.

The LT1073 switching regulator and associated components supply the necessary high voltage. The LT1073 forms a flyback voltage boost regulator. Further voltage step-up is obtained from a diode-capacitor voltage step-up network. L1 periodically receives charge and its flyback discharge delivers high voltage events to the step-up network. A portion of the step-up network's DC output is fed back to the LT1073 via the 10M, 24k divider, closing a control loop.

The regulator's 90V output is applied to Q1 via the 1M-2pF combination. Q1, a 40V breakdown device, non-destructively avalanches when C1 charges high enough. The result is a quickly rising, very fast pulse across R4. C1

discharges, Q1's collector voltage falls and breakdown ceases. C1 then recharges until breakdown again occurs. This action causes free running oscillation at about 200kHz.^{1,2} Figure D2 shows the output pulse. A 1GHz sampling oscilloscope (Tektronix 556 with 1S1 sampling plug-in) measures the pulse at 10V high with about a 1ns base. Rise time is 350ps, with fall time also indicating 350ps. There is a slight hint of ring after the falling edge, but it is well controlled. The figures may actually be faster, as the 1S1 is specified with a 350ps limit.³

Q1 may require selection to get avalanche behavior. Such behavior, while characteristic of the device specified, is not guaranteed by the manufacturer. A sample of 50 Motorola 2N2369s, spread over a 12 year date code span, yielded 82%. All good devices switched in less than 650ps. C1 is selected for a 10V amplitude output. Value spread is typically 2pF-4pF. Ground plane type construction with

Note 1: This method of generating fast pulses borrows heavily from the Tektronix type 111 Pretrigger Pulse Generator. See References 8 and 25.

Note 2: This circuit replaces the tunnel diode based arrangement shown in AN13, Appendix D. While AN13's circuit works well, it generates a smaller, more irregularly shaped pulse and the tunnel diodes have become quite expensive.

Note 3: Just before going to press the pulse was measured at Hewlett-Packard Laboratories with a HP-54120B 12GHz sampling oscilloscope. Rise and fall times were 216ps and 232ps, respectively. Photo available on request.

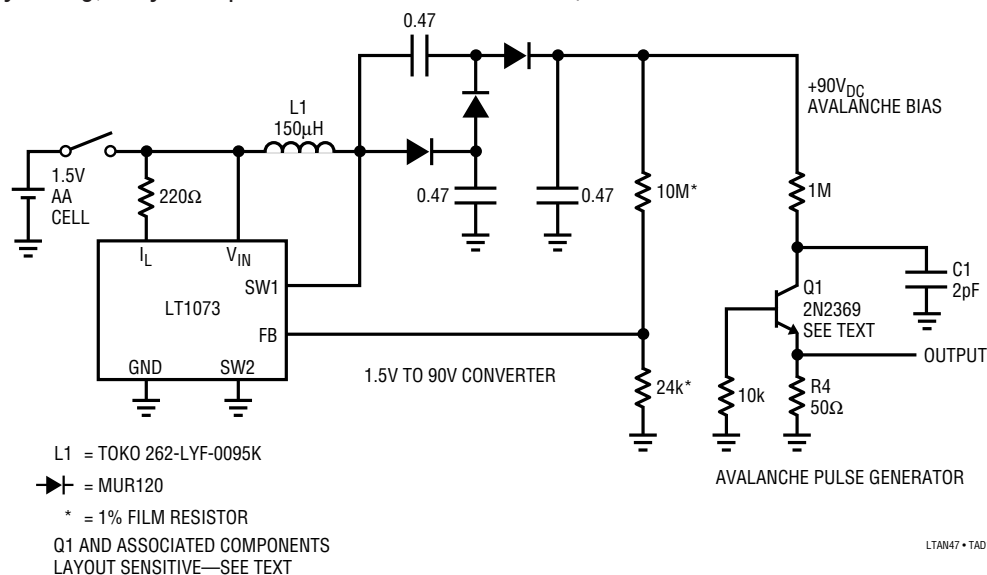


Figure D1. 350ps Rise/Fall Time Avalanche Pulse Generator

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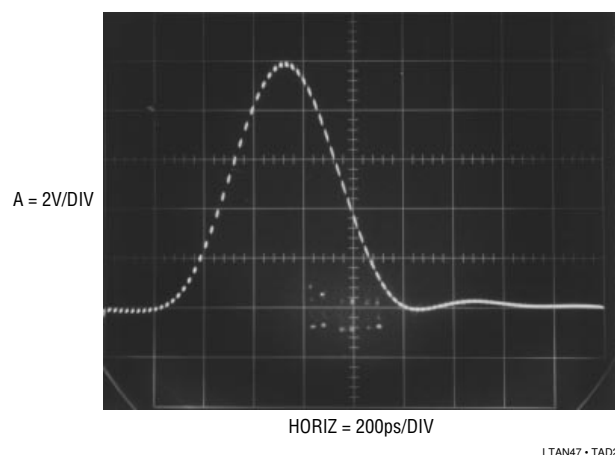


Figure D2. The Avalanche Pulse Generator's Output Monitored on a 1GHz Sampling Oscilloscope

high speed layout techniques are essential for good results from this circuit. Current drain from the 1.5V battery is about 5mA.

Figure D3 shows the physical construction of the actual generator. Power, supplied from a separate box, is fed into the generator's enclosure via a BNC connector. Q1 is mounted *directly* at the output BNC connector, with grounding and layout appropriate for wideband operation. Lead lengths, particularly Q1's and C1's, should be experimented with to get best output pulse purity. Figure D4 is the complete unit.



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Figure D3. Details of the Avalanche Pulse Generator's Head. 90V_{DC} Enters at Lower Right BNC, Pulse Exits at Top Left BNC. Note Short Lead Lengths Associated with Output



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Figure D4. The Packaged Avalanche Pulser. 1.5V-90V Converter is in the Black Box. Avalanche Head is at Left

APPENDIX E

An Ultra-Fast High Impedance Probe

Under most circumstances the 1pF-2pF input capacitance and 10M Ω resistance of FET probes is more than adequate for difficult probing situations. Occasionally, however, very high input resistance with high speed is needed. At some sacrifice in speed and input capacitance compared to commercial probes, it is possible to construct such a probe. Figure E1 shows schematic details. A1, a 350MHz hybrid FET buffer, forms the electrical core of the probe. This device is a low input capacitance, wideband FET source follower driving a fast bipolar output stage. The input of the probe goes to this device via a 51 Ω resistor, reducing the possibility of oscillations in the follower input stage when the probe sees low AC impedance. A1's output drives a guard shield around the probe's input line, reducing effective input capacitance to about 4pF. A ground referred shield encircles the guard shield, reducing pickup and making high quality ground connections to the circuit under test easy. A1 drives the output BNC cable to feed the oscilloscope. Normally, it is undesirable to back terminate the cable at A1 because the oscilloscope will see only half of A1's output. While a back termination provides the best signal dynamics, the resulting attenuation is a heavy penalty. The RC damper shown can be trimmed for best edge response while maintaining an unattenuated output.

What can't be seen in the schematic is the probe's physical construction. Very careful construction is required to

maintain low input capacitance, low bias current and wide bandwidth. The probe head is particularly critical. Every effort should be made to minimize the length of wire between A1's input and the probe tip. In our lab, we have found that discarded pieces of broken 10X probes, particularly attenuator boxes and probe heads, provide an excellent packaging basis for this probe.¹ Figure E2 shows the probe head. Note the compact packaging. Additionally, A1's package is arranged so that it's (not insubstantial) dissipated heat is transferred to the probe case body when the snap-on cover (shown in photo) is in place. This reduces A1's substrate temperature, keeping bias current down. A1's input is directly connected to the probe head to minimize parasitic capacitance. The power supply for A1, located in a separate enclosure, is fed in through separate wires. A1's output is delivered to the oscilloscope via conventional BNC hardware.

Figure E3 shows the probe output (Trace B) responding to an input (Trace A) as monitored on a 350MHz oscilloscope (Tektronix 485). Measured specifications for our version of this probe include a rise time of 6ns, 6ns delay and 58MHz bandwidth. The delay time contribution is about evenly split between the amplifier and cable. Input capacitance is about 4pF without the probe hook tip and 7pF with the hook tip. Input bias current measured 400pA and gain error about 5%. (A1 is an open loop device.)

Note 1: This is not to encourage or even accept the breakage of probes. The author regards the breakage of oscilloscope probes as the lowest possible human activity. The sole exception to this condemnation is poor quality probes, which should be destroyed as soon as their deficiencies are discovered.

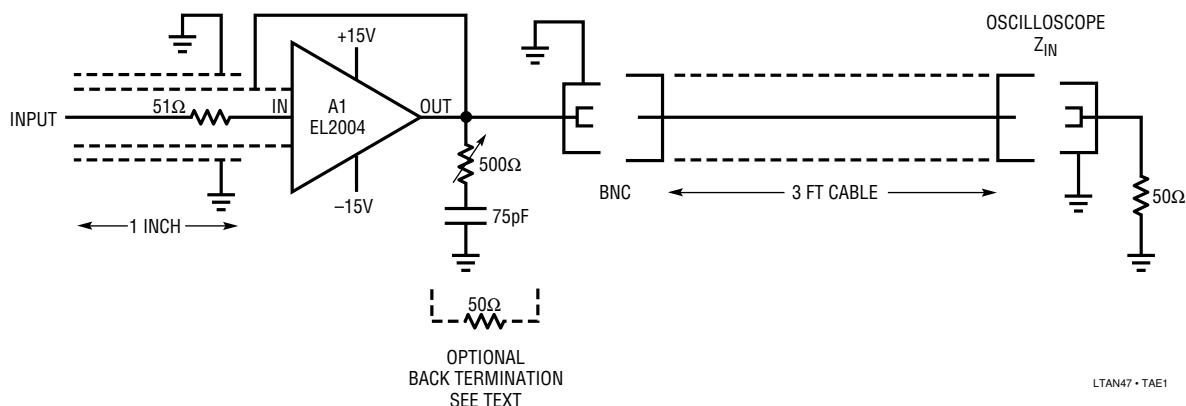
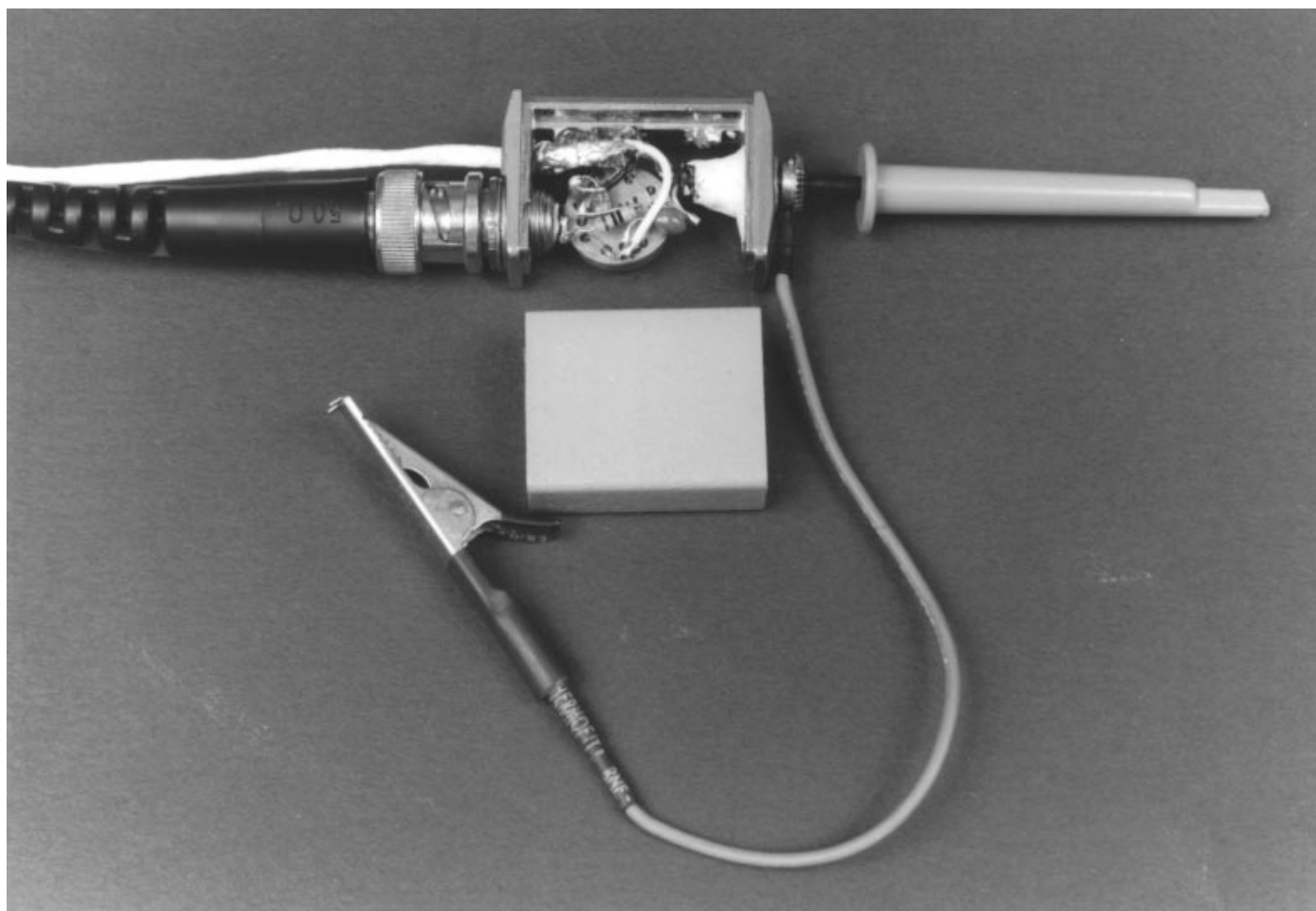
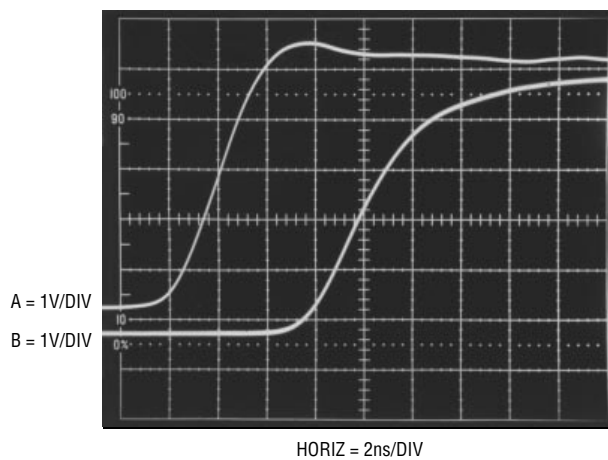


Figure E1. Ultra Fast Buffer Probe Schematic



LTAN47 • TAE2

Figure E2. Physical Layout of Ultra Fast Buffer Probe



LTAN47 • TAE3

Figure E3. Probe Response (Trace B) to Input Pulse (Trace A)

APPENDIX F

Additional Comments on Breadboarding

This section contains, in visual form, commentary on some of the breadboards of the circuits described in the text. The breadboards appear in roughly corresponding

order to their text presentation and comments are brief but hopefully helpful. The bit pushers have commented software; why not commented hardware?

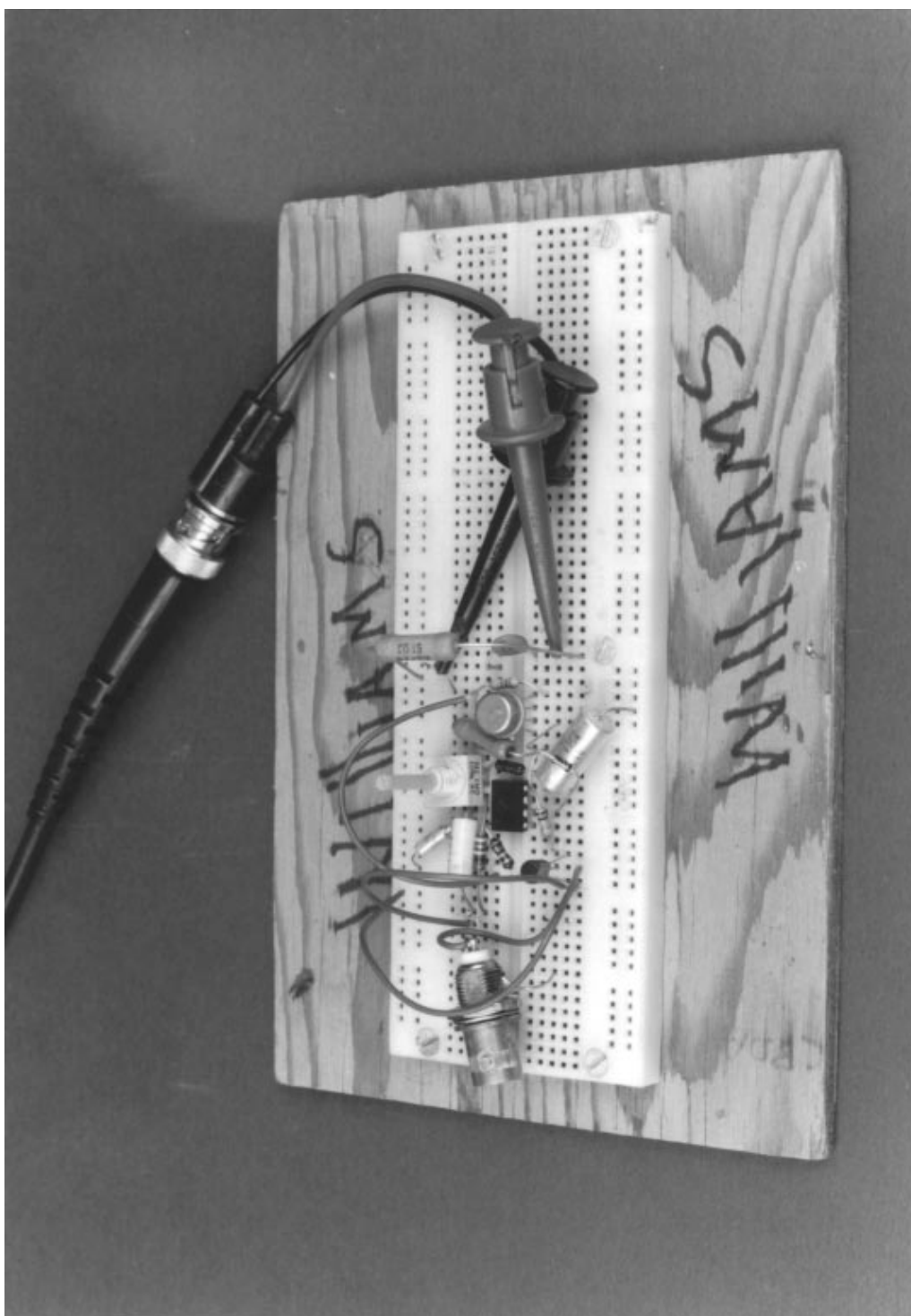
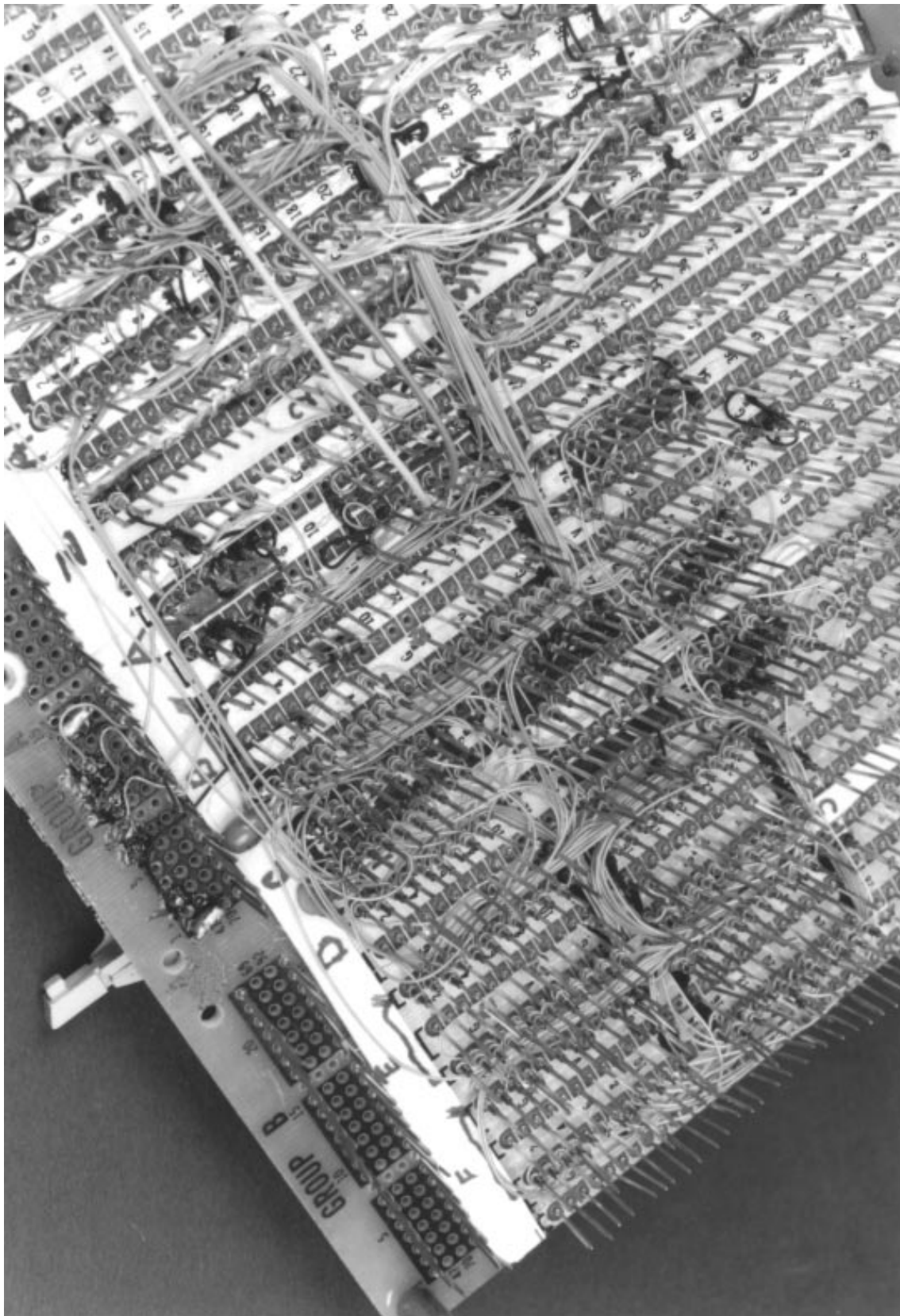
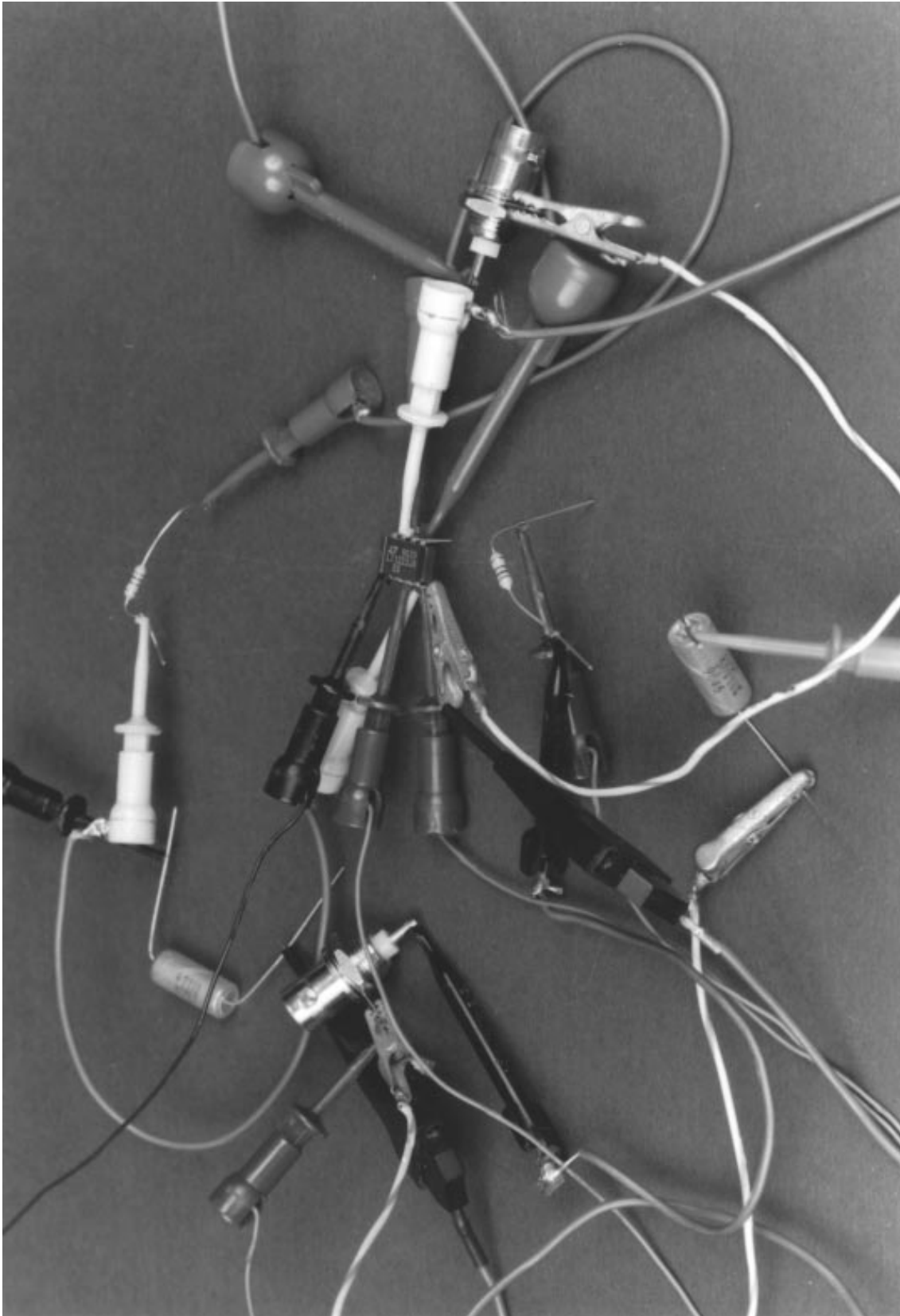


Figure F1. No



LTAN47 • TAF2

Figure F2. No



LTAN47-TAF3

Figure F3. No

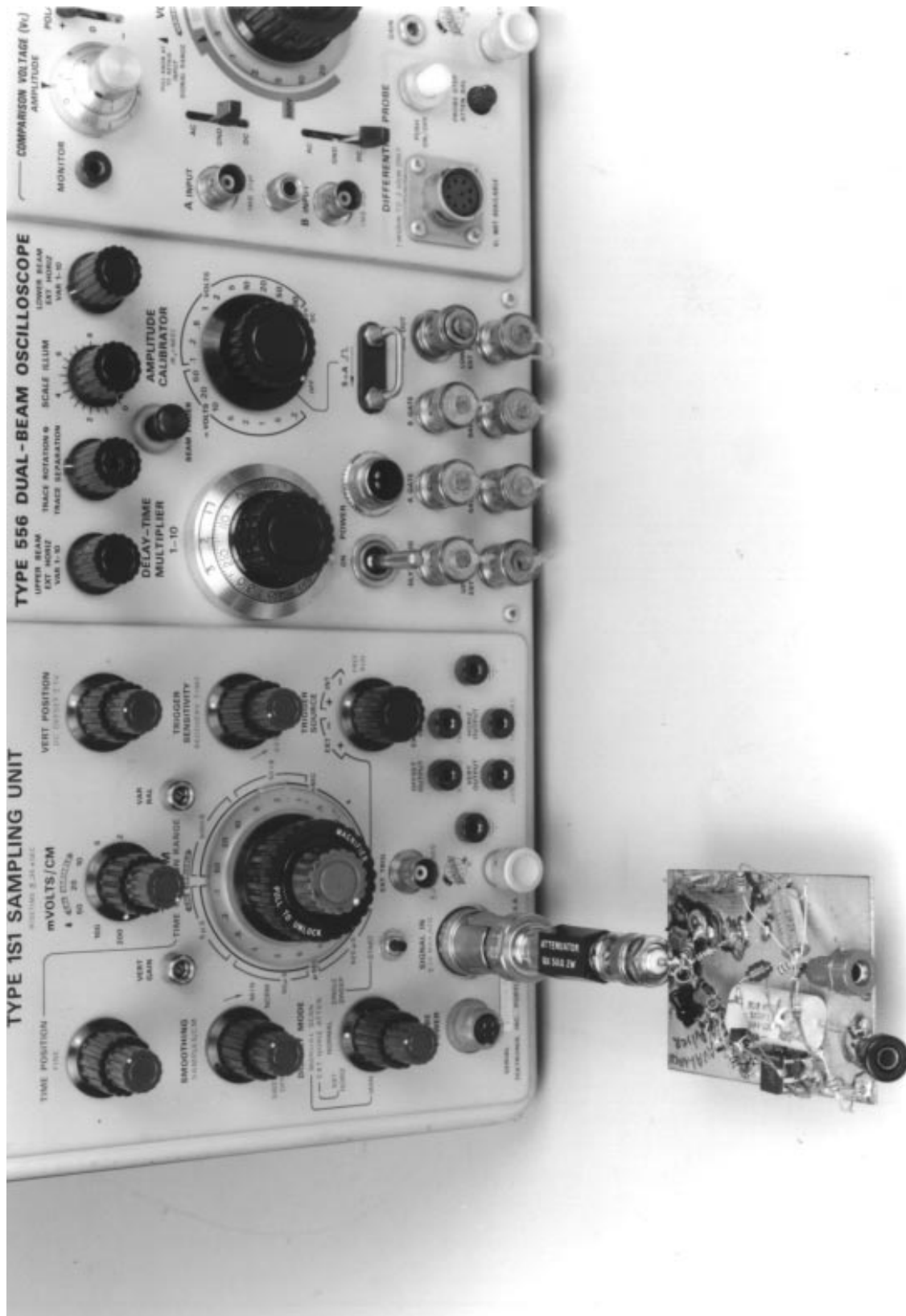
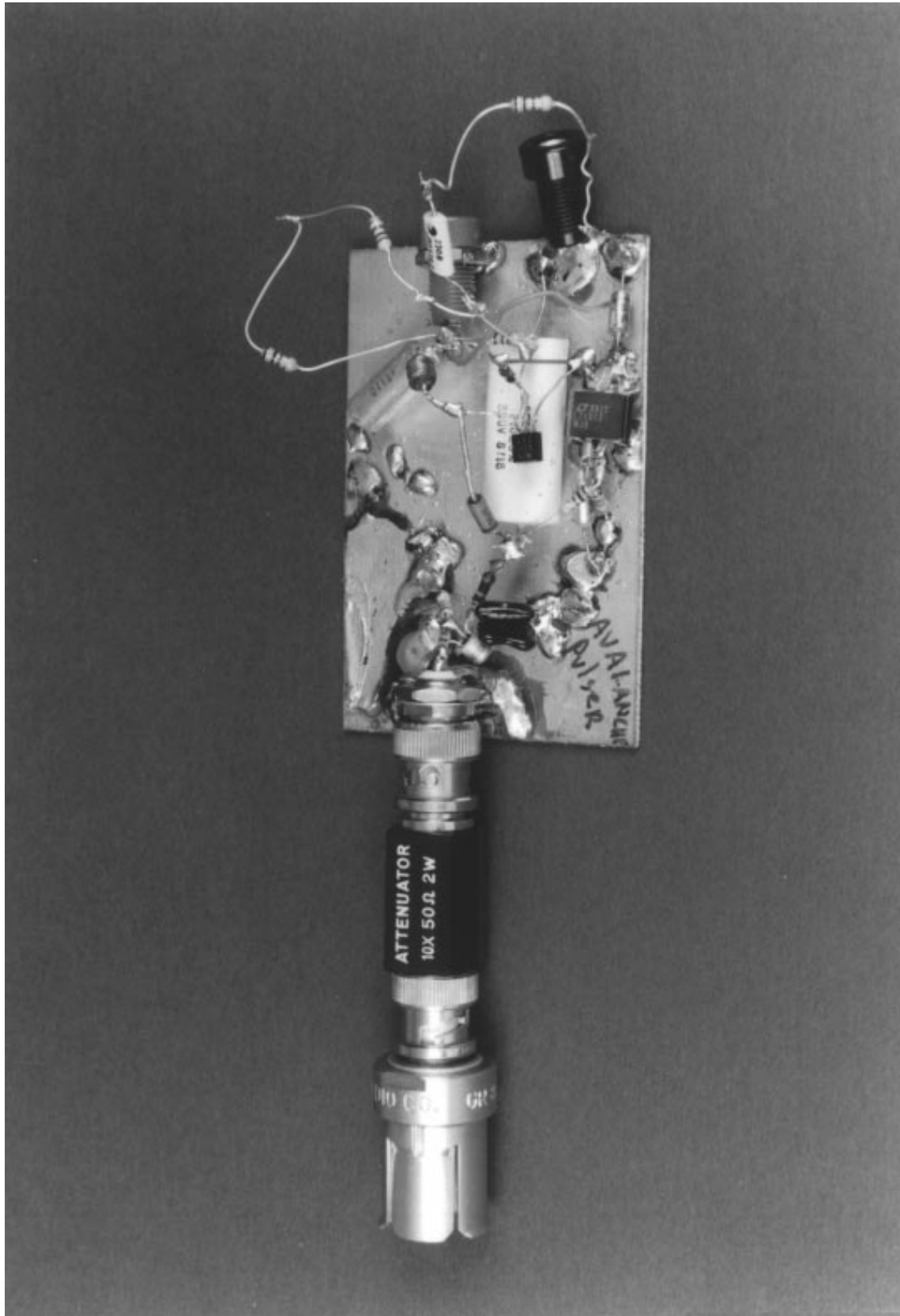
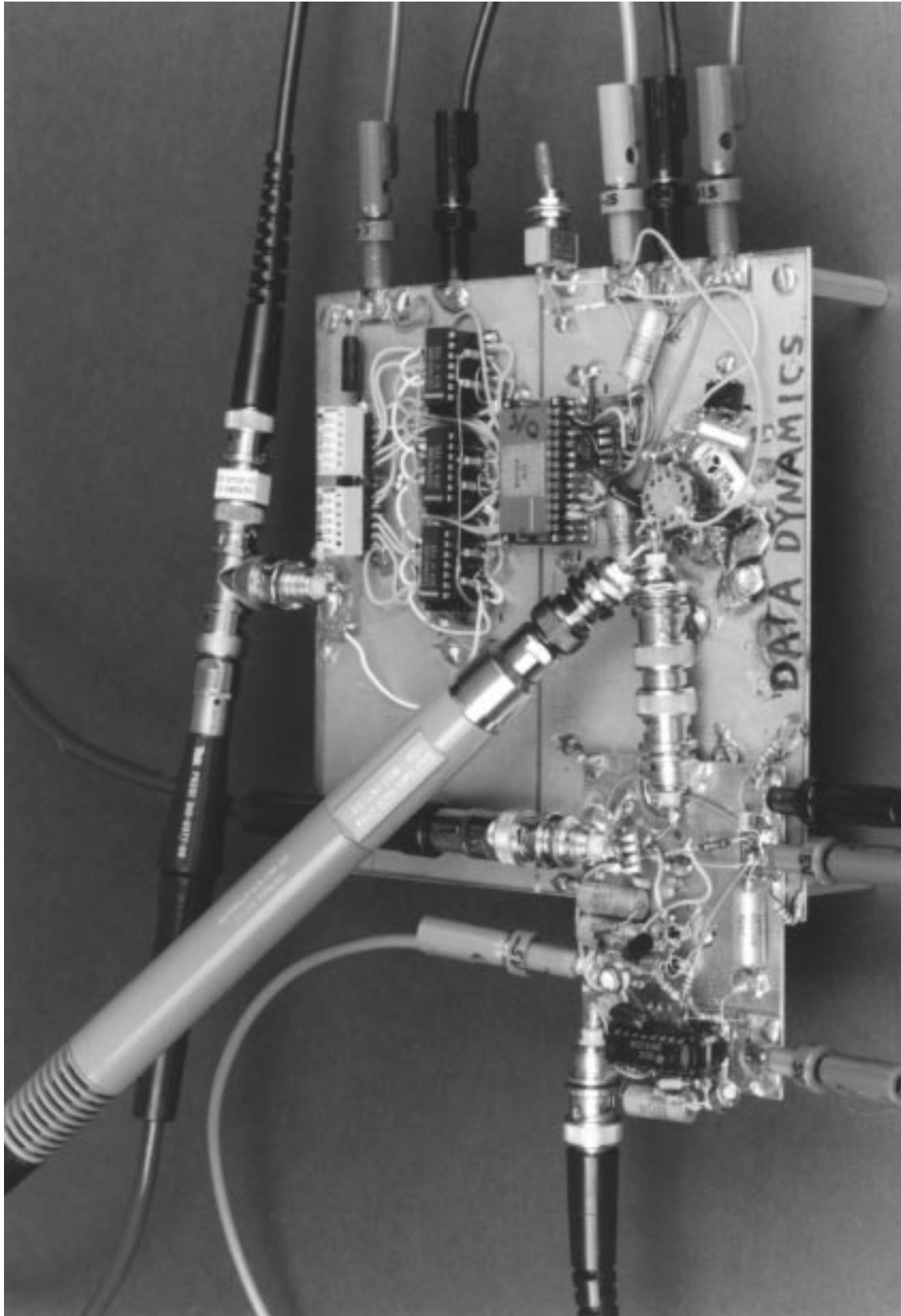


Figure F4. Prototype Avalanche Pulser Under Test. Direct Connection to Oscilloscope Eliminates Cable or Probe Effects



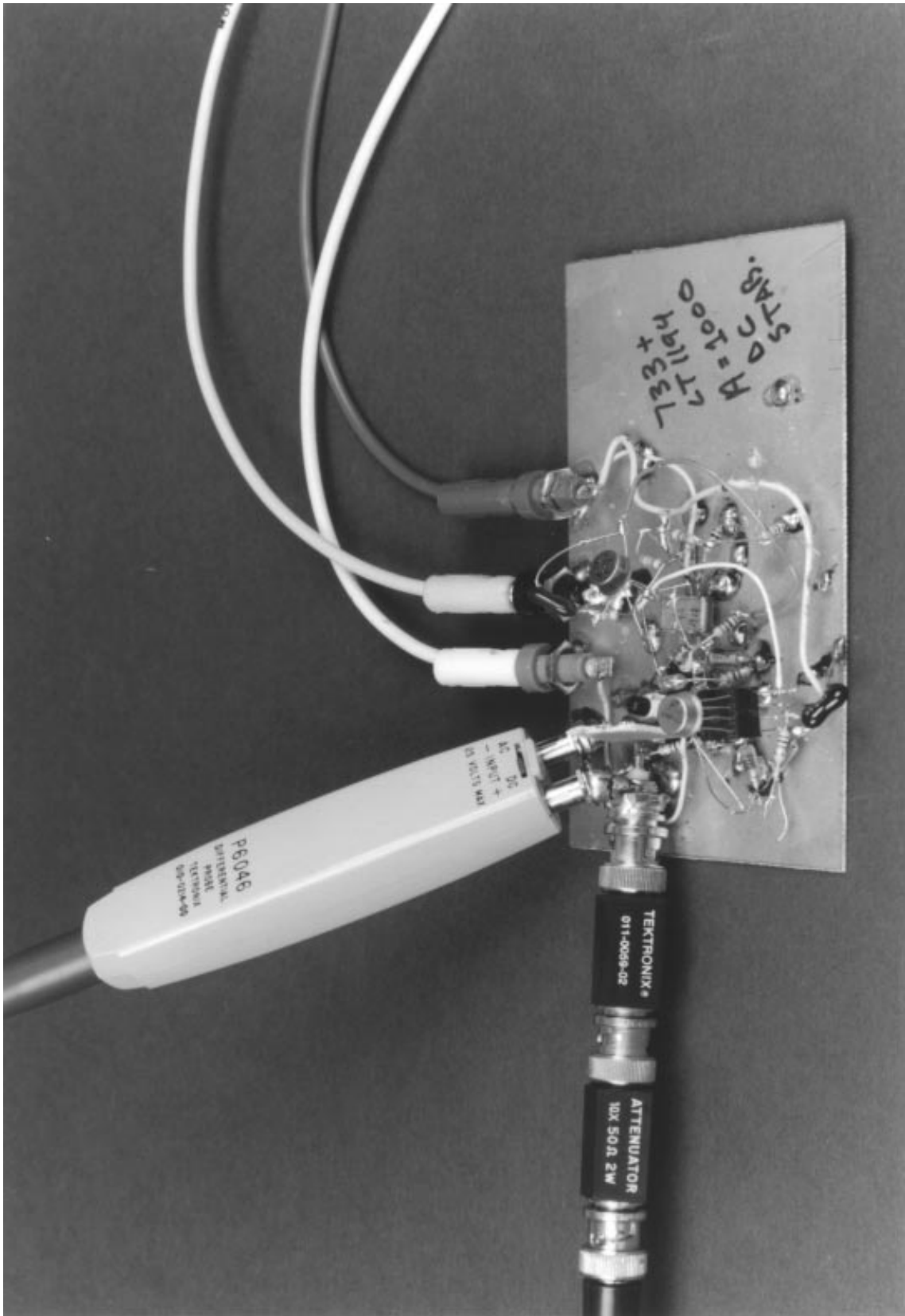
LTAN47-TAF5

Figure F5. Close-Up of Prototype Avalanche Pulse Generator. DC Bias Generator (Right Side of Board) is Carelessly Wired, but Pulse Forming Circuitry (Left Side of Board, by BNC Connector) is Carefully and Tightly Wired



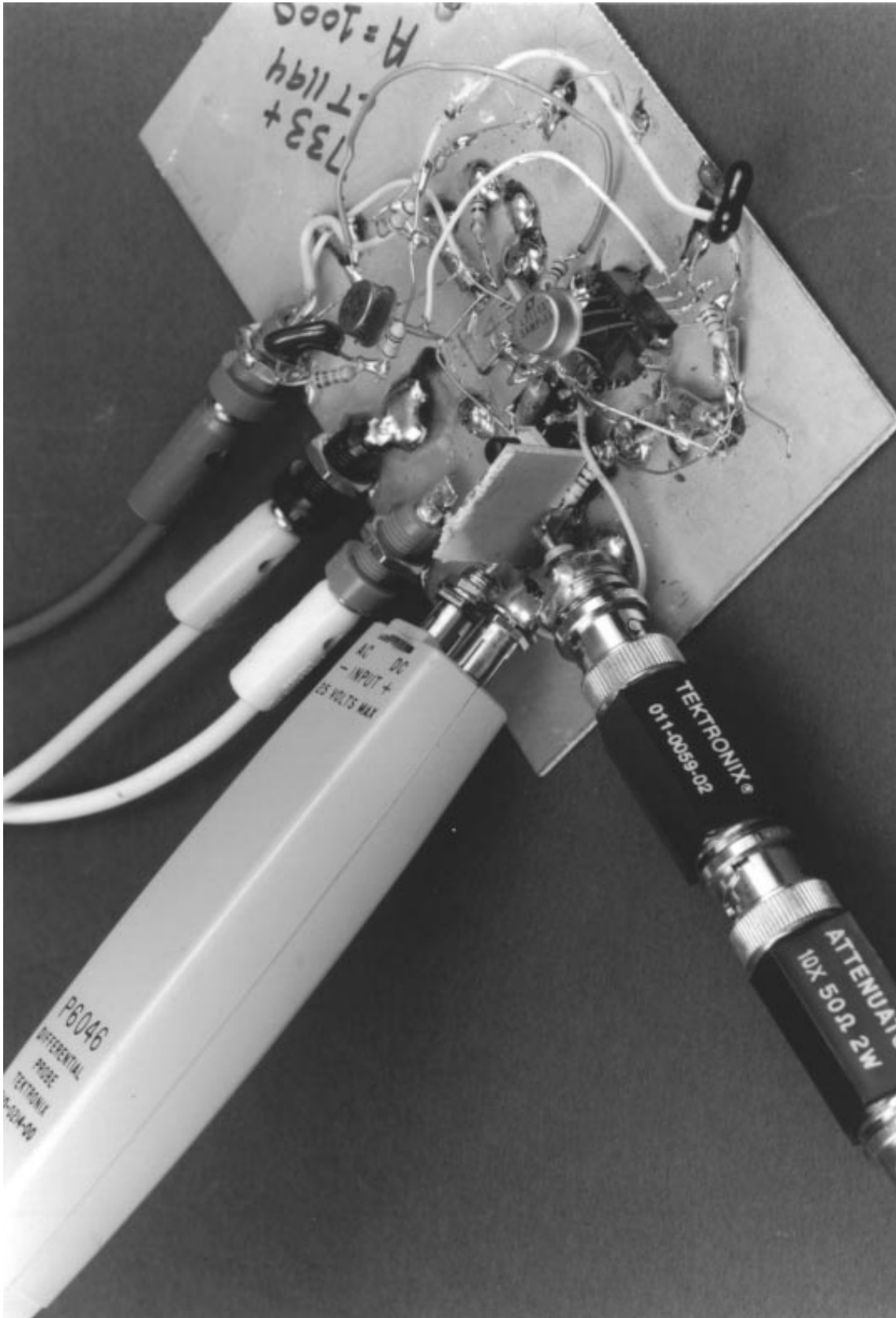
LTAN17-TAF6

Figure F6. The Settling Time Test Fixture Described in Appendix B. DAC and Amplifier are in Center Right of Photo. Note Break in Clad Separating Analog and Digital Grounds and Attention to Layout in Switching Bridge (Lower Left). Switching Bridge is Returned Separately to Ground — Its Board is Mechanically Stood-Off From Main Board by 10M Ω Resistors. Output Section, Driving the Large P6032 Follower Probe, is at Lower Right



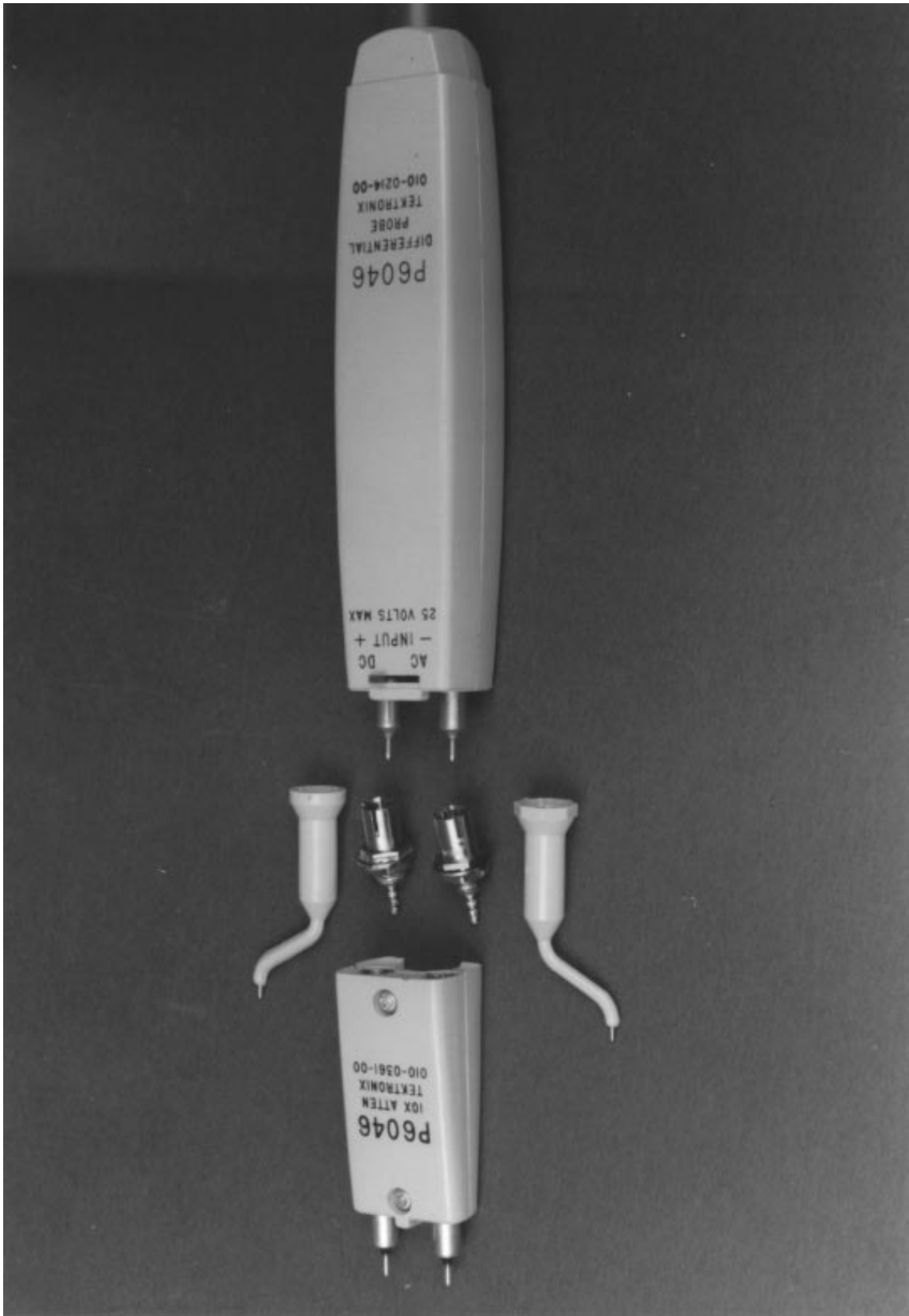
LTAN7 • TAF7

Figure F7. The X1000 38MHz Differential Amplifier. DC and Low Frequency Electronics use Sockets (Foreground) and Air Wire Techniques (Center Right) for Easy and Fast Breadboarding. Wideband Circuitry Hugs the Ground Plane, and is Clustered Near the Input BNC



LTAN7 • TAF8

Figure F8. Input Detail of X1000 Differential Amplifier. Clad Shield (Center Right) Prevents BNC Radiation from Corrupting Low Level Circuitry. Differential Probe Verifies Fidelity of 2.5mV Pulse Out of the X100 Attenuator Stacked Sections. Note DIP Packages Hugging Ground Plane, While Cans Operating at Low Frequency are Carelessly Wired



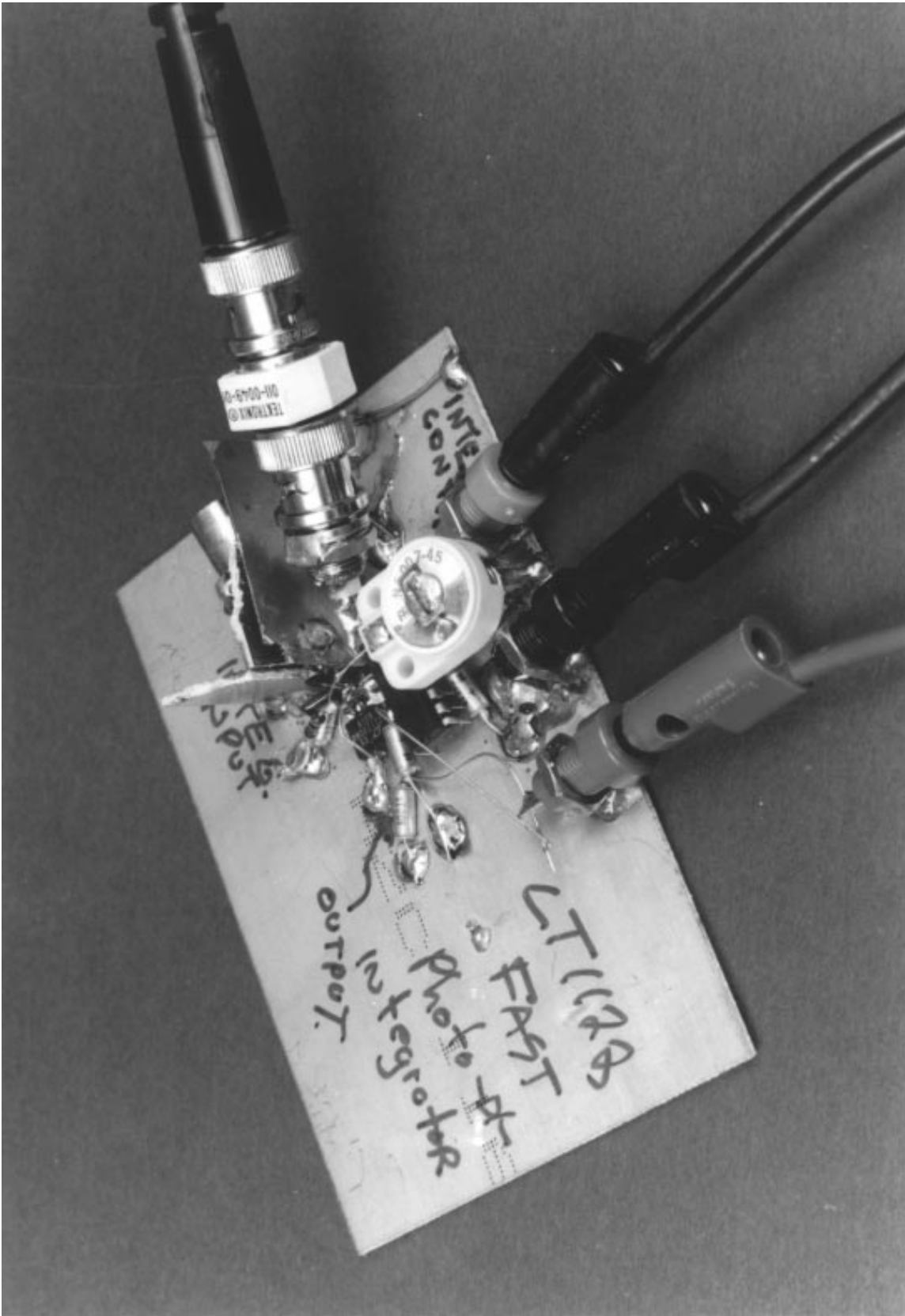
LTAN17 • TAF9

Figure F9. The Differential Probe and Its 10X Attenuator. Offset Probe Tips are Convenient for Making Differential Connections, but Sockets Maintain a True Coaxial Environment and are Preferred



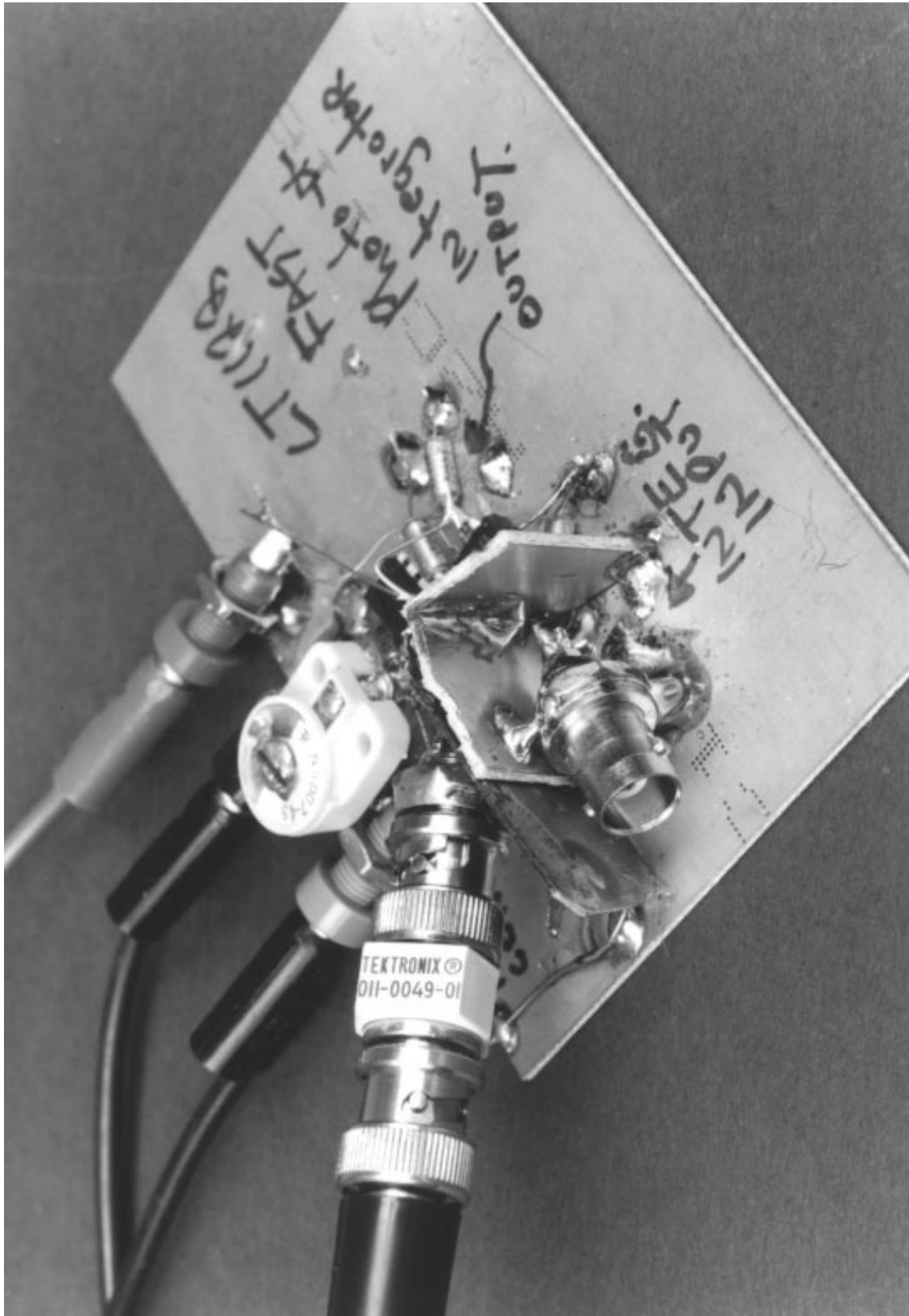
LTAN47-TAF10

Figure F10. The Photodiode Amplifier Layout Emphasizes Low Capacitance at Amplifier (Located Below Trimmer Capacitor, Photo Center Upper Left). Vertical Guard Shield Breaks Up BNC Radiation; was Used When Photo Input was Simulated with a Pulse Generator



LTAN47-TAF11

Figure F11. Fast Photo Integrator Under Test with Pulse Generator Simulating a Photo Input. BNC Radiation is Controlled with Extensive Shielding at Integrator Input (Just Visible Upper Center). Control Input (Cable Connected BNC) is Less Critical; Does Not Require Shielding



LT1128 F12

Figure F12. Photo Integrator Details. Integrator Input BNC is Fully Shielded From Integrator Amp — 1pF Coupling From BNC Output to Summing Point will Cause Excessive Peaking. Amplifier and Switch ICs are Just Visible



LTAN47-TAF13

Figure F13. The Adaptive Trigger Fiber Optic Receiver. BNC Photo-Simulation Input and Fiber Optic Line Both Connected. Low Frequency Wiring is Haphazardly Constructed While High Frequency Sections are Tight and Hug the Ground Plane. Note Vertical Shield at Photo-Simulation Input BNC

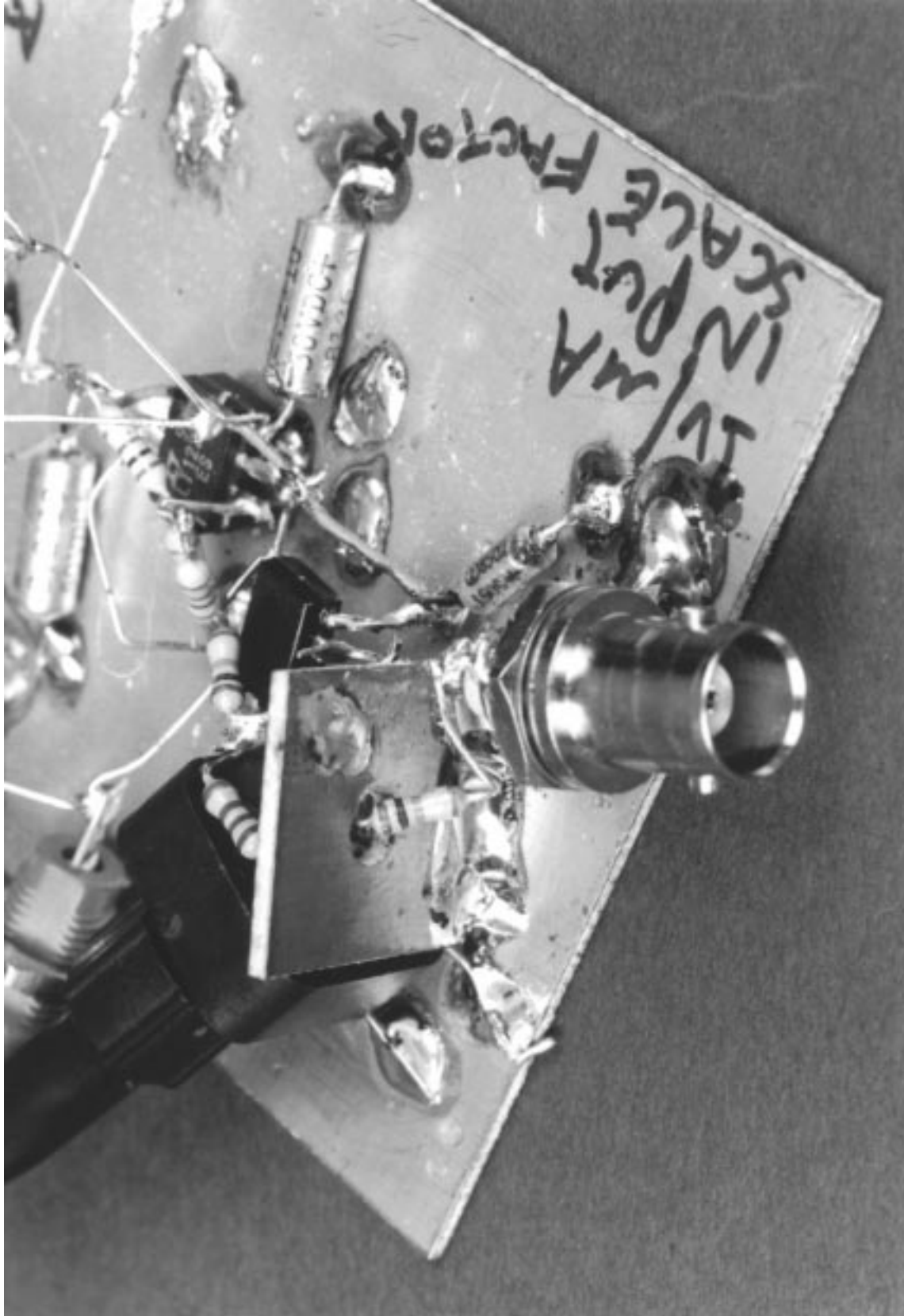
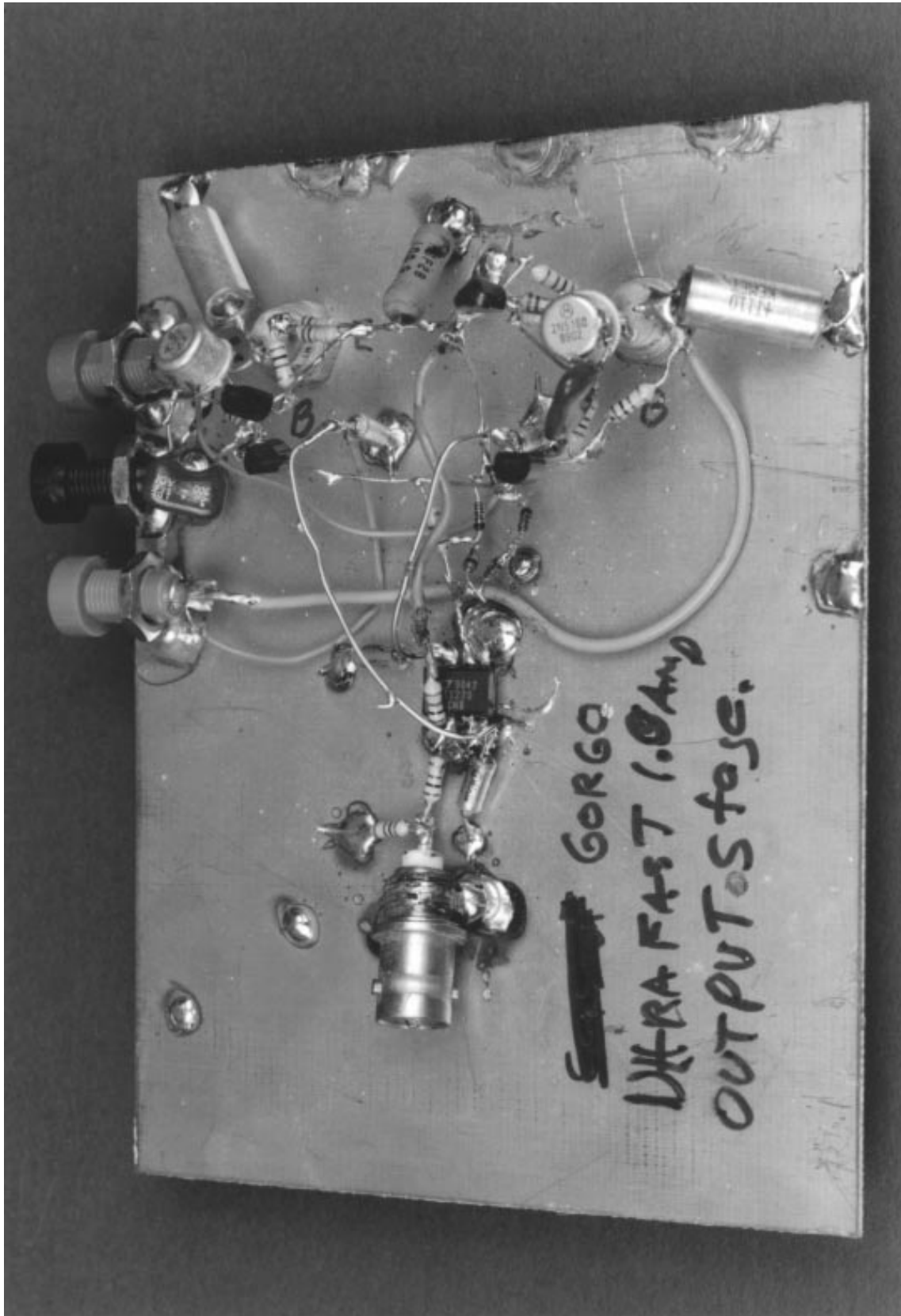
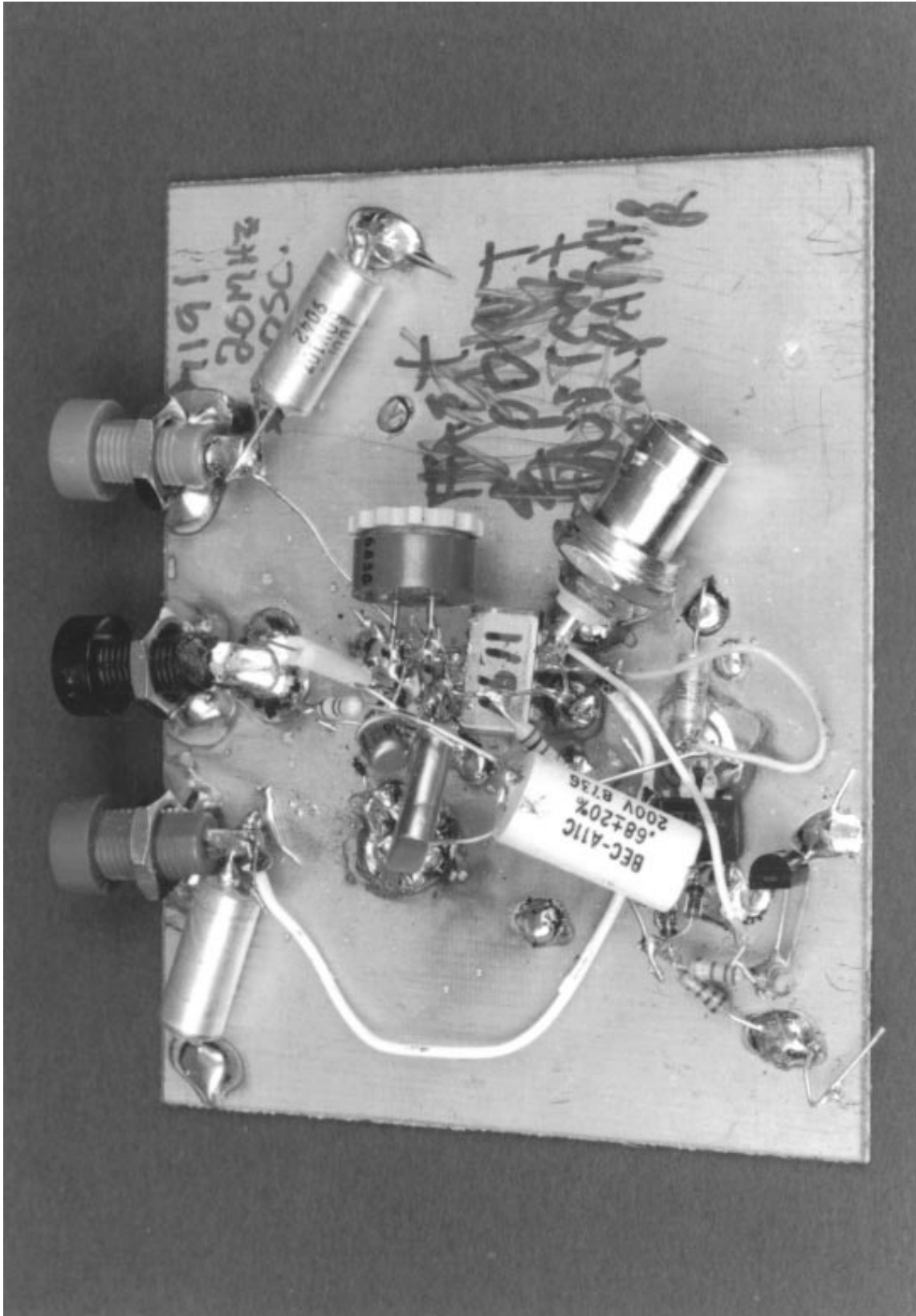


Figure F14. Detail of the Fiber Optic Receiver's Photo-Simulation BNC Input. Resistor From BNC is Routed Through a Small Hole in Vertical Shield, Minimizing Capacitance. Another Resistor on the Shield's Other Side Divides Effects of Residual Capacitance to Keep Summing Point Clean



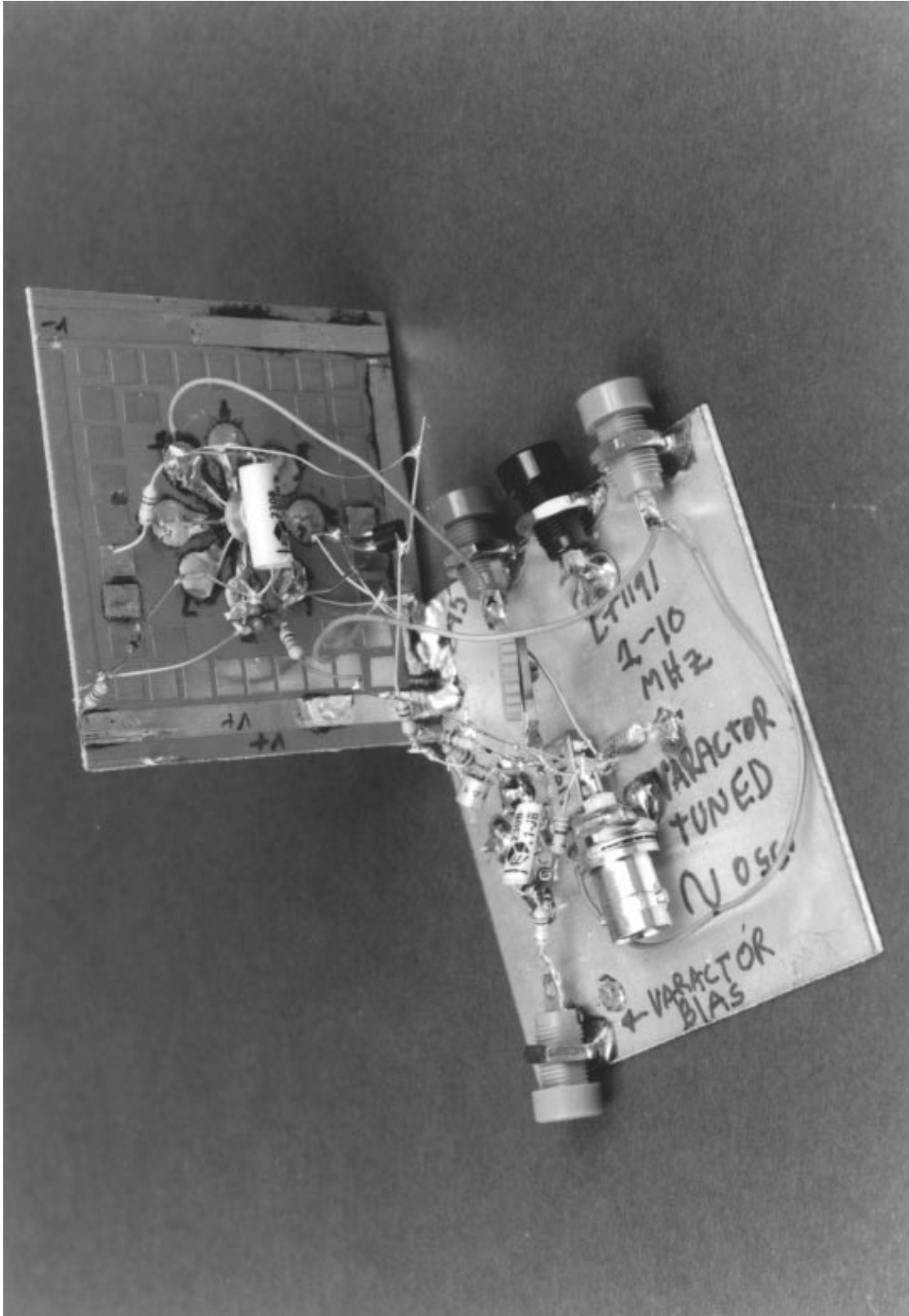
LT4N47-TAF15

Figure F15. The 1A Booster. Note Heavy Bypassing Right at the Output Power Transistors (Both Stud-Mounted to Clad). Local Compensation Capacitors (Right Side Upper and Lower) Have Short Leads



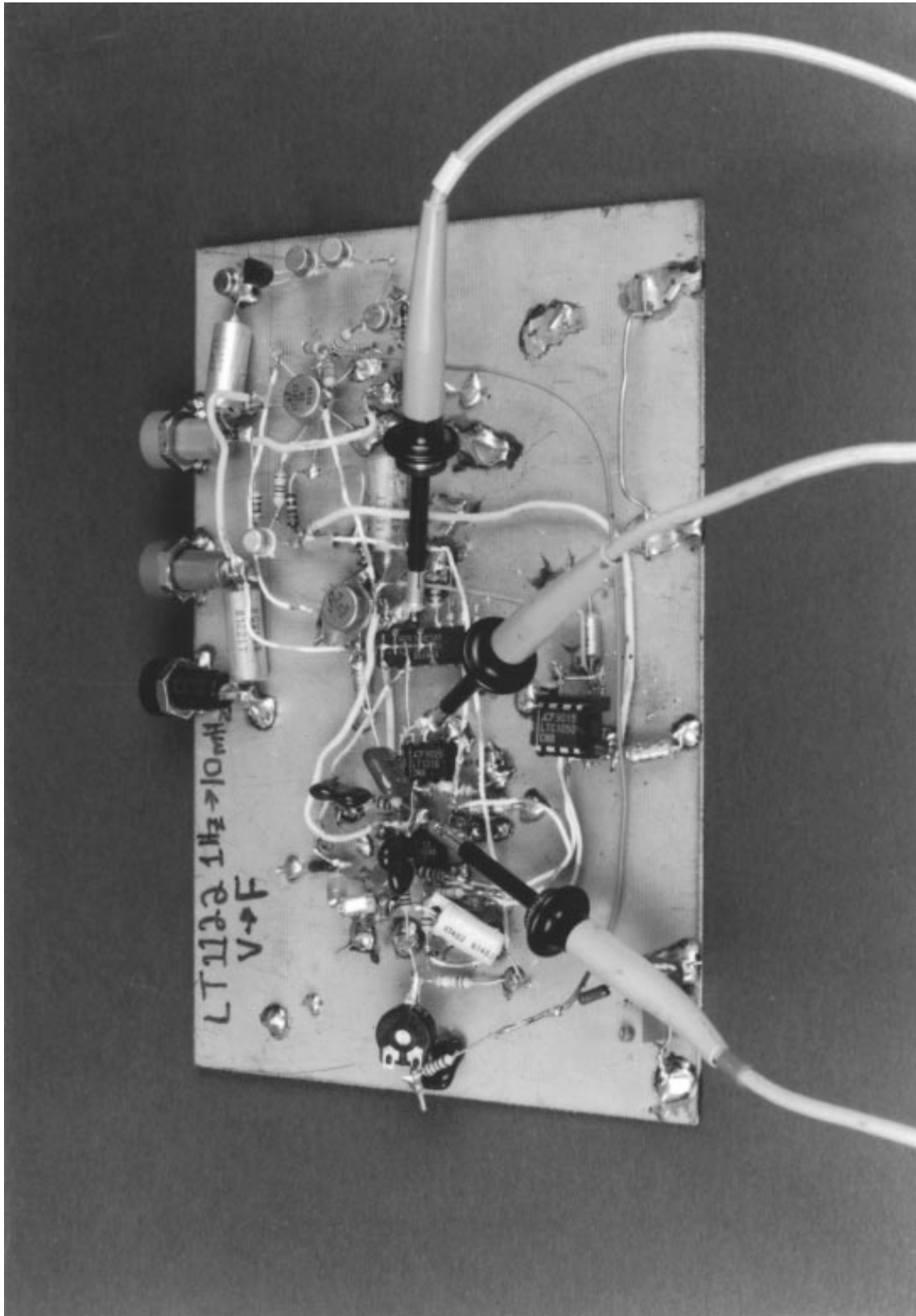
LTAN47-TAF16

Figure F16. 20MHz Sine Wave Crystal Oscillator. DC-AGC Section is at Lower Left, Oscillator is in Center. Control FET is Located at Oscillator Amplifier. Slow Gate Control Signal Arrives via Long-Leaded Resistor, (Photo Center Upper Left)



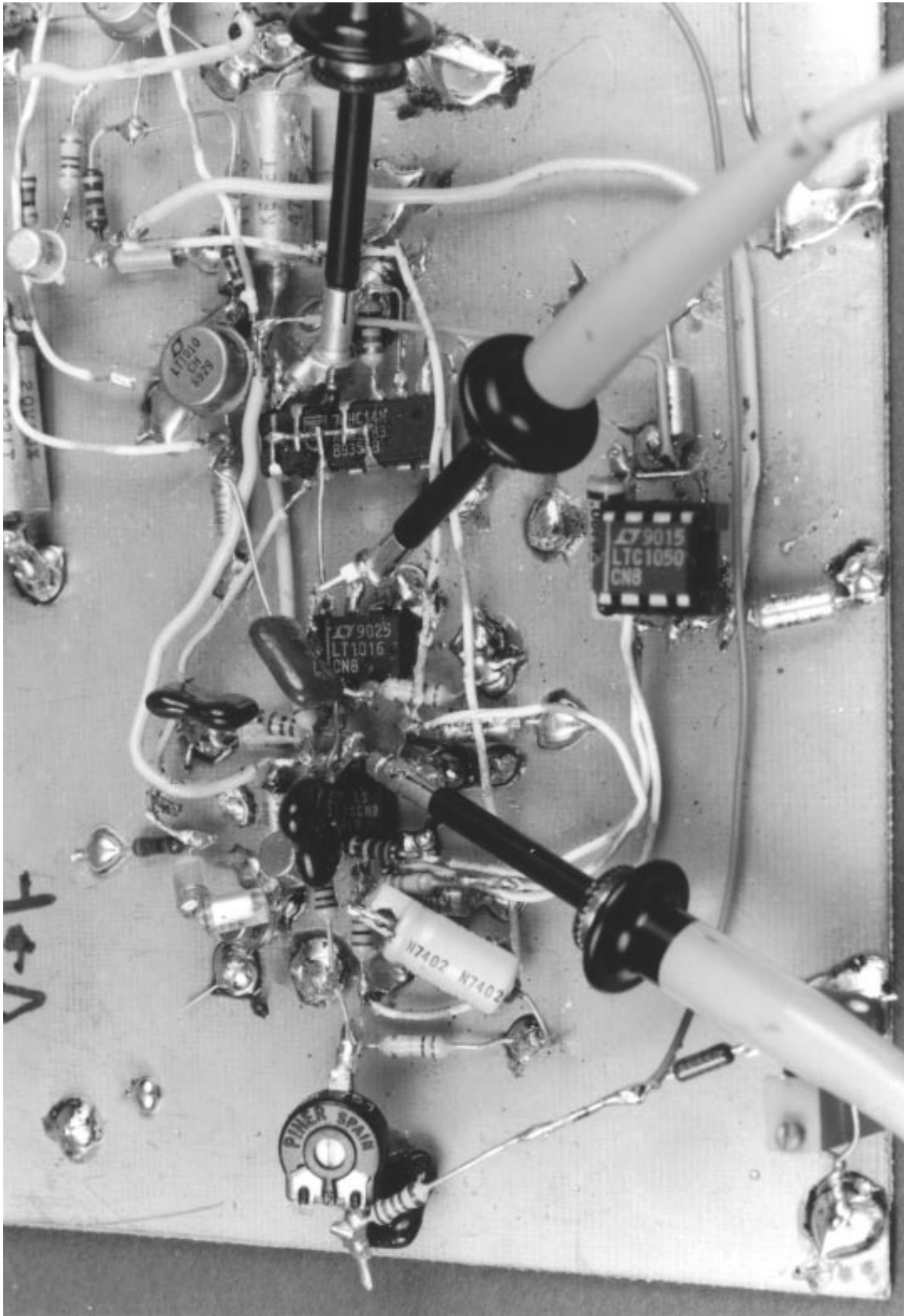
LTAN47-TAF17

Figure F17. The Varactor Tuned Wien Bridge, DC-AGC Section is on Vertical Board — High Frequency Section Hugs the Ground Plane. Control FET (Center Left), Located at Oscillator, is Biased From a Long Line Originating on AGC Board. Note FET Gate Resistor is Located at FET, Not DC Board. Oscillator Output Receives Reverse Treatment



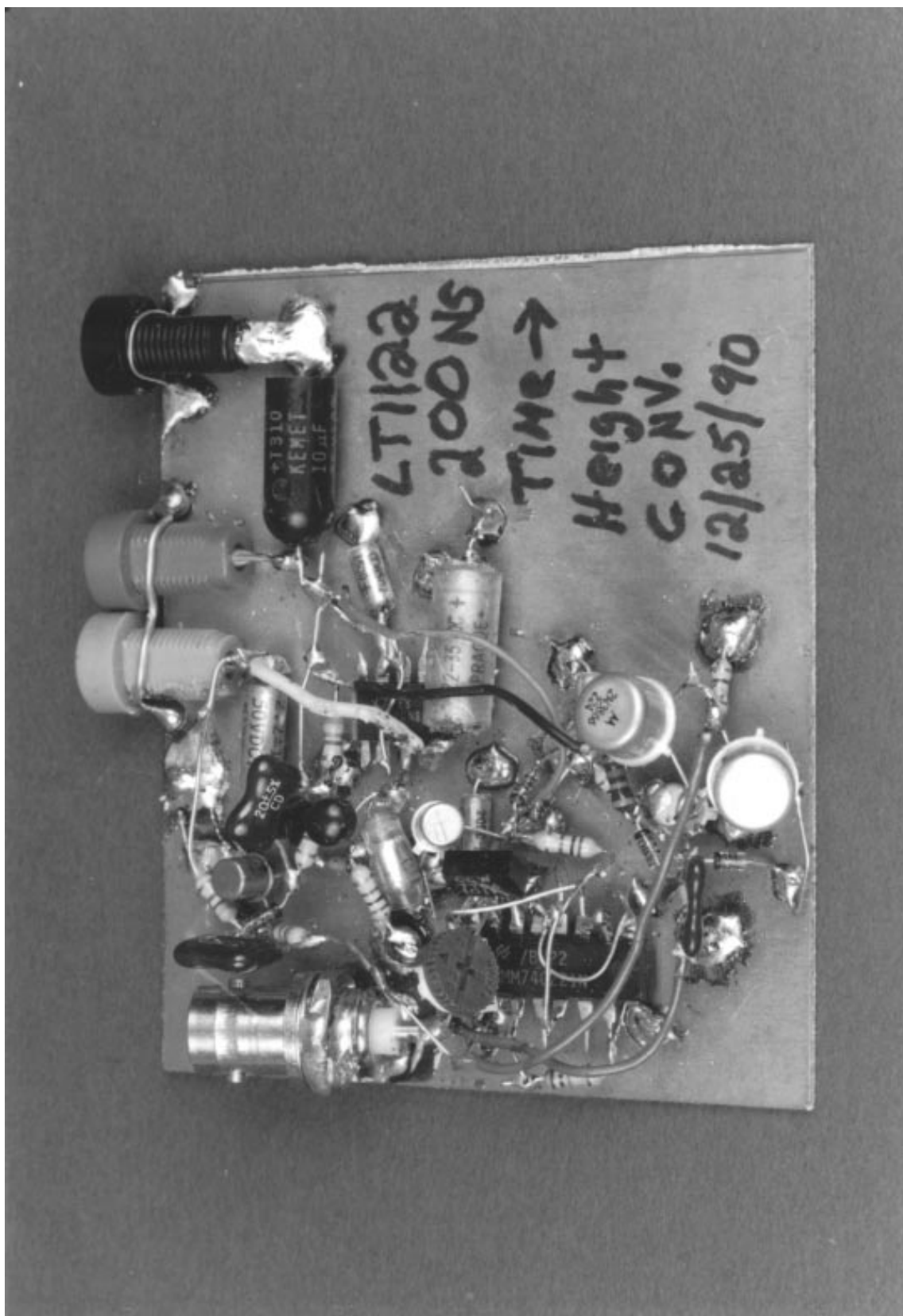
LTAN47-TAF18

Figure F18. The 1Hz-10MHz V to F Breadboard with Probes Attached. DC Servo Amplifier is Socketed, with Long Leads. Reference Section, Starting at Breadboard (Upper Right), Works Toward Reference Switch (Large DIP at Board Center). Note Very Tight Layout in Amplifier-Comparator Region (Board Left Center)



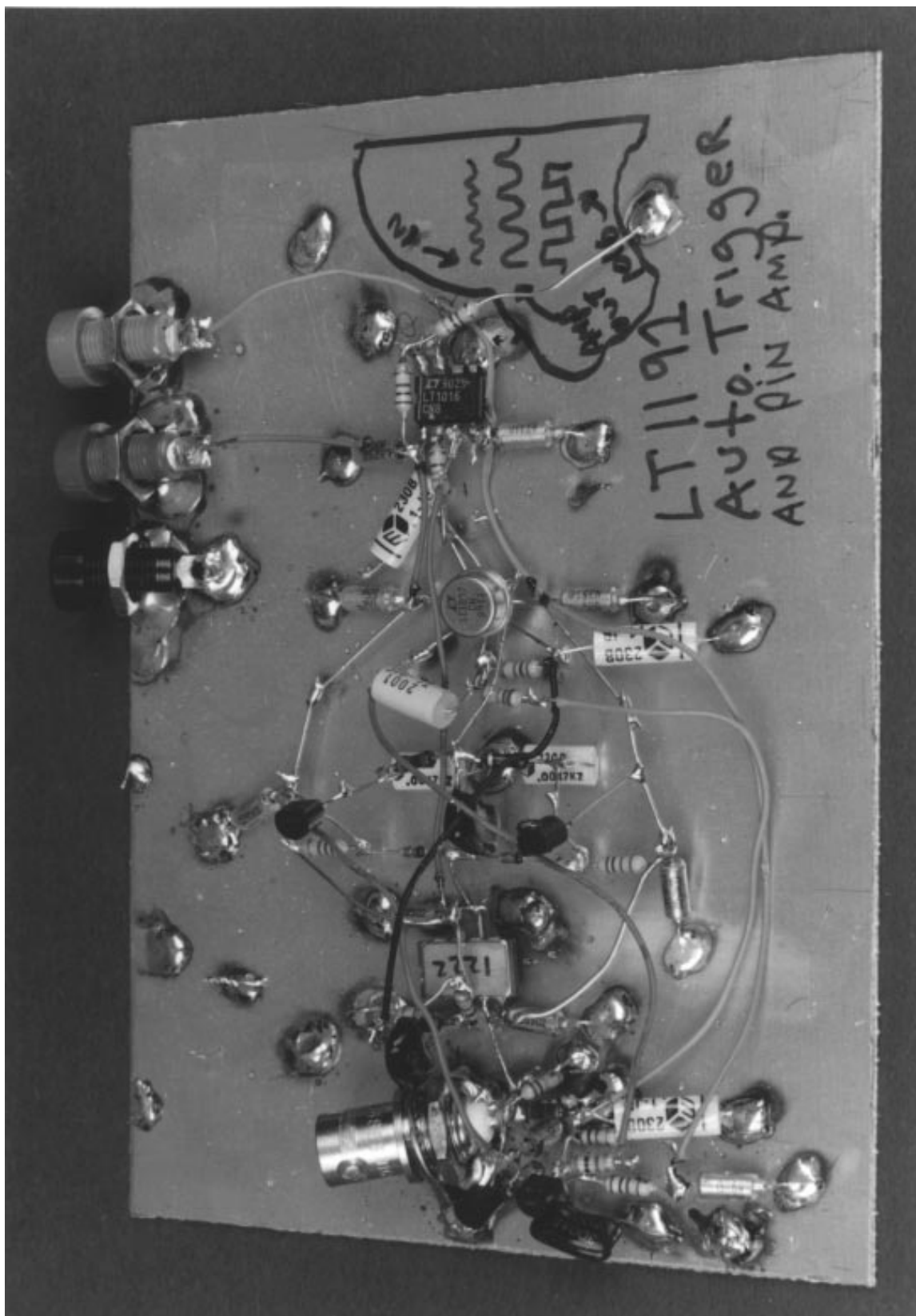
LTAN47-TAF19

Figure F19. Details of 1Hz-10MHz V to F High Speed Section. LT1122 Integrator is Just Visible Under its Associated Discrete Components. Summing Point (Left Side of Amplifier) is Layout's Electrical Center. LT1016 Wiring is Also Very Tight Except for its Output Which Goes to Reference Switch. DC Servo Amplifier Sleeps in its Socket. Note Probe Tip Connectors



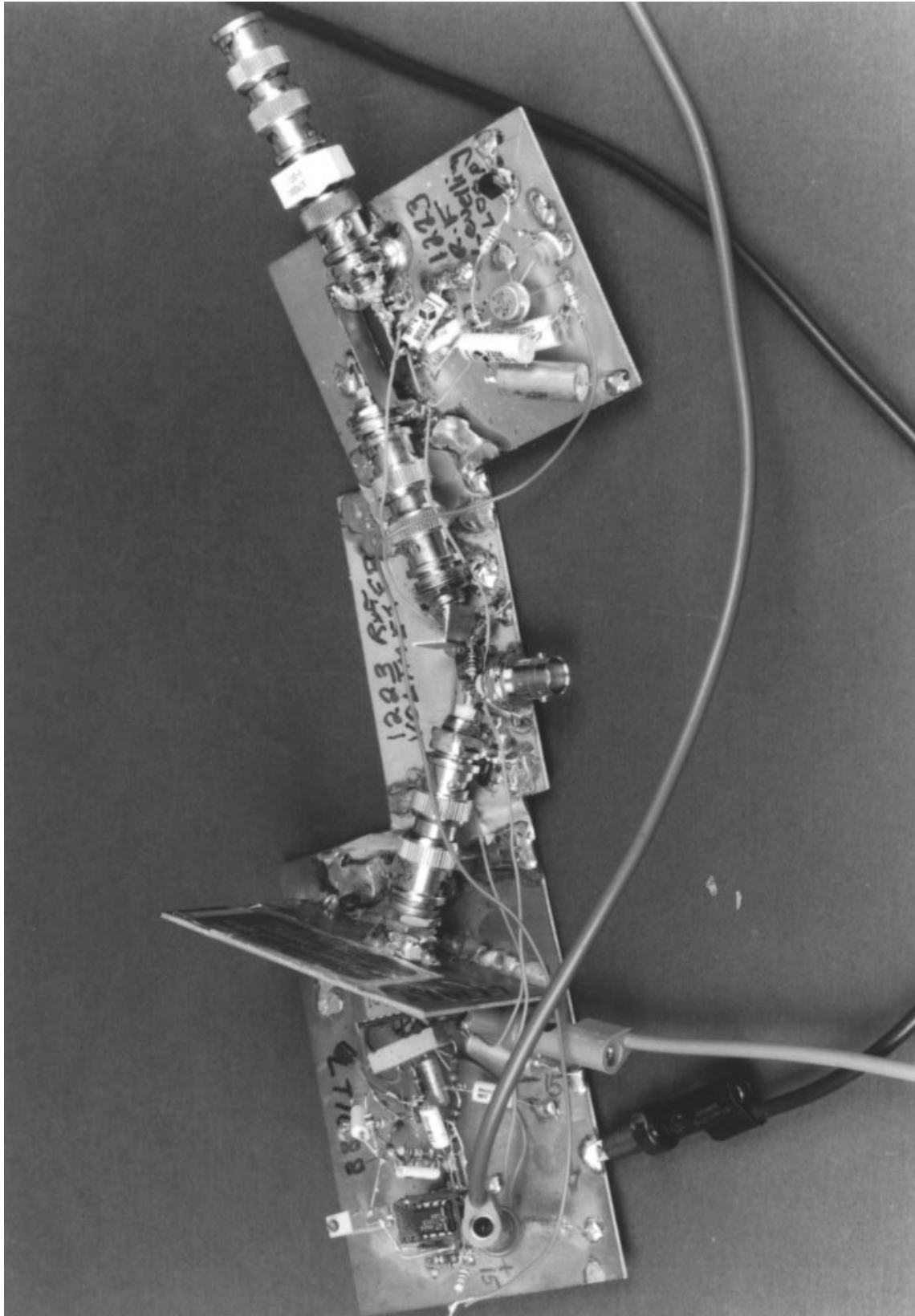
LTAN47-TAF20

Figure F20. The Time-to-Height Converter. Switched Current Source, (Board Center Left), has Very Tight Layout. Follower Amplifier is at Board Upper Center. Major Components (in Order from Top to Bottom), Include Current Source Switch Transistor, Current Source Transistor (Black Case), Integrator Capacitor (Silver) and Reset Transistor. Note Short Connection to Amplifier Input Pin



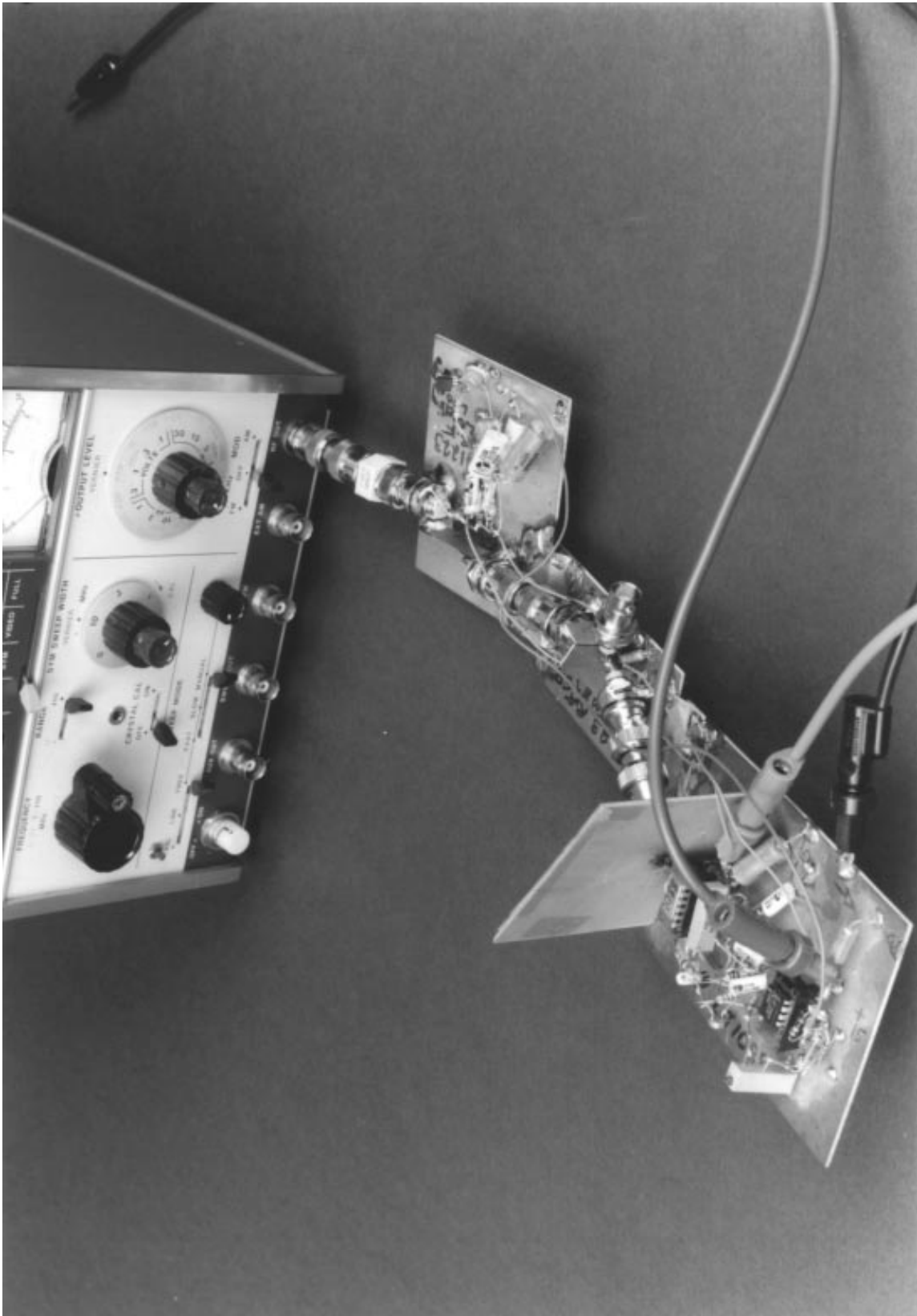
LTAN47-TAF21

Figure F21. The Automatic Trigger. Low Frequency Automatic Level Section is Spread Out, (Right Side of Board). Wideband Circuitry Hugs Ground Plane and is Located Near Input BNC. Amplifier's Low Impedance, Fast Output Feeds LT1016 Output Comparator Over a Relatively Long Wire Run, Routed Through Insensitive Section of DC Circuitry



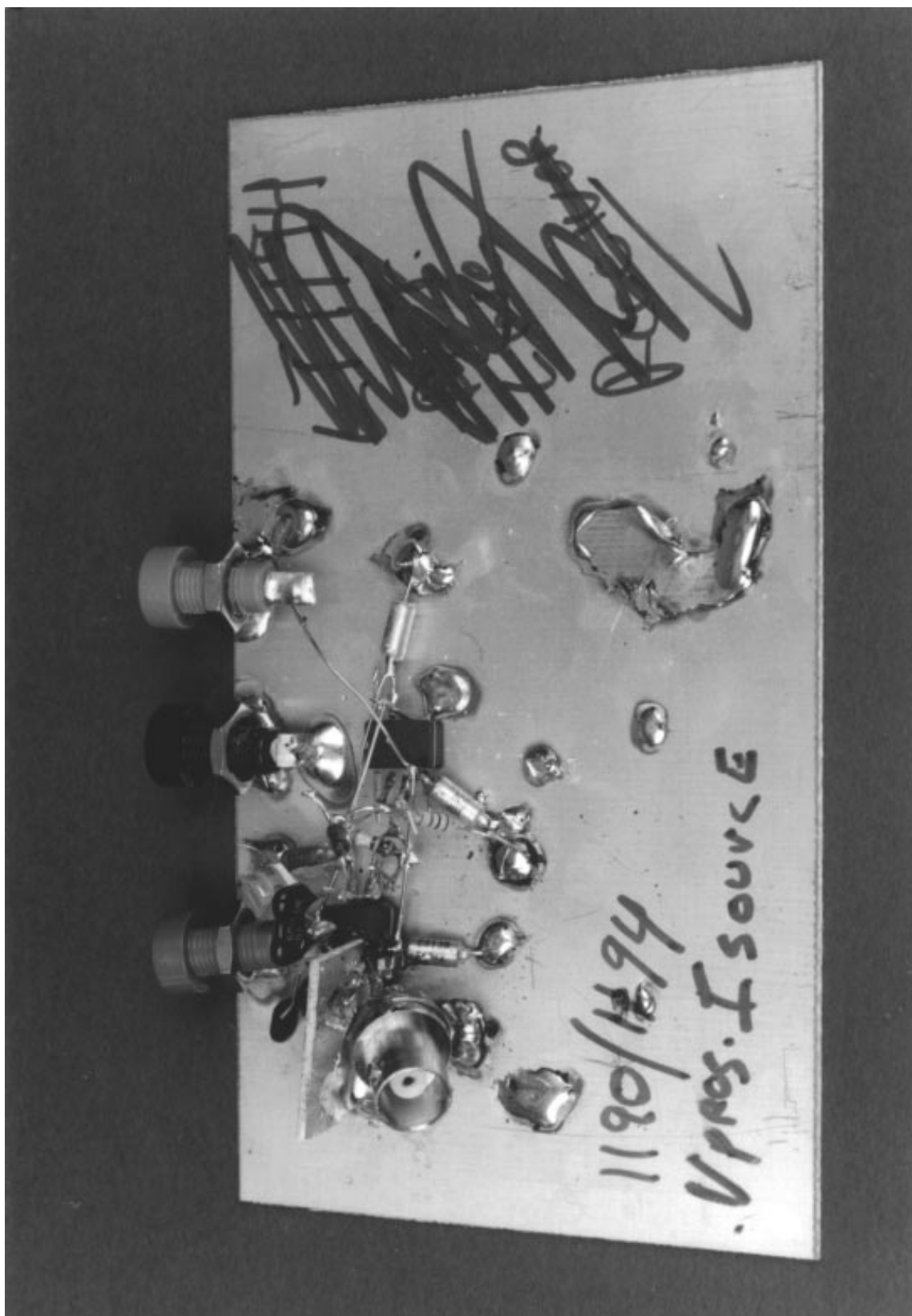
LTAN47-TAF22

Figure F22. The RF Leveling Loop. The RMS to DC Converter (Left Board) was Built First, Then the RF Pre-Amp (Center Board) and Finally the Multiplier-Servo Board. Each Board's Performance was Verified Before Joining Them. Note Copper Tape Maintaining Ground Plane Integrity Between Boards



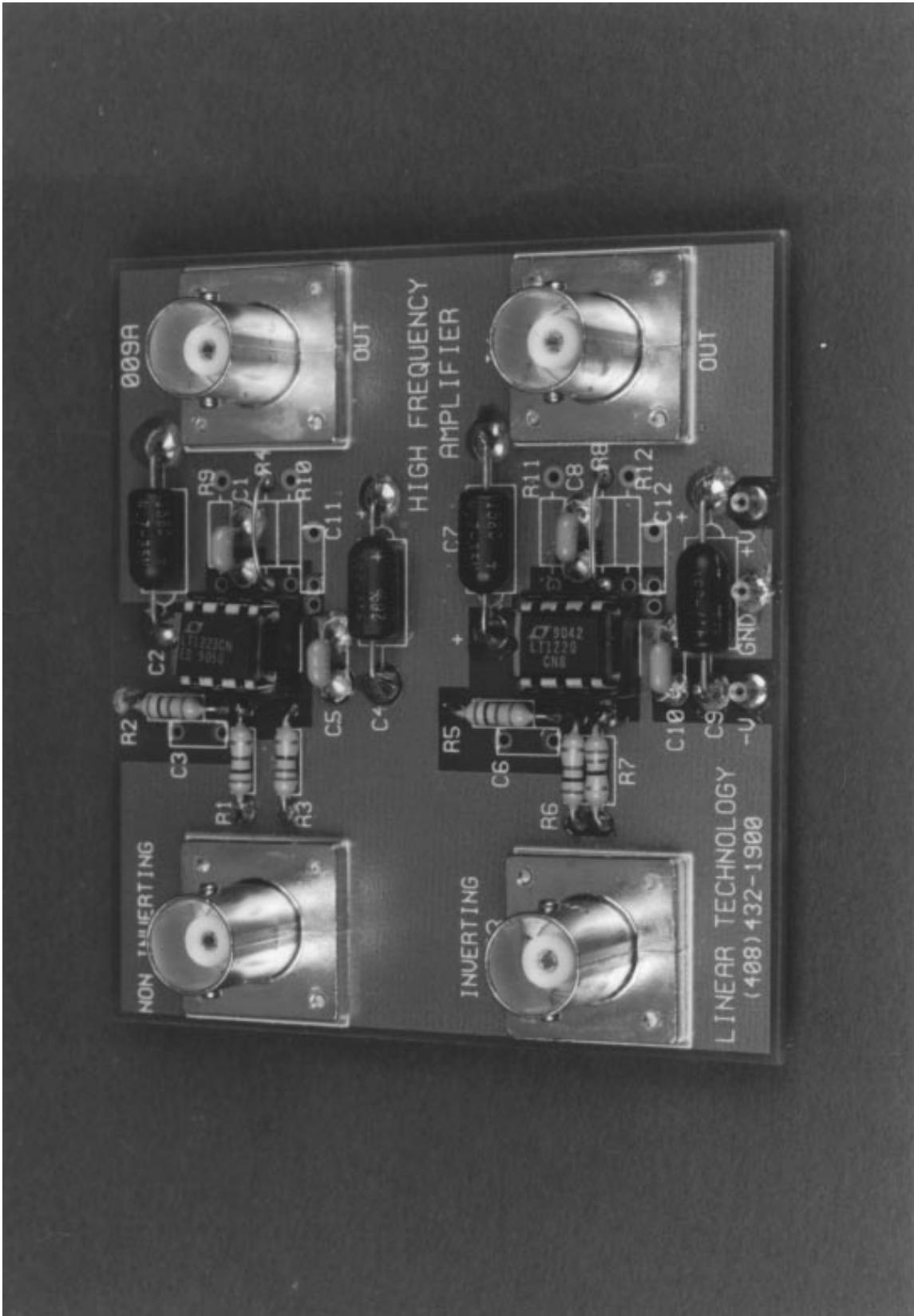
LTAN47-TAF23

Figure F23. RF Leveling Loop Attached Directly to the Test Generator. Cable Uncertainties are Eliminated Because There is No Cable



LTAN47-TAF24

Figure F24. The Voltage Controlled Current Source. Vertical Shield (Upper Left) Absorbs Input BNC Radiation. Amplifier-Loop Compensation Components are Located Directly at Amplifier (Behind Shield)



LTAN17-TAF25

Figure F25. The Good Life. The High Frequency Amplifier Demonstration Board Discussed in Appendix I. Sockets are a Compromise Between Best Performance and Flexibility

APPENDIX G

FCC Licensing and Construction Permit Applications for Commerical AM Broadcasting Stations

In accordance with the application for Figure 116's circuit, and our law-abiding nature, find facsimiles of the appro-

priate FCC applications below. The complete forms are available by writing to:

Federal Communications Commission
Washington, D.C. 20554

Federal Communications Commission
Washington, D.C. 20554

FCC 301

Approved by OMB
3000-0037
Expires 2/29/92
See Page 25 for information
regarding public burden estimate

APPLICATION FOR CONSTRUCTION PERMIT FOR COMMERCIAL BROADCAST STATION

For COMMISSION Fee Use Only	FEE NO:	For APPLICANT Fee Use Only
	FEE TYPE:	Is a fee submitted with this application? ☐ Yes ☐ No
	FEE AMT:	If fee exempt (see 47 C.F.R. Section 1.1112), indicate reason therefor (check one box):
	ID SEQ:	☐ Noncommercial educational licensee
		☐ Governmental entity
		FOR COMMISSION USE ONLY
		FILE NO.

Section I - GENERAL INFORMATION

1. Name of Applicant	Send notices and communications to the following person at the address below:
Street Address or P.O. Box	Name
City State ZIP Code	Street Address or P.O. Box
Telephone No. (include area code)	City State ZIP Code
	Telephone No. (include area code)

2. This application is for: ☐ AM ☐ FM ☐ TV

(a) Channel No. or Frequency	(b) Principal Community
	City State

(c) Check one of the following boxes:

☐ Application for NEW station

☐ MAJOR change in licensed facilities; call sign: _____

☐ MINOR change in licensed facilities; call sign: _____

☐ MAJOR modification of construction permit; call sign: _____

File No. of construction permit: _____

☐ MINOR modification of construction permit; call sign: _____

File No. of construction permit: _____

☐ AMENDMENT to pending application; Application file number: _____

NOTE: It is not necessary to use this form to amend a previously filed application. Should you do so, however, please submit only Section I and those other portions of the form that contain the amended information.

3. Is this application mutually exclusive with a renewal application? ☐ Yes ☐ No

If Yes, state:

Call letters	Community of License
City	State

Figure G1

Federal Communications Commission
Washington, D.C. 20554

APPLICATION FOR NEW BROADCAST STATION LICENSE

(Carefully read instructions before filing out Form)

RETURN ONLY FORM TO FCC

Approved by OMB
3000-0038
Expires 9/30/90

For COMMISSION Fee Use Only	FEE NO:	For APPLICANT Fee Use Only
	FEE TYPE:	Is a fee submitted with this application? ☐ Yes ☐ No
	FEE AMT:	If No, indicate reason therefor (check one box):
	ID SEQ:	☐ Nonfeeable application
		Fee Exempt (See 47 C.F.R. Section 1.1112)
		☐ Noncommercial educational licensee
		☐ Governmental entity

SECTION I - GENERAL DATA

Legal Name of Applicant	Mailing Address
	City State ZIP Code
	Telephone No. (include area code)

1. Facilities authorized by construction permit

This application is for: ☐ Commercial ☐ Noncommercial

☐ AM Directional ☐ AM Non-Directional ☐ FM Directional ☐ FM Non-Directional ☐ TV

Call Letters	Community of License	Construction Permit File No.	Modification of Construction Permit File No(s)	Expiration Date of Last Construction Permit
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2. Is the station now operating pursuant to automatic program test authority in accordance with 47 C.F.R. Section 73.1620? ☐ Yes ☐ No

If No, explain.

3. Have all the terms, conditions, and obligations set forth in the above described construction permit been fully met? ☐ Yes ☐ No

If No, state exceptions.

4. Apart from the changes already reported, has any cause or circumstance arisen since the grant of the underlying construction permit which would result in any statement or representation contained in the construction permit application to be now incorrect? ☐ Yes ☐ No

If Yes, explain.

5. Has the permittee filed its Ownership Report (FCC Form 323) or ownership certification in accordance with 47 C.F.R. Section 73.3615(b)? ☐ Yes ☐ No

If No, explain.

Figure G2

Figure G1-G2. The FCC Forms Appropriate for Figure 116's Circuit

APPENDIX H

About Current Feedback

Contrary to some enthusiastic marketing claims, current feedback isn't new. In fact, it is much older than "normal" voltage feedback, which has been so popularized by op amps. The current feedback connection is *at least* 50 years old, and probably much older. William R. Hewlett used it in 1939 to construct his now famous sine wave oscillator.¹ "Cathode feedback" was widely applied in RF and wideband instrument design throughout the 30's, 40's and 50's. It was a favorite form of feedback, if for no other reason than there wasn't any place else left to feed back to!

In the early 1950's G.A. Philbrick Researches introduced the K2-W, the first commercially available packaged operational amplifier. This device, with its high impedance differential inputs, permitted the voltage type feedback so common today. Although low frequency instrumentation types were quick to utilize the increased utility afforded by high impedance feedback nodes, RF and wideband designers hardly noticed. They continued to use cathode feedback, called (what else?) emitter feedback in the new transistor form.

Numerous examples of the continued use of current feedback in RF and wideband instruments are found in designs dating from the 1950's to the present.² With ostensibly easier to use voltage type feedback a reality during this period, particularly as monolithic devices became cheaper, why did discrete current feedback continue to be used? The reason for the continued popularity of current techniques was (and is) bandwidth. Current feedback is simply much faster. Additionally, within limits, a current feedback based amplifier's bandwidth does not degrade as closed loop gain is increased. This is a significant advantage over voltage feedback amplifiers, where bandwidth falls as closed loop gain is increased.

Note 1: See Appendix C, "The Wien Bridge and Mr. Hewlett", in Reference 19. See also References 20 through 24 and 46.

Note 2: See the "General Electric Transistor Manual", published by G.E. in 1964. See also operating and service manuals for the Hewlett-Packard 3400A RMS Voltmeter, 1120A FET probe, and the Tektronix P6042 current probe.

Relatively recently, current based designs have become available as general purpose, easy to use monolithic and hybrid devices. This brings high speed capability to a much wider audience, hopefully opening up new applications. So, while the technique is not new, marketing claims notwithstanding, the opportunity is. Although current based designs have poorer DC performance than voltage amplifiers, their bandwidth advantage is undeniable. What's the magic?

Current Feedback Basics

William H. Gross

The distinctions of how current feedback amplifiers differ from voltage feedback amplifiers are not obvious at first, because, from the outside, the differences can be subtle. Both amplifier types use a similar symbol, and can be applied on a first order basis using the same equations. However, their behavior in terms of gain bandwidth trade-offs and large signal response is another story.

Unlike voltage feedback amplifiers, small signal bandwidth in a current feedback amplifier isn't a straight inverse function of closed loop gain, and large signal response is closer to ideal. Both benefits are because the feedback resistors determine the amount of current driving the amplifier's internal compensation capacitor. In fact, the amplifier's feedback resistor (R_f) from output to inverting input works with internal junction capacitances to set the closed loop bandwidth. Even though the gain set resistor (R_g) from inverting input to ground works with the R_f to set the voltage gain, just as in a voltage feedback op amp, the closed loop bandwidth does not change. The explanation of this is fairly straightforward. The equivalent gain bandwidth product of the CFA is set by the Thevenin equivalent resistance at the inverting input and the internal compensation capacitor. If R_f is held constant and gain changed with R_g , the Thevenin resistance changes by the same amount as the gain. From an overall loop standpoint, this change in feedback attenuation will produce a change in noise gain, and a proportionate reduction of open loop bandwidth (as in a conventional op amp). With current feedback, however, the key point is that changes in Thevenin resistance also produce compensatory changes

in open loop bandwidth, unlike a conventional fixed gain bandwidth amplifier. As a result, the net closed loop bandwidth of a current-fed-back amplifier remains the same for various closed loop gains.

Figure H1 shows the LT1223 voltage gain vs frequency for five gain settings driving 100Ω . Shown for comparison is a plot of the fixed 100MHz gain bandwidth limitation that a voltage feedback amplifier would have. It is obvious that for gains greater than one, the LT1223 provides 3-20 times more bandwidth.

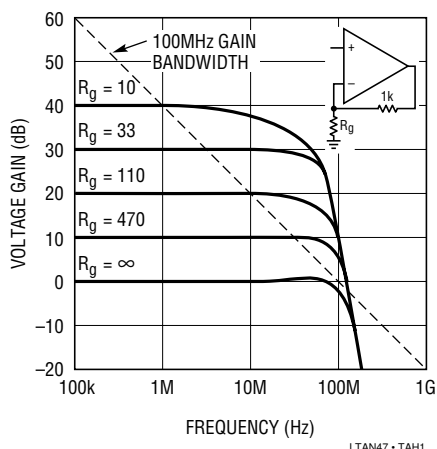


Figure H1. Voltage Gain vs Frequency for Current Feedback Amplifier (Family of Curves) and a Conventional Voltage Amplifier (Straight Line)

Because the feedback resistor determines the compensation of the LT1223, bandwidth and transient response can be optimized for almost every application. When operating on $\pm 15V$ supplies, R_f should be $1k\Omega$ or more for stability, but on $\pm 5V$, the minimum value is 680Ω , because the junction capacitors increase with lower voltage. For either case, larger feedback resistors can also be used, but will slow down the LT1223 (which may be desirable in some applications).

The LT1223 delivers excellent slew rate and bandwidth with better DC performance than previous current feed-

back amplifiers (CFAs). On $\pm 15V$ supplies with a $1k$ feedback resistor, the small signal bandwidth is $100MHz$ into a 400Ω load and $75MHz$ into 100Ω . The input will follow slew rates of $250V/\mu s$ with the output generating over $500V/\mu s$, and output slew rate is well over $1000V/\mu s$ for large input overdrive. Input offset voltage is $3mV$ (max), and input bias current is $3\mu A$ (max). A $10k\Omega$ pot, connected to pins 1 and 5 with wiper to V^+ , provides optional offset trimming. This trim shifts inverting input current about $\pm 10\mu A$, effectively producing input voltage offset.

The LT1223 also has shutdown control, available at pin 8. Pulling more than $200\mu A$ from pin 8 drops the supply current to less than $3mA$, and puts the output into a high impedance state. The easy way to force shutdown is to ground pin 8 using an open collector (drain) logic stage. An internal resistor limits current, allowing direct interfacing with no additional parts. When pin 8 is open, the LT1223 operates normally.

The difference in operating characteristics between op amps and CFAs result in slight differences in common circuit configurations. Figure H2 summarizes some popular circuit types, showing differences between op amps and CFAs. Gain can be set with either R_{IN} or R_f in an op amp, while a CFA's feedback resistor (R_f) is fixed. Op amp bandwidth is controllable with a feedback capacitor; for a CFA, bandwidth must be limited at the input. A feedback capacitor is never used. In an integrator, the $1k$ resistor must be included in the CFA so its negative input sees the optimal impedance. Finally, (not shown) there is no correlation between bias currents of a CFA's inputs. Because of this, source impedance matching will not improve DC accuracy. Matching input source impedances aids offset performance in op amps that do not have internal bias current cancellation.

Application Note 47

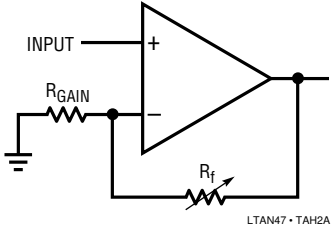
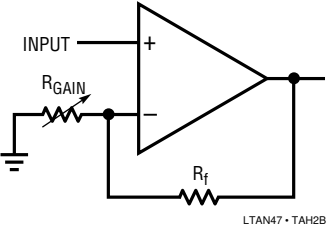
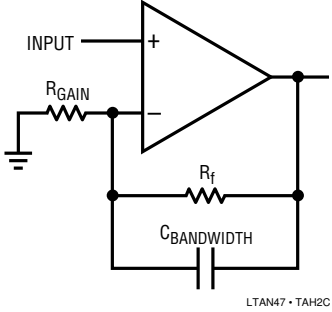
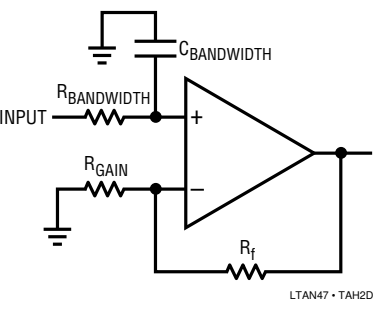
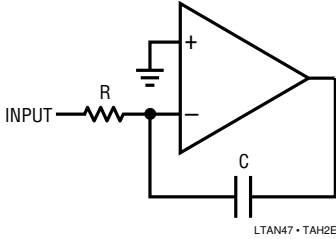
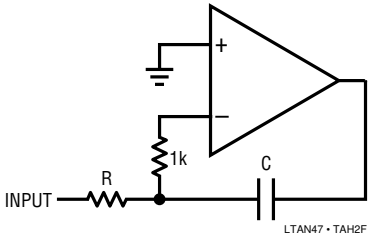
CIRCUIT TYPE	VOLTAGE FEEDBACK OP AMP	CURRENT FEEDBACK AMPLIFIER
SETTING GAIN	 LTAN47 • TAH2A	 LTAN47 • TAH2B
LIMITING BANDWIDTH	 LTAN47 • TAH2C	 LTAN47 • TAH2D
INTEGRATOR	 LTAN47 • TAH2E	 LTAN47 • TAH2F

Figure H2. Some Practical Differences in Applying Current and Voltage Amplifiers

APPENDIX I

High Frequency Amplifier Evaluation Board

LTC demo board 009 (photo, Figure I1, schematic, Figure I2) is designed to simplify the evaluation of high speed operational amplifiers. It includes both an inverting and non-inverting circuit, and extra holes are provided to allow the use of board-mounted BNC or SMA connectors. The two circuits are independent with the exception of shared power supply and ground connections.

Layout is a primary contributor to the performance of any high speed amplifier. Poor layout techniques adversely affect the behavior of a finished circuit. Several important layout techniques, all used in demo board 009, are described below:

1. Top side ground plane. The primary task of a ground plane is to lower the impedance of ground connections. The inductance between any two points on a uniform sheet of copper is less than the inductance of a thin, straight trace of copper connecting the same two points. The ground plane approximates the characteristics of a copper sheet and lowers the impedance at key points in the circuit, such as the grounds of connectors and supply bypass capacitors.
2. Ground plane voids. Certain components and circuit nodes are very sensitive to stray capacitance. Two good examples are the summing node of the op amp and the feedback resistor. Voids are put in the ground plane in these areas to reduce stray ground capacitance.
3. Input/output matching. The width of the input and output traces is adjusted to a stripline impedance of 50Ω . Note that the terminating resistors (R3 and R7) are connected to the end of the input lines, not at the connector. While stripline techniques aren't absolutely necessary for the demo board, they are important on larger layouts where line lengths are longer. The short lines on the demo board can be terminated in 50Ω , 75Ω , or 93Ω without adversely affecting performance.

4. Separation of input and output grounds. Even though the ground plane exhibits a low impedance, input and output grounds are still separated. For example, the termination resistors (R3 and R7) and the gain-setting resistor (R1) are grounded in the vicinity of the input connector. Supply bypass capacitors (C1, C2, C4, C5, C7, C8, C9 and C10) are returned to ground in the vicinity of the output connectors.

The circuit board is designed to accommodate standard 8-pin miniDIP, single operational amplifiers such as the LT1190 and LT1220 families. Both voltage and current feedback types can be used. Pins 1, 5 and 8 are outfitted with extra holes for use in adjusting DC offsets, compensation, or, in the case of the LT1223 and LT1190/1/2, for shutting down the amplifier.

If a current feedback amplifier such as the LT1223 is being evaluated, omit C3/C6. R4 and R6 are included for impedance matching when driving low impedance lines. If the amplifier is supposed to drive the line directly, or if the load impedance is high, R4 and R8 can be replaced by jumpers. Similarly R10 and R12 can be used to establish a load at the output of the amplifier.

Low profile sockets may be used for the op amps to facilitate changing parts, but performance may be affected above 100MHz.

High speed operational amplifiers work best when their supply pins are bypassed with R_f-quality capacitors. C1, C5, C8 and C10 should be 10nF disc ceramic or other capacitors with a self-resonant frequency greater than 10MHz. The polarized capacitors (C2, C4, C7, and C9) should be $1\mu\text{F}$ to $10\mu\text{F}$ tantalums. Most 10nF ceramics are self-resonant well above 10MHz and $4.7\mu\text{F}$ solid tantalums (axial leaded) are self-resonant at 1MHz or below. Lead lengths are critical; the self-resonant frequency of a $4.7\mu\text{F}$ tantalum drops by a factor of 2 when measured through 2" leads. Although a capacitor may become inductive at high frequencies, it is still an effective bypass component above resonance because the impedance is low.

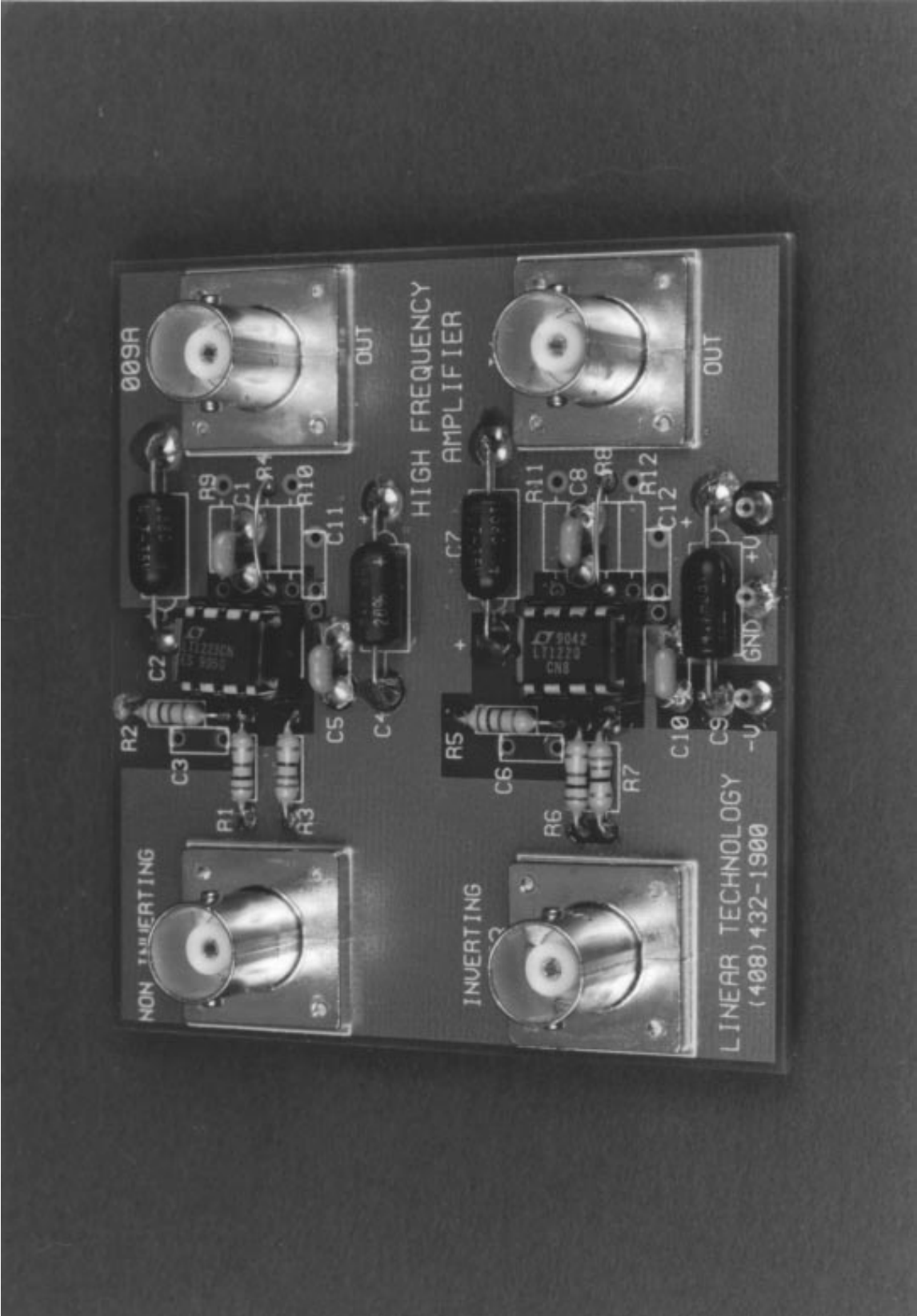


Figure 11. The Enticing LTC High Frequency Amplifier Demonstration Board. Sockets are Not Optimal, but Allow Trying Different Amplifiers

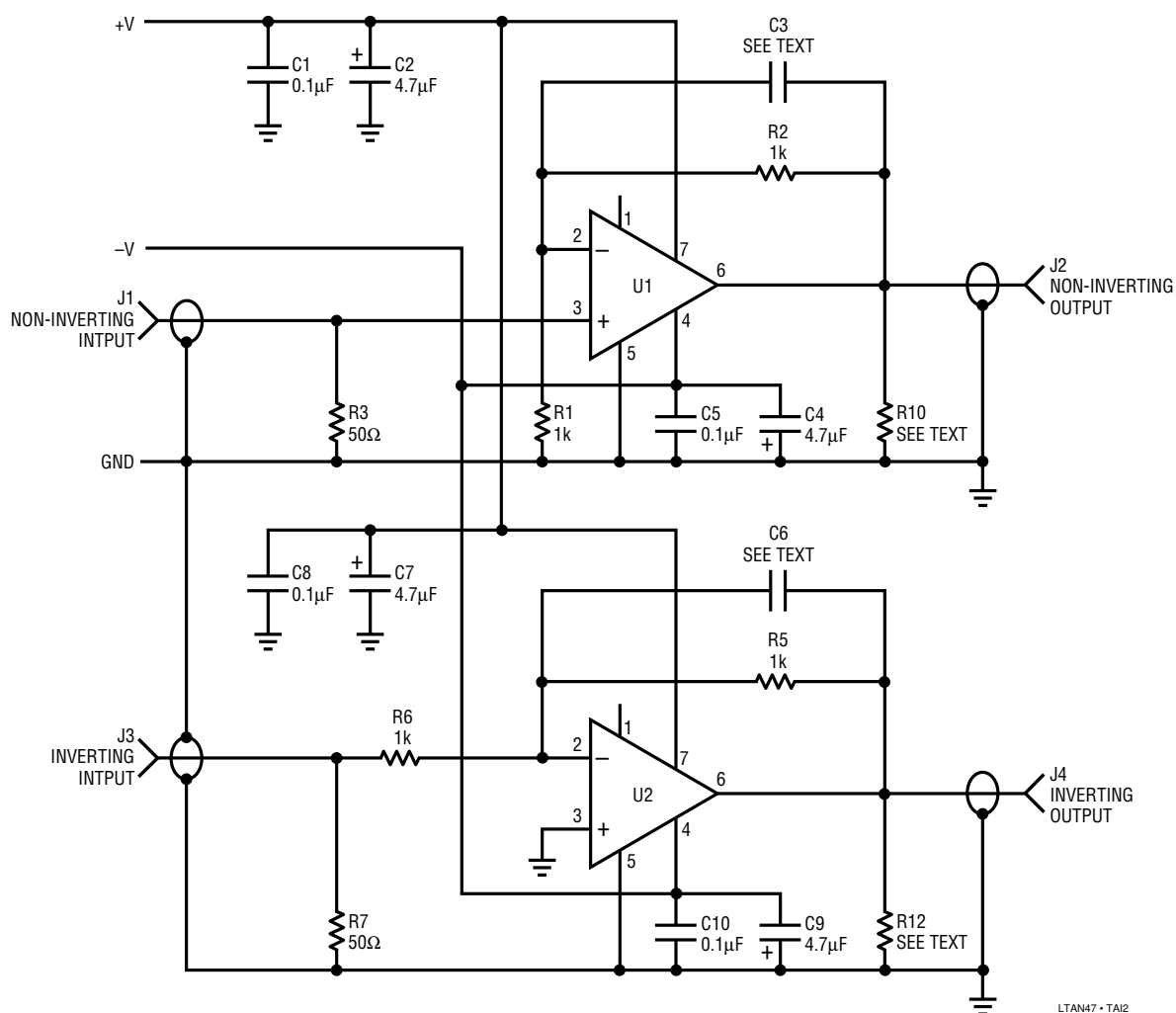


Figure I2. High Frequency Amplifier Demonstration Board Schematic

APPENDIX J

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The Contributions of Edsel Murphy to the Understanding of the Behavior of Inanimate Objects

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Abstract—Consideration is given to the effects of the contributions of Edsel Murphy to the discipline of electronics engineering. His law is stated in both general and special form. Examples are presented to corroborate the author's thesis that the law is universally applicable.

I. INTRODUCTION

IT HAS LONG BEEN the consideration of the author that the contributions of Edsel Murphy, specifically his general and special laws delineating the behavior of inanimate objects, have not been fully appreciated. It is deemed that this is, in large part, due to the inherent simplicity of the law itself.

It is the intent of the author to show, by references drawn from the literature, that the law of Murphy has produced numerous corollaries. It is hoped that by noting these examples, the reader may obtain a greater appreciation of Edsel Murphy, his law, and its ramifications in engineering and science.

As is well known to those versed in the state-of-the-art, Murphy's Law states that "If anything can go wrong, it will." Or, to state it in more exact mathematical form:

$$1 + 1^{*4} = 2 \quad (1)$$

where *4 is the mathematical symbol for hardly ever.

Some authorities have held that Murphy's Law was first expounded by H. Cohen¹ when he stated that "If anything can go wrong, it will — during the demonstration." However, Cohen has made it clear that the broader scope of Murphy's general law obviously takes precedence.

To show the all-pervasive nature of Murphy's work, the author offers a small sample of the application of the law in electronics engineering.

II. GENERAL ENGINEERING

II.1. A patent application will be preceded by one week by a similar application made by an independent worker.

II.2. The more innocuous a design change appears, the further its influence will extend.

II.3. All warranty and guarantee clauses become void upon payment of invoice.

II.4. The necessity of making a major design change

increases as the fabrication of the system approaches completion.

II.5. Firmness of delivery dates is inversely proportional to the tightness of the schedule.

II.6. Dimensions will always be expressed in the least usable term. Velocity, for example, will be expressed in furlongs per fortnight.²

II.7. An important Instruction Manual or Operating Manual will have been discarded by the Receiving Department.

II.8. Suggestions made by the Value Analysis group will increase costs and reduce capabilities.

II.9. Original drawings will be mangled by the copying machine.³

III. MATHEMATICS

III.1. In any given miscalculation, the fault will never be placed if more than one person is involved.

III.2. Any error that can creep in, will. It will be in the direction that will do the most damage to the calculation.

III.3. All constants are variables.

III.4. In any given computation, the figure that is most obviously correct will be the source of error.

III.5. A decimal will always be misplaced.

III.6. In a complex calculation, one factor from the numerator will always move into the denominator.

IV. PROTOTYPING AND PRODUCTION

IV.1. Any wire cut to length will be too short.

IV.2. Tolerances will accumulate unidirectionally toward maximum difficulty of assembly.

IV.3. Identical units tested under identical conditions will not be identical in the field.

IV.4. The availability of a component is inversely proportional to the need for that component.

IV.5. If a project requires n components, there will be $n-1$ units in stock.⁴

IV.6. If a particular resistance is needed, that value will not be available. Further, it cannot be developed with any available series or parallel combination.⁵

IV.7. A dropped tool will land where it can do the most damage. (Also known as the law of selective gravitation.)

IV.8. A device selected at random from a group having 99% reliability, will be a member of the 1% group.

IV.9. When one connects a 3-phase line, the phase

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sequence will be wrong.⁶

IV.10. A motor will rotate in the wrong direction.⁷

IV.11. The probability of a dimension being omitted from a plan or drawing is directly proportional to its importance.

IV.12. Interchangeable parts won't.

IV.13. Probability of failure of a component, assembly, subsystem or system is inversely proportional to ease of repair or replacement.

IV.14. If a prototype functions perfectly, subsequent production units will malfunction.

IV.15. Components that must not and cannot be assembled improperly will be.

IV.16. A dc meter will be used on an overly sensitive range and will be wired in backwards.⁸

IV.17. The most delicate component will drop.⁹

IV.18. Graphic recorders will deposit more ink on humans than on paper.¹⁰

IV.19. If a circuit cannot fail, it will.¹¹

IV.20. A fail-safe circuit will destroy others.¹²

IV.21. An instantaneous power-supply crowbar circuit will operate too late.¹³

IV.22. A transistor protected by a fast-acting fuse will protect the fuse by blowing first.¹⁴

IV.23. A self-starting oscillator won't.

IV.24. A crystal oscillator will oscillate at the wrong frequency — if it oscillates.

IV.25. A pnp transistor will be an npn.¹⁵

IV.26. A zero-temperature-coefficient capacitor used in a critical circuit will have a TC of $-750 \text{ ppm}/^{\circ}\text{C}$.

IV.27. A failure will not appear till a unit has passed Final Inspection.¹⁶

IV.28. A purchased component or instrument will meet its specs long enough, and only long enough, to pass Incoming Inspection.¹⁷

IV.29. If an obviously defective component is replaced in an instrument with an intermittent fault, the fault will reappear after the instrument is returned to service.¹⁸

IV.30. After the last of 16 mounting screws has been removed from an access cover, it will be discovered that the wrong access cover has been removed.¹⁹

IV.31. After an access cover has been secured by 16 hold-down screws, it will be discovered that the gasket has been omitted.²⁰

IV.32. After an instrument has been fully assembled, extra components will be found on the bench.

IV.33. Hermetic seals will leak.

V. SPECIFYING

V.1. Specified environmental conditions will always be exceeded.

V.2. Any safety factor set as a result of practical experience will be exceeded.

V.3. Manufacturers' spec sheets will be incorrect by a factor of 0.5 or 2.0, depending on which multiplier gives the most optimistic value. For salesmen's claims these factors will be 0.1 or 10.0.

V.4. In an instrument or device characterized by a number of plus-or-minus errors, the total error will be the sum of all errors adding in the same direction.

V.5. In any given price estimate, cost of equipment will exceed estimate by a factor of 3.²¹

V.6. In specifications, Murphy's Law supersedes Ohm's.

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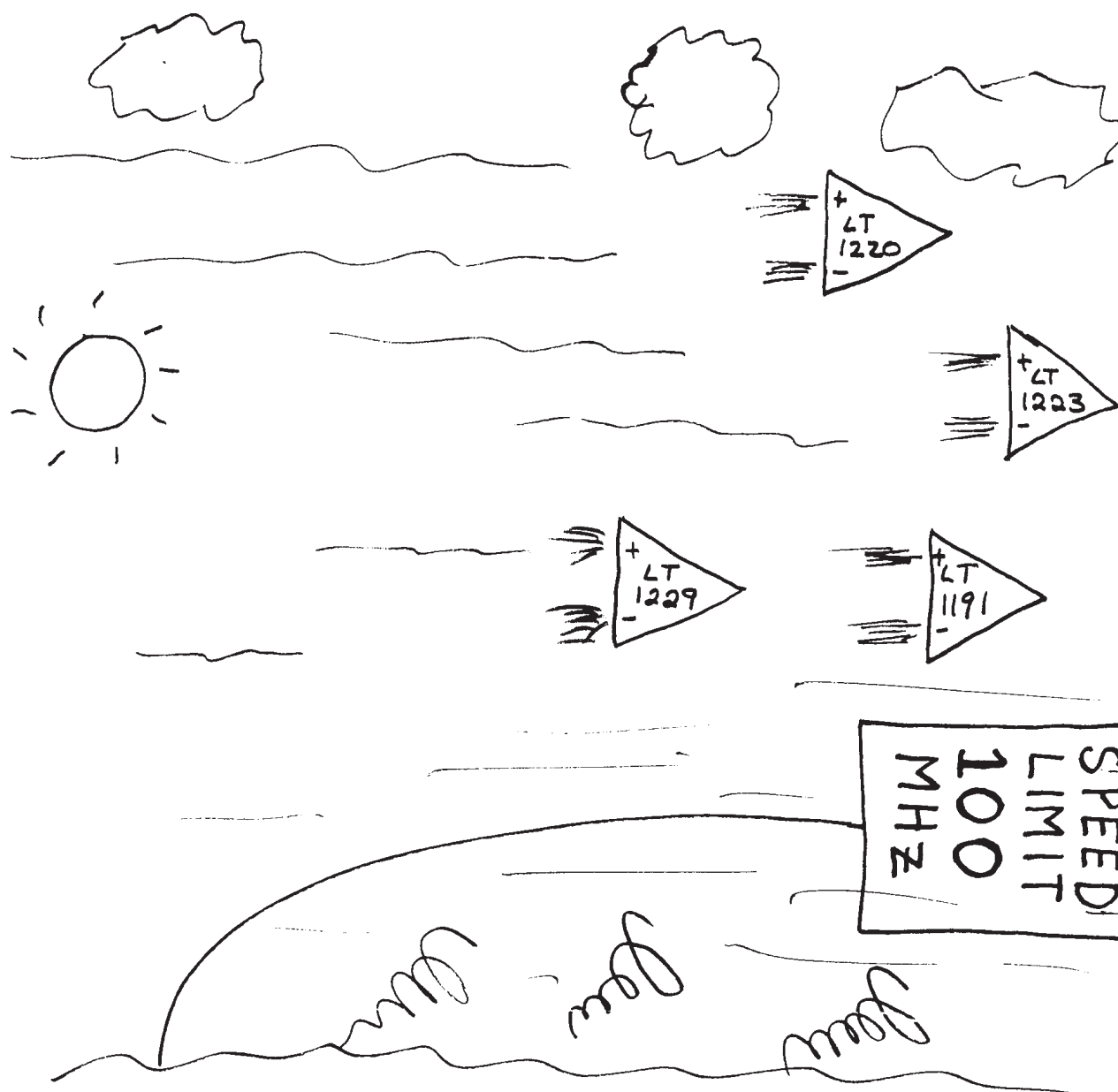
*In some cases where no reference is given, the source material was misplaced during preparation of this paper (another example of Murphy's Law). In accordance with the law, these misplaced documents will turn up on the date of publication of this paper.



The man who developed one of the most profound concepts of the twentieth century is practically unknown to most engineers. He is a victim of his own law. Destined for a secure place in the engineering hall of fame, something went wrong.

His real contribution lay not merely in the discovery of the law but more in its universality and in its impact. The law itself, though inherently simple, has formed a foundation on which future generations will build.

Application Note 47



(I WISH THIS APP. NOTE WENT AS
FAST AS LTC AMPLIFIERS)

Weller
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